

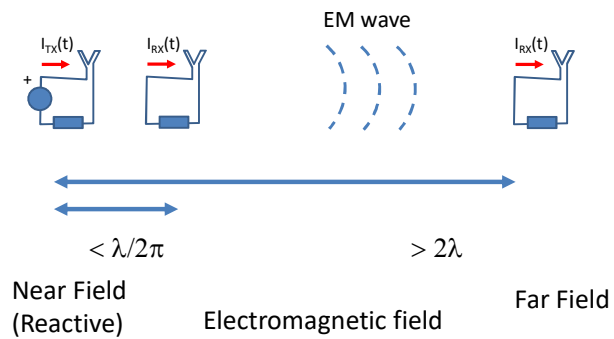
Mobile Internet

10.- Sumario

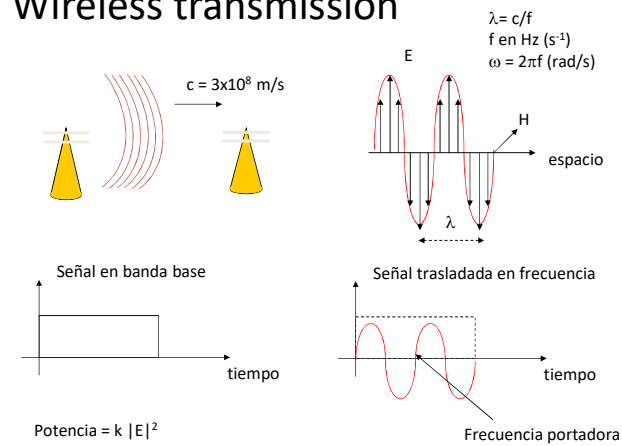
Mobile Internet

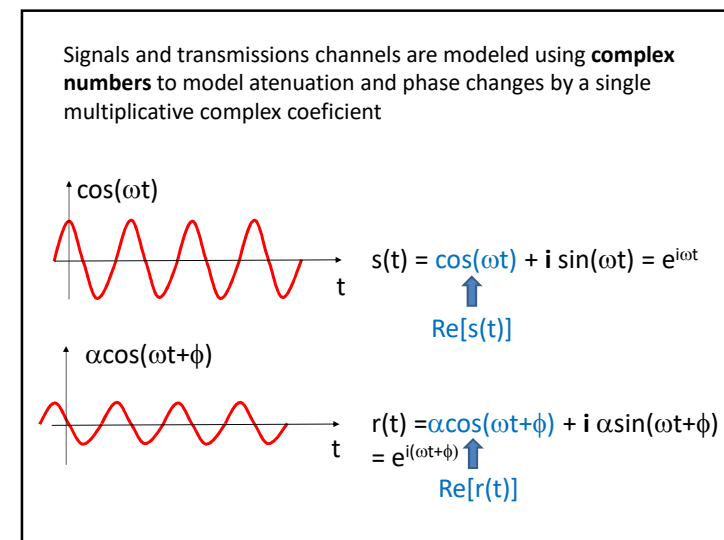
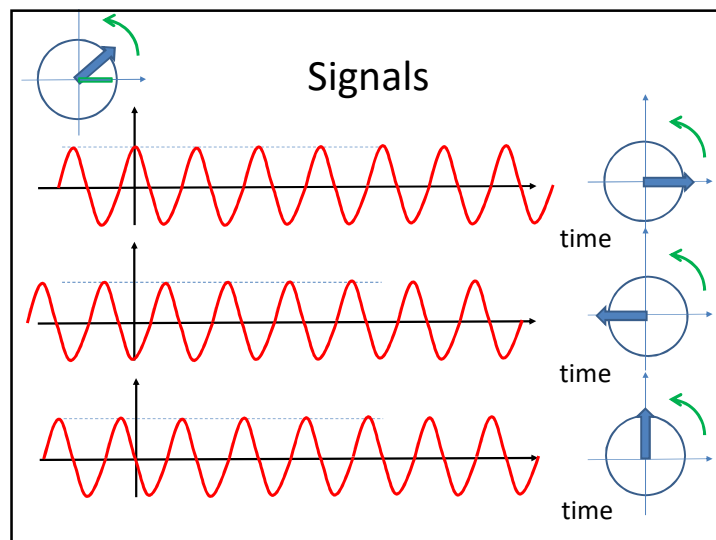
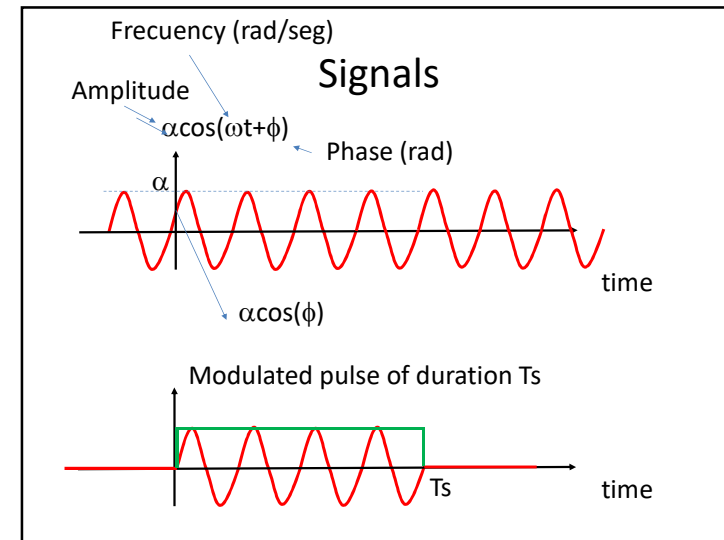
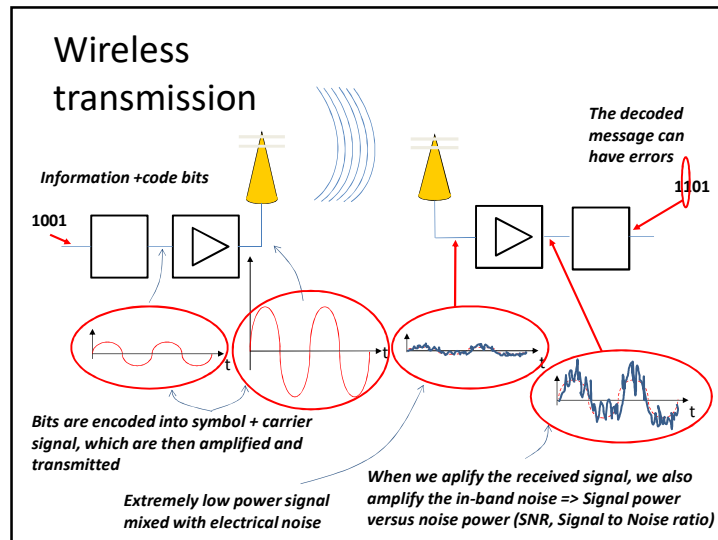
2.- Wireless transmission

Physical principles: Near/Far EM Field

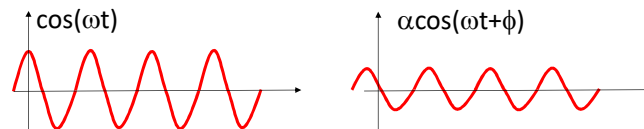


Wireless transmission





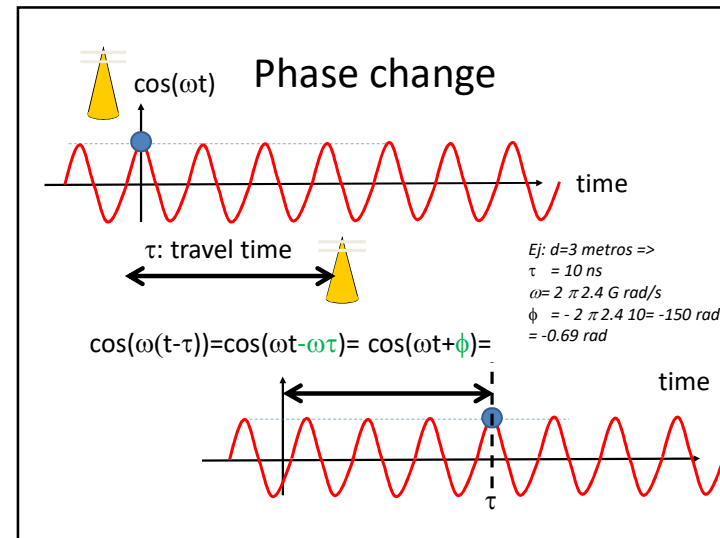
Signals and transmissions channels are modeled using **complex numbers** to model attenuation and phase changes by a single multiplicative complex coefficient



$$h = \alpha e^{i\phi}$$

$$r(t) = s(t)h = e^{i\omega t} \alpha e^{i\phi} = \alpha e^{i(\omega t + \phi)}$$

Channel effect: modify amplitude and phase



Reminder: dB

- Loss:
 - P_{tx}/P_{rx} (No units)
 - $10 \log(P_{tx}/P_{rx})$ (decibels, dB)
- Eg:
 - P_{tx} : 100 mW,
 - P_{rx} : 10 nW
 - Loss: $10.000.000 = 70$ dB

Reminder: dBm

- Power in dBm:
 - $10 \log(P/1 \text{ mW})$ (dBm)
- Eg:
 - $P = 100 \text{ mW} = 20 \text{ dBm}$
 - $P = 10 \text{ nW} = -50 \text{ dBm}$
 - Loss (dB) = $P_{tx} \text{ (dBm)} - P_{rx} \text{ (dBm)} = 70 \text{ dB}$

Wireless Transmission (Land)

$$\frac{P_r}{P_t} = k \frac{1}{d^2}$$

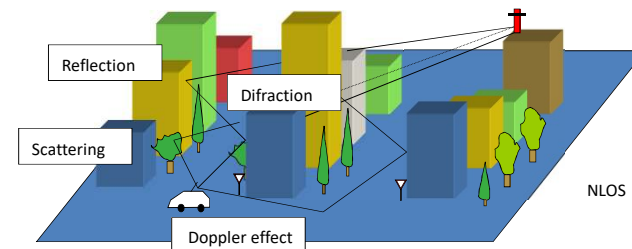
In land transmi^{ón}, we have different attenuation exponents with distance

$$\frac{P_r}{P_t} = k \frac{1}{d^\alpha} \quad 1.5 \leq \alpha \leq 6$$

In many examples we will use $\alpha=4$

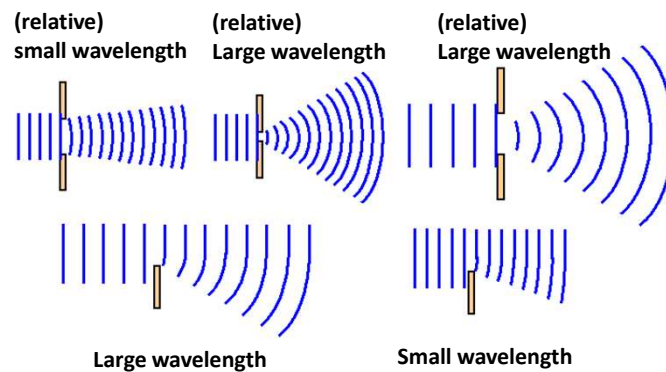
Wireless transmission

- Fading, shadowing, *multipath fading*, Doppler effect, changing channel conditions, etc



Diffraction

Better coverage for lower frequencies (i.e. larger wavelengths)



Source: https://www.tes.com/lessons/FO9uRawQJ_pHww/p1-diffraction-gcse-science

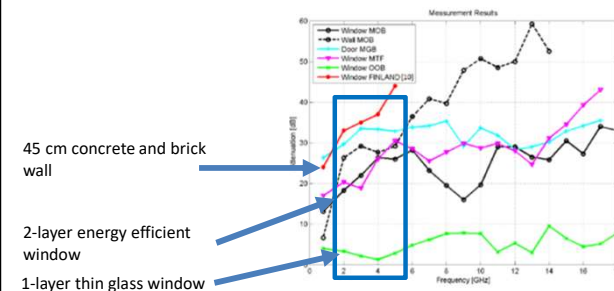
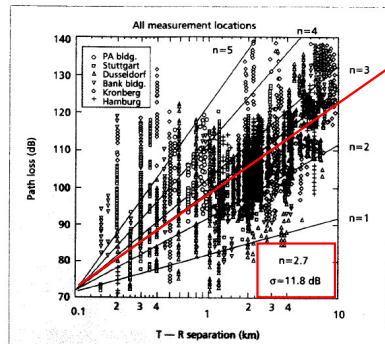


Fig. 6. Measured attenuation for the different scenarios in the range from 800 MHz to 18 GHz.

Source: Radio Propagation into Modern Buildings: Attenuation Measurements in the Range from 800 MHz to 18 GHz, Ignacio Rodríguez, Han C. Nguyen, Mels T. K. Jørgensen, Troels B. Sørensen and Peter Mogensen

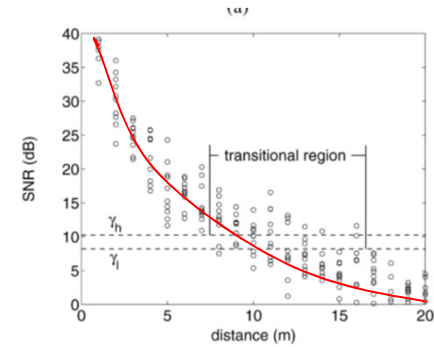
Atenuation



■ Figure 3. Scatter plot of path loss vs. distance for measurements in five German cities. The path loss is referenced to a free space reference measurement at $d_0 = 100$ m. Note that σ has a value of 11.8 dB due to the large spread of PL values relative to the straight-line fit.

$1/d^n +$
Log-Normal...

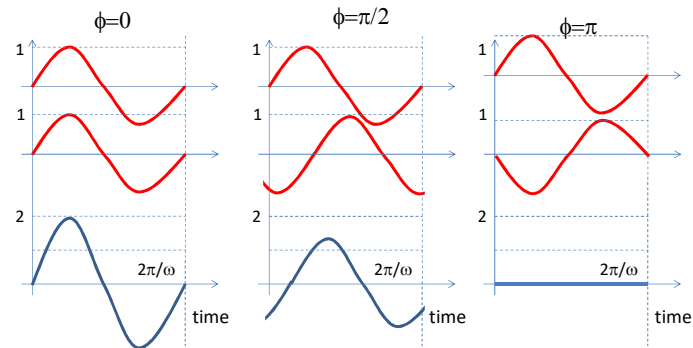
Atenuation



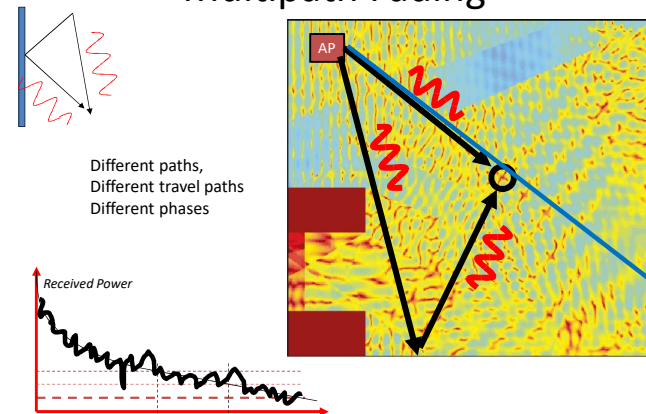
Ref: M Zúñiga, B. Krishnamachari, "An Analysis of Unreliability and Asymmetry in Low-Power Wireless Links", 2007

Adding sinusoids with different phases

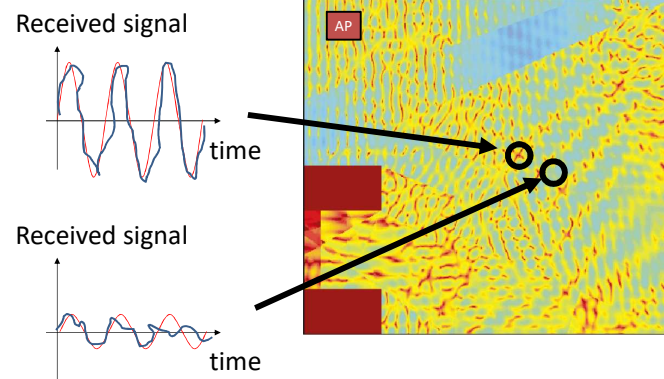
$$\sin(\omega t) + \sin(\omega t + \phi) = 2 \cos(\phi/2) \sin(\omega t + \phi/2)$$



Multipath Fading

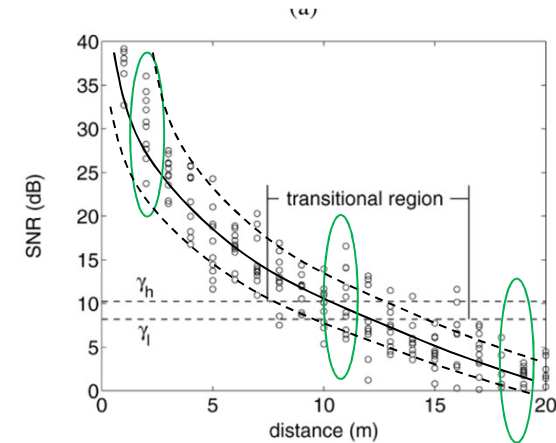


Signal to Noise ratio (SNR)

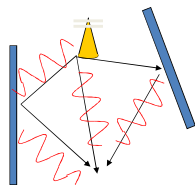


Atenuation

Ref: M Zúñiga, B. Krishnamachari, "An Analysis of Unreliability and Asymmetry in Low-Power Wireless Links", 2007



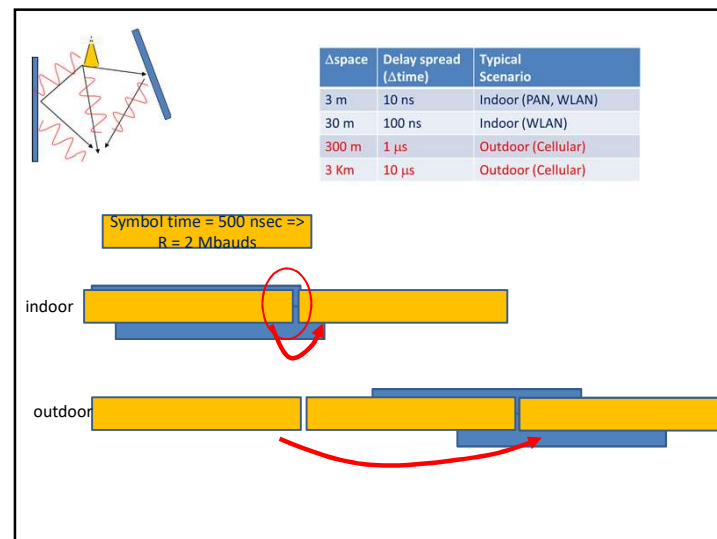
Delay spread in indoor/outdoor



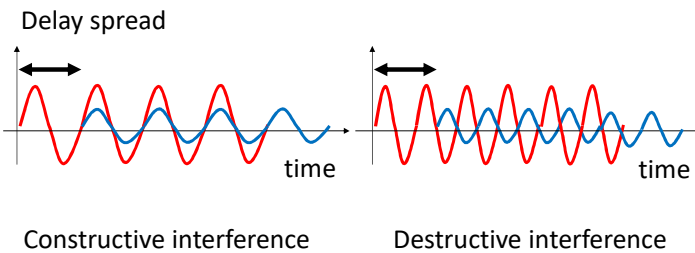
The **delay spread** is the maximum difference between the arrival times of different multipath signals.

Δ space	Delay spread (Δ time)	Typical Scenario
3 m	10 ns	Indoor (PAN, WLAN)
30 m	100 ns	Indoor (WLAN)
300 m	1 μ s	Outdoor (Cellular)
3 Km	10 μ s	Outdoor (Cellular)

Symbol time = 500 nsec

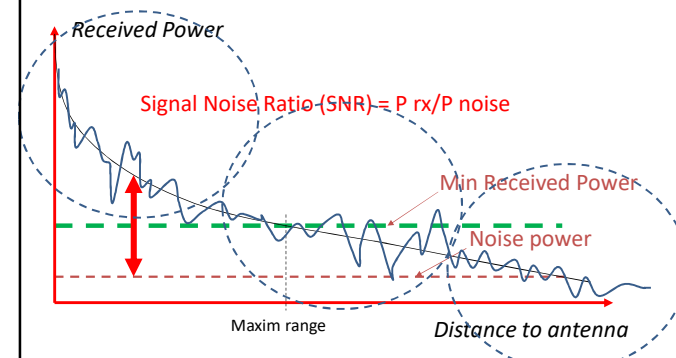


Same delay spread, different fc



Radio transmission

Received power > minimum threshold

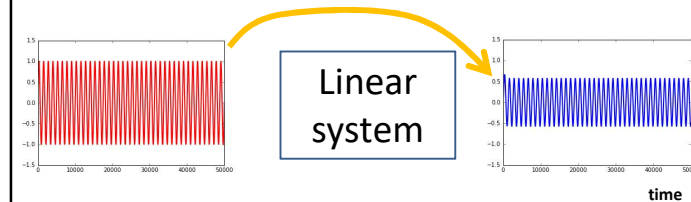


Frequency response

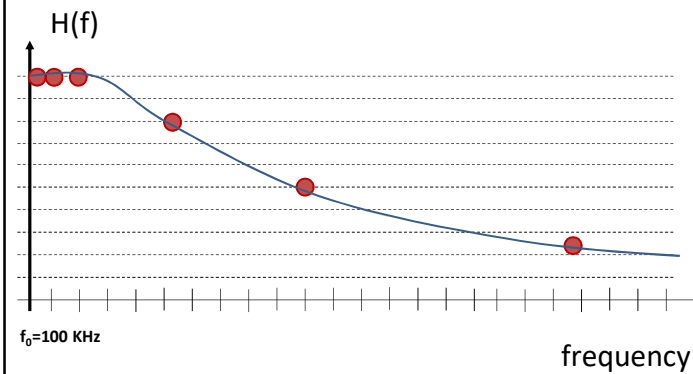
Response to sinusoidal signals

- When the input of a time-invariant linear system is a sinusoidal signal of a given frequency, the response is **another sinusoidal signal of the same frequency**, but different amplitude and phase (**depending on the frequency**)

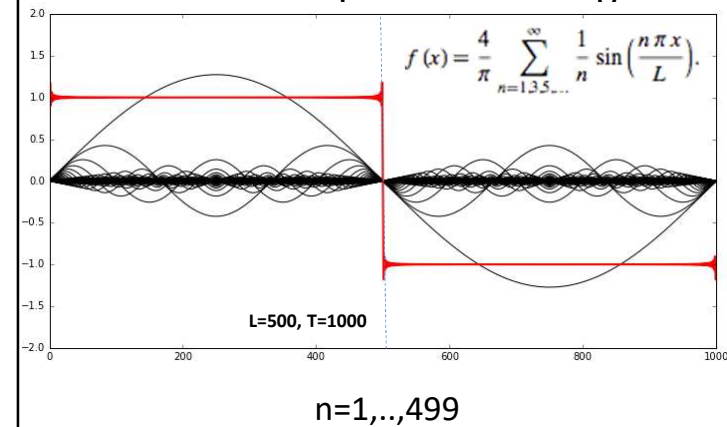
$$\text{output amplitude} = \text{input amplitude} \times H(f)$$



Amplitude frequency response

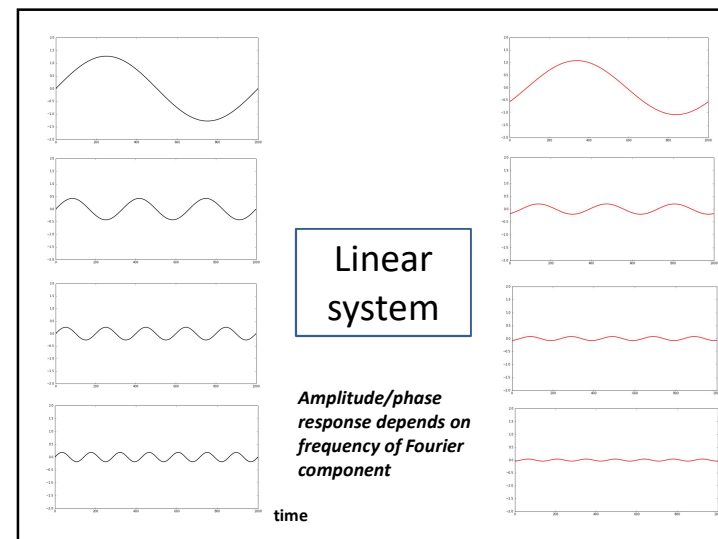


Fourier decomposition of a signal



Main idea behind signal filtering

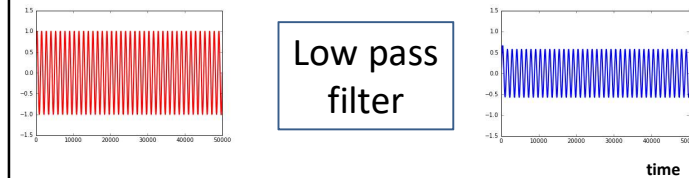
Exploit **Fourier decomposition** and **superposition property of linear systems** to predict the output signal



Noise & signal filtering

Filtering

- Low pass filters: attenuates/rejects high frequency components of a signal



Low-pass filters

- Simple example of **low-pass FIR** (Finite Impulse Response) filter

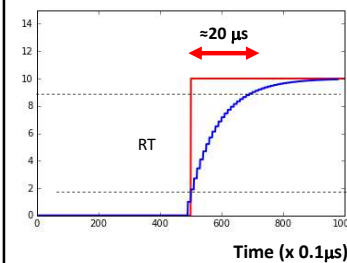
$$y[t] = (1/N) * (x[t-(N-1)] + \dots + x[t-2] + x[t-1] + x[t])$$

- Simple example of **low-pass IIR** (Infinite Impulse response) filter

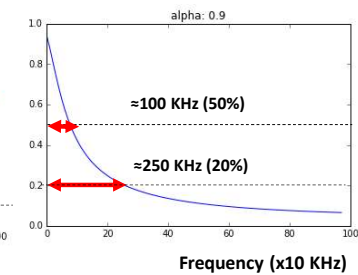
$$y[t] = \alpha * y[t-1] + (1-\alpha) * x[t]$$

Time and frequency response

Rising Time (e.g. 10 to 90%)



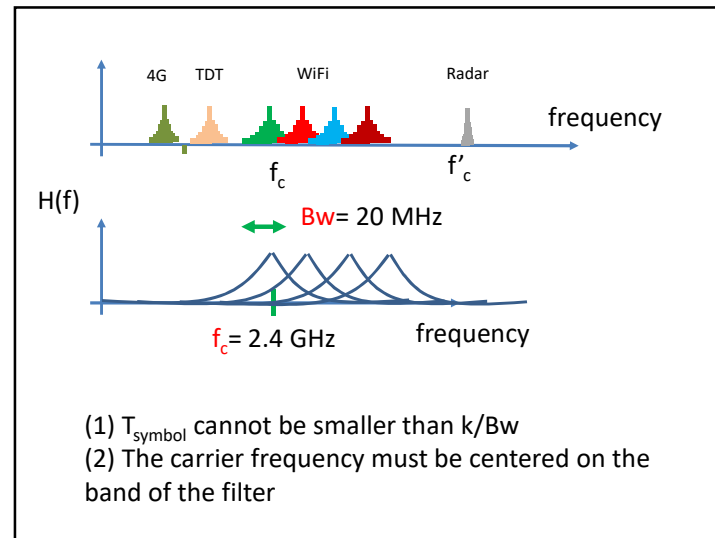
Bandwidth (e.g. 50%, 20%)



Time and frequency response

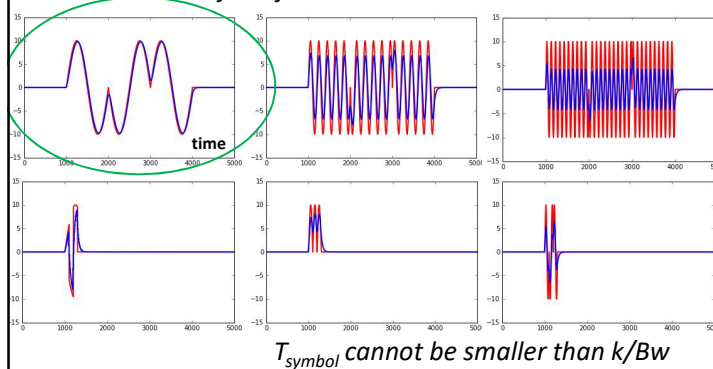
- There is a relationship between Bandwidth (Bw) and Rising Time (RT).
- The exact formula depends on the thresholds we use to define bandwidth and rising times
- In general:

$$Bw = \frac{k}{RT}$$



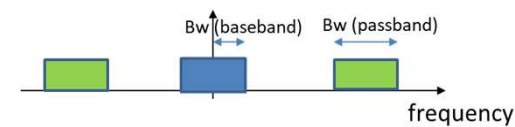
Putting all together, Passband

The carrier frequency must be centered on the band of the filter

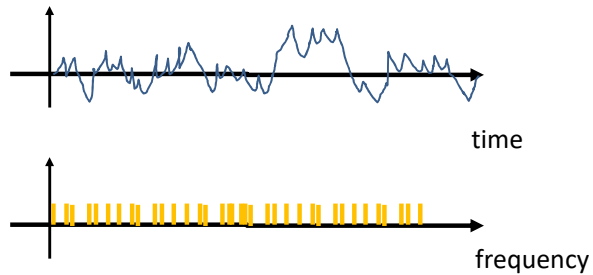


Bw and symbols per second

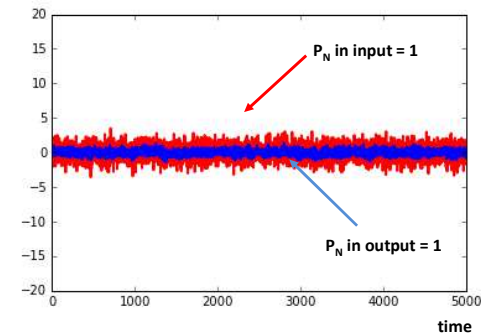
- **For radiofrequencies (passband)** it doubles, so $R=Bw$ (e.g. 1kbaud of rate takes 1 kHz of band, in 1 MHz of band it is possible to transmit 1Mbaud).
- True rates in practice for a given band are usually lower (e.g. 10-50%).



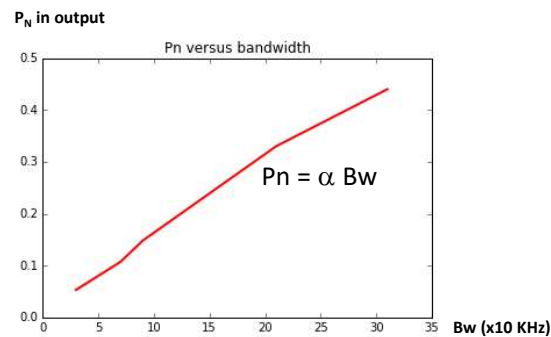
White noise



Noise power versus bandwidth



Noise power versus bandwidth



Electronic Noise

- In **electronic circuits**

*Densidad espectral
de potencia de ruido*

$$P_N (\text{output}) = N_0 \times Bw$$

$$N_0 = -174 \text{ dBm/Hz} \approx 4 \times 10^{-18} \text{ mW/Hz} \\ \approx 4 \times 10^{-21} \text{ W/Hz}$$

E.g. Wifi: $Bw=20 \text{ MHz} \Rightarrow P_N \approx -101 \text{ dBm}$

For a BPSK or QPSK receiver $\text{SNR}_{\text{min}} = 7 \text{ dB}$, in theory the minimum receive power is -94 dBm , although in practice it must be higher (e.g. -84 dBm)

E_b/N_0 and R/Bw

$$SNR = P_{Rx}/P_N$$

We can express P_{Rx} and P_N as:

$$\begin{cases} P_{Rx} = E_b R \\ P_N = N_0 Bw \end{cases}$$

Does not include code bits!!

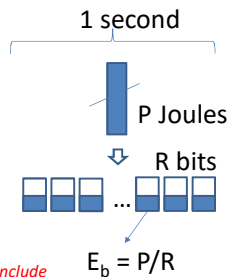
Obtaining

$$SNR = (E_b/N_0) (R/Bw)$$

Spectral efficiency
No units: (b/s)/(1/s)

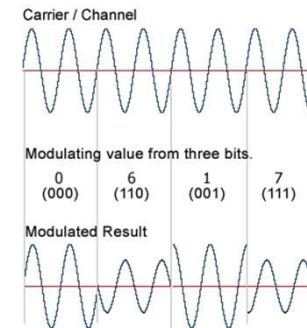
Energy per bit to noise power spectral density ratio

No units: J/(W/s)



Multilevel modulation

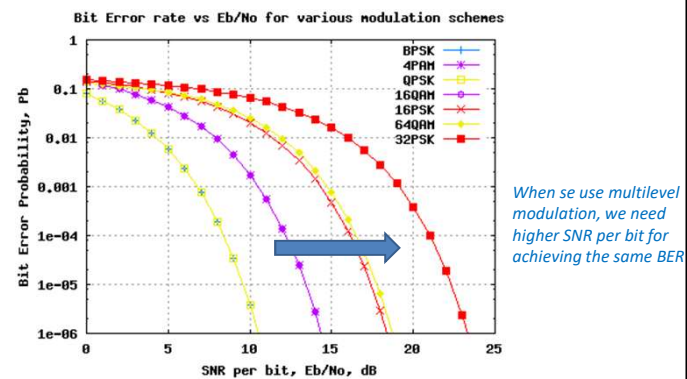
DIGITAL QAM (8QAM)



Note: Only four (0, 6, 1 and 7) out of the eight possible modulation states (0-7) are shown in this illustration.

Source: <https://www.pcmag.com/encyclopedia/term/50007/qam>

Multilevel modulation



HAMMING BINARY BLOCK CODE WITH $k=4$ AND $n=7$

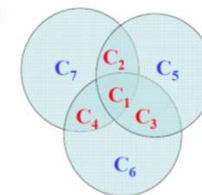
- Message digits: $C_1 C_2 C_3 C_4$
- Code word: $C_1 C_2 C_3 C_4 C_5 C_6 C_7$

Parity Check Equations:

$$C_1 \oplus C_2 \oplus C_3 \oplus C_5 = 0$$

$$C_1 \oplus C_3 \oplus C_4 \oplus C_6 = 0$$

$$C_1 \oplus C_2 \oplus C_4 \oplus C_7 = 0$$



The circles represent the equations.

Parity Check Matrix:

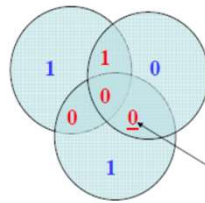
$$\begin{bmatrix} 1 & 1 & 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 1 & 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 1 & 0 & 0 & 1 \end{bmatrix}$$

There is an even number of 1's in each circle.

Source: <http://circuit.ucsd.edu/~yhk/ece154c-spr17/pdfs/ErrorCorrectionI.pdf>

HAMMING (7,4) CODE: DECODING

- Transmitted code word: 0 1 1 0 0 1 1
- Example 1: Received block with one error in a message bit.
0 1 0 0 0 1 1



By counting 1's in each circle we find:
 There is an error in right circle.
 There is an error in bottom circle.
 There is no error in left circle.
 Therefore the error is in the third digit!

Source: <http://circuit.ucsd.edu/~yhk/ece154c-spr17/pdfs/ErrorCorrectionI.pdf>

Channel capacity

- Using *multilevel modulation* we increase R/Bw but we need more SNR for the same BER
- Using *error correcting codes* we decrease R/Bw but we need lower SNR for the same BER
- Fundamental limits for R/Bw versus E_b/N_0 for achieving arbitrary low levels of BER**
 => C. Shannon

Channel Capacity (AWGN)

$$\frac{C}{Bw} = \log_2 \left(1 + \frac{E_b R}{N_0 Bw} \right)$$

$$\frac{2^{R/Bw} - 1}{R/Bw} \leq E_b/N_0$$

N_0 : Spectral noise power density at receiver (W/Hz=J)
 E_b : Energy per bit at receiver (J/b)
 Bw : Bandwidth (Hz)
 C, R : Capacity, Rate (information bits) (b/s);
 $R < C$

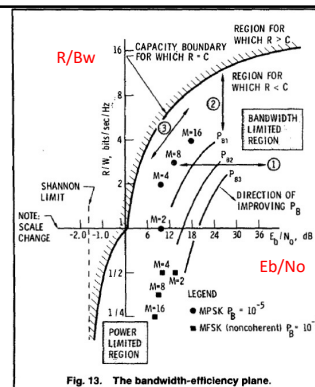
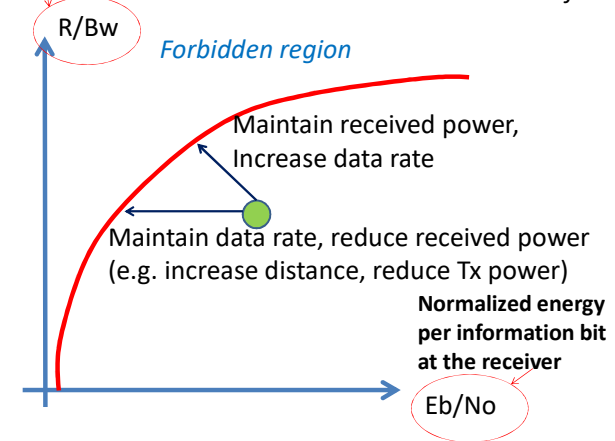


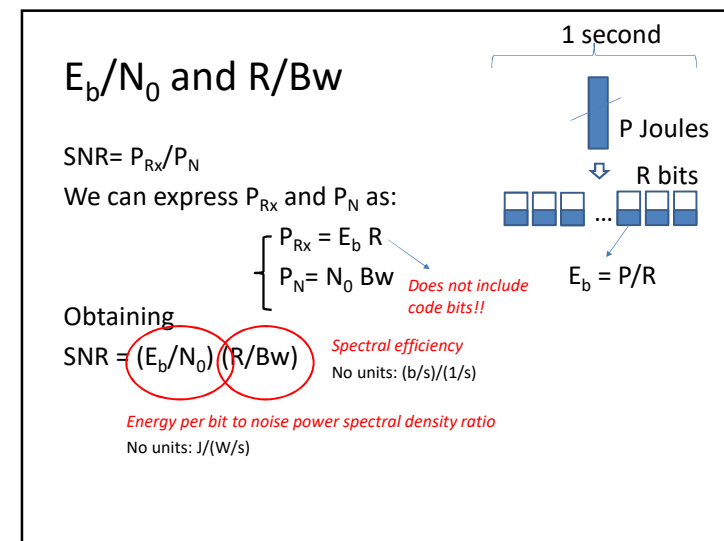
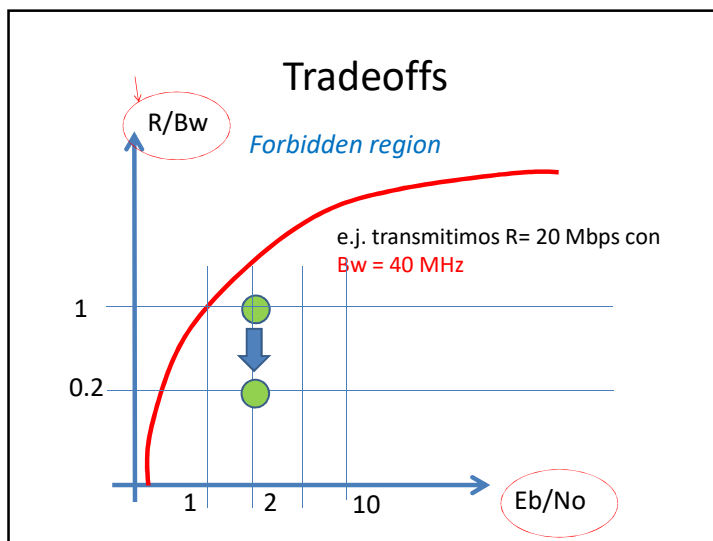
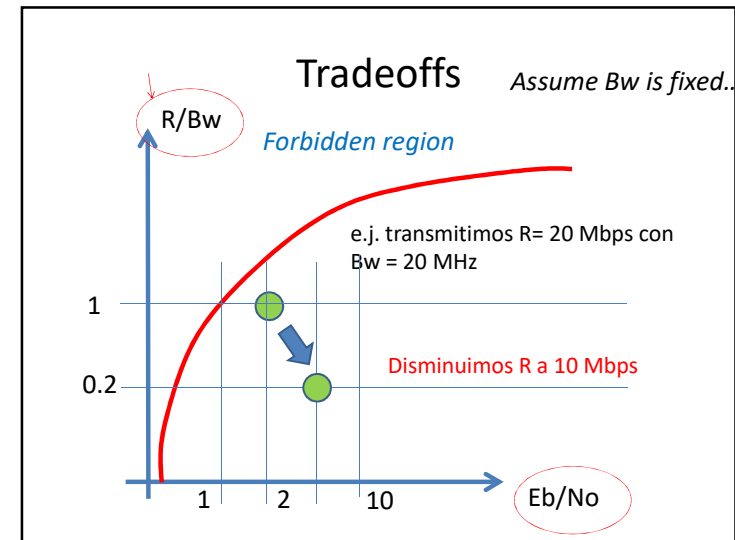
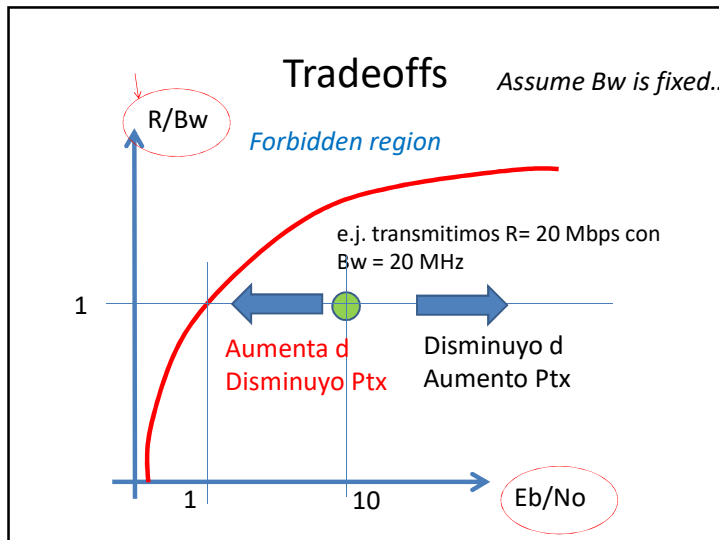
Fig. 13. The bandwidth-efficiency plane.

Spectral efficiency

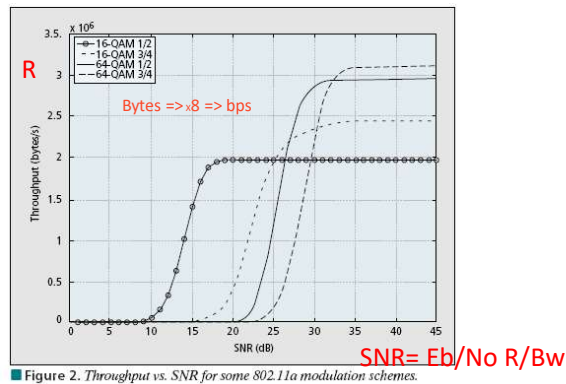
Tradeoffs

Assume Bw is fixed..





e.g. 802.11a rate adaptation



Advanced transmission techniques

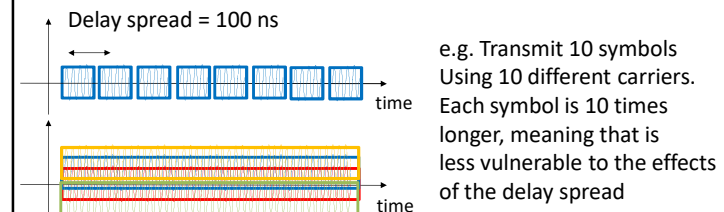
- Modern wireless transmission systems use more sophisticated transmission techniques in order to increase data rate or range:
 - OFDM**: Solves some problems that appear when we use higher bandwidth in a wireless channel (e.g. 802.11agn, 4G)
 - Multiple Antenna In, Multiple Antenna Out (MIMO)**: increased capacity, meaning faster transmission or larger range (e.g. 4G, 802.11n)

Bw and R in different technologies

Tech	Bw	R	R/Bw	Ptx	d
4G (2x2)	5, 20 MHz	20 Mbps	1	100 mW	1Km
802.11b	20 MHz	54 Mbps	3	60 mW	100 m
802.11n (2x2)	20 MHz	150 Mbps	7	60 mW	100 m
BLE	2 MHz	125 Kbps	0.2	10 mW	10 m
802.15.4	2 MHz	115 Kbps	0.2	1 mW	10 m
LoRA	125 KHz	50 Kbps (xDC)	0.4	25 mW DC=1%	1 Km
NBIoT	180 KHz	60 Kbps (xDC)	0.3	100 mW DC nolimit	1 Km

Multicarrier modulation

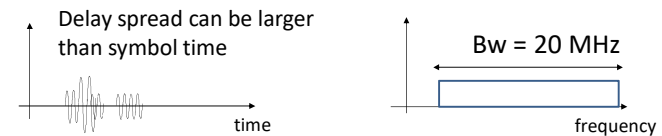
- Idea: instead of transmitting consecutive many short symbols using a common carrier, **transmit many simultaneous long symbols using multiple carriers.**



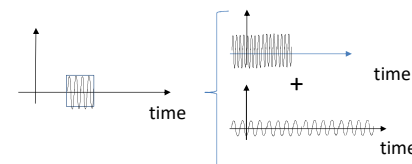
OFDM: diversity

- The different subchannels can suffer different multipath fading.
- The explanation to this comes from the fact that **the same delay spread can lead to a destructive or constructive interference depending on the frequency of the carrier.**
- We can use some subcarriers to estimate the multipath fading using pilot tones

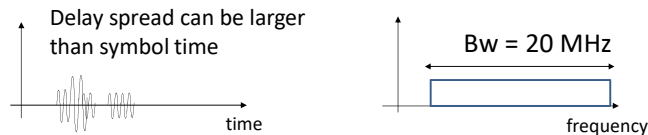
Wideband Wireless channels



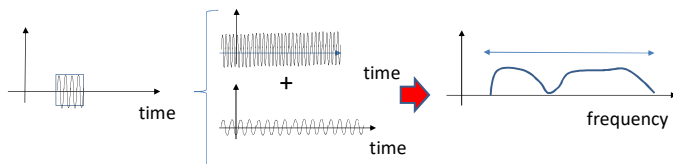
The Fourier components of short symbols span many different frequencies and each component can suffer different multipath attenuations



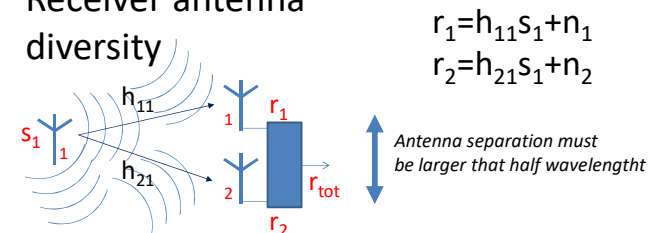
Wideband Wireless channels



This means that wideband wireless channels suffer a **frequency selective multipath fading**



Receiver antenna diversity

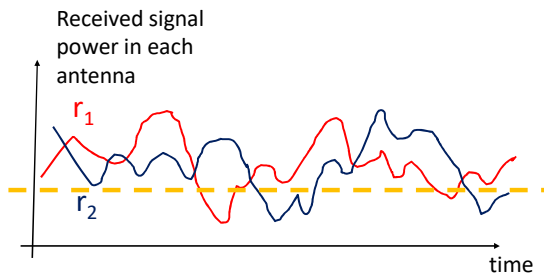


Due to the receiver antenna separation, there are different travel times for the interfering paths. This means that **the signals received in antennas 1 and 2 suffer in general different multipath fading.**

Noise not included in the drawing...

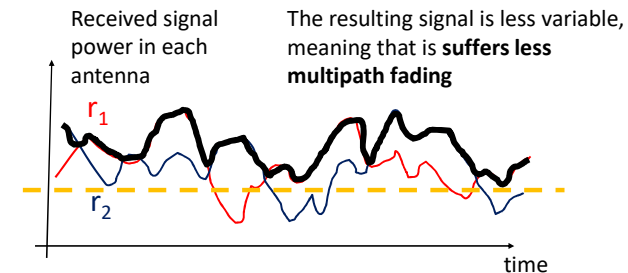
Receiver antenna diversity: Selection combining

Use the antenna with the **strongest signal** (i.e. largest SNR) to receive the packet and ignore the others.

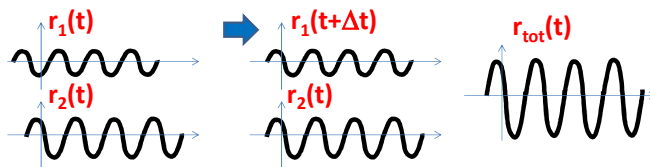


Selection combining: Diversity gain

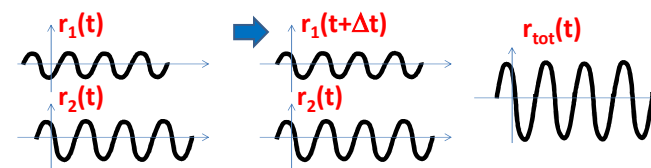
Use the antenna with the **strongest signal** (i.e. largest SNR) to receive the packet and ignore the others.



Receiver antenna diversity: Maximal Ratio Combining (MRC)



Better option: Add both signals **in phase** (coherently) : delay one signal so that both signals are in phase before they are added



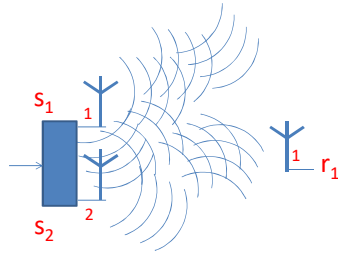
We add both signal **in phase** (coherently) => We add amplitudes. e.g_ when $A_1=A_2 \Rightarrow A_{tot}=2 A_1 \Rightarrow P_{tot}=4 P_1$
For the noise, we have **incoherent sum** (add powers) =>
 $PN_{tot} = 2P_{N1}$

$$SNR_{tot} = SNR_1 + SNR_2$$

Additionally, we can give more weight in the sum to the signal with better SNR

Transmitter antenna diversity

$$r_1 = h_{11}s_1 + h_{12}s_2 + n_1$$

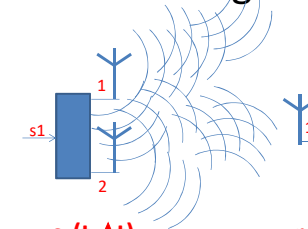


We can also use an equivalent to MRC, called **beamforming**. The transmitter delays one of the signal to change the phase such that they combine constructively at the receiver's antenna,

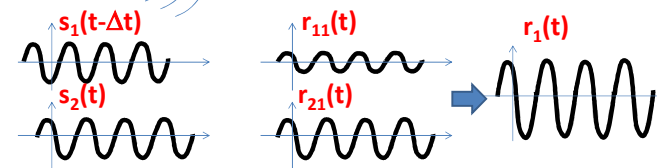
Noise not included in the drawing...

Beamforming

$$r_1 = h_{11}s_1 + h_{12}s_2 + n_1$$



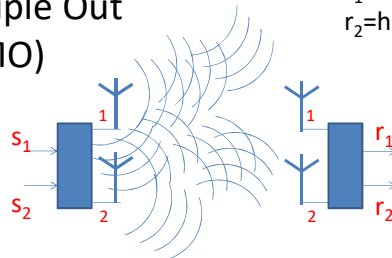
The **received signal is less variable** (diversity gain)
Feedback is needed to adjust Δt



Noise not included in the drawing...

Multiple In Multiple Out (MIMO)

$$\begin{aligned} r_1 &= h_{11}s_1 + h_{12}s_2 + n_1 \\ r_2 &= h_{21}s_1 + h_{22}s_2 + n_2 \end{aligned}$$



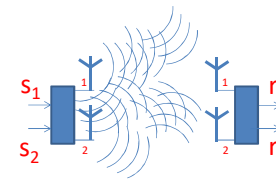
Transmit same symbol in both Tx antennas, meaning that received signal is less variable (**diversity gain**)

Transmit different symbol through each Tx antenna (**multiplexing gain**)

Direct mapping MIMO

$$r = Hs + n$$

$$\begin{aligned} r_1 &= h_{11}s_1 + h_{12}s_2 + n_1 \\ r_2 &= h_{21}s_1 + h_{22}s_2 + n_2 \end{aligned}$$



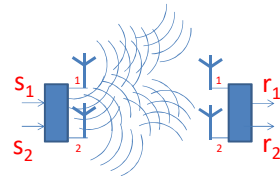
$$s = \begin{bmatrix} s_1 \\ s_2 \end{bmatrix} = \begin{bmatrix} a_1 \cos(wt) \\ a_2 \cos(wt) \end{bmatrix}$$

$$r = \begin{bmatrix} r_1 \\ r_2 \end{bmatrix}$$

$$H = \begin{bmatrix} h_{11} & h_{12} \\ h_{21} & h_{22} \end{bmatrix}$$

In general the elements of H are complex numbers

Direct mapping MIMO



$$s = \begin{bmatrix} (+1)\cos(wt) \\ (-1)\cos(wt) \end{bmatrix}$$

$$H = \begin{bmatrix} 0.1 & 0.2 \\ -0.3 & 0.1 \end{bmatrix}$$

$$r = \begin{bmatrix} (-0.1)\cos(wt) \\ (-0.4)\cos(wt) \end{bmatrix}$$

$$H^{-1} = \begin{bmatrix} 1.429 & -2.857 \\ 4.286 & 1.429 \end{bmatrix}$$

$$\text{Estimation of } s = \begin{bmatrix} 0.9999 \\ -1.0002 \end{bmatrix}$$

Here we are not taking noise into account

Direct mapping MIMO

$$s = \begin{bmatrix} (+1)\cos(wt) \\ (-1)\cos(wt) \end{bmatrix}$$

$$H = \begin{bmatrix} 0.1 & 0.2 \\ -0.3 & 0.1 \end{bmatrix}$$

$$r = \begin{bmatrix} (-0.1)\cos(wt) + (-0.01)\cos(wt) \\ (-0.4)\cos(wt) + (0.02)\cos(wt) \end{bmatrix}$$

$$H^{-1} = \begin{bmatrix} 1.429 & -2.857 \\ 4.286 & 1.429 \end{bmatrix}$$

$$\text{Estimation of } s = \begin{bmatrix} 0.928 \\ -1.014 \end{bmatrix}$$

Taking noise into account

Due to noise we cannot solve exactly the linear system of equations => Use SVD

Assume that Tx and Rx know the channel (H) =>

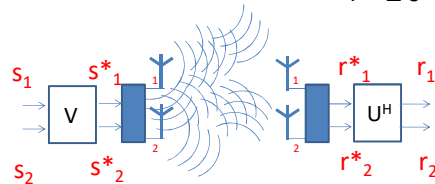
Singular Value Decomposition

$$\text{SVD: } H = U \Sigma V^H$$

$$s^* = V s \quad \text{"Transmit precoding"}$$

$$r = U^H (r^* + n) \quad \text{"Receiver shaping"}$$

$$r = \Sigma s + n' \Rightarrow r_i = \sigma_i s_i + n'_i$$



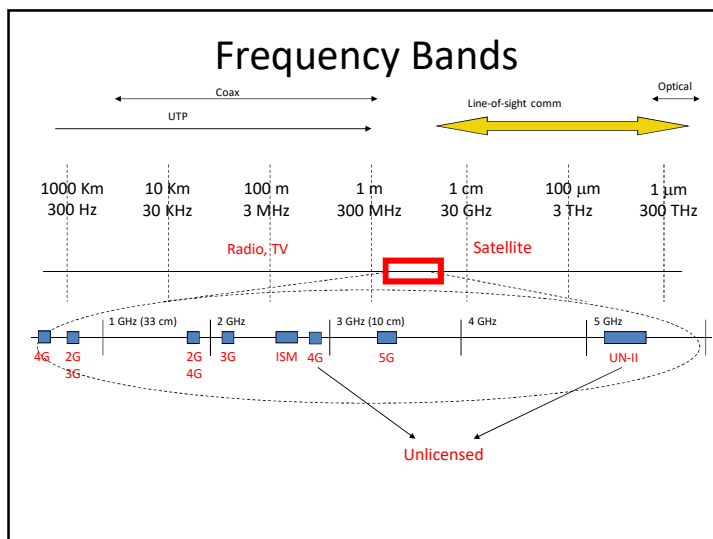
MIMO Capacity increase, ideal case

Method	Capacity (bits/sec)
SISO	$B \log_2(1 + \rho)$
Diversity (1xN or Nx1)	$B \log_2(1 + \rho N)$
Diversity (NxN)	$B \log_2(1 + \rho N^2)$
Multiplexing	$BN \log_2(1 + \rho)$

Improve SNR

Table 1: Capacity for wireless links in an ideal channel. Diversity and multiplexing respectively improve SNR and capacity. For an $N \times N$ link, diversity is better to use at low SNR, multiplexing at high SNR. These best-case gains will vary in real channels.

Source: "802.11 with Multiple Antennas for dummies", D. Halperin, W. Wu, A. Sheth, D. Wetherall



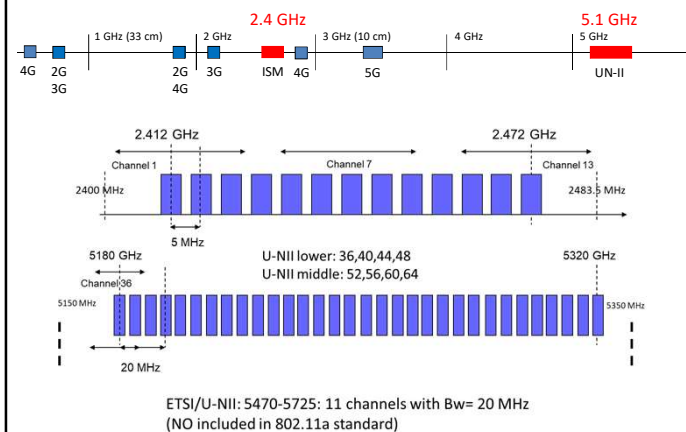
Legislación española (Ley General Telecom., '03)

- El espectro radioeléctrico es un *bien de dominio público*, cuya titularidad, gestión, planificación, administración y control corresponden al Estado.
- El Gobierno reglamentará:
 - Niveles de emisión radioeléctrica tolerables
 - Cuadro Nacional de Atribución de Frecuencias (CNAF)
 - Procedimientos de otorgamiento de derechos de uso del dominio público radioeléctrico
 - Habilitación para el ejercicio de los derechos de uso
 - Adecuada utilización del espectro radioeléctrico

Legislación española (Ley General Telecom., '03)

- Licencias y asignación de espectro:
 - ITU -> CEPT y ETSI -> Ministerio Ind Tur & Com- > SETSI - > AER,CMT
 - Cuadro Nacional de Atribución de Frecuencias (CNAF) español
 - NOTA: La CMT NO regula el espectro, sino los servicios y la calidad con la que deben ser provistos
- Bandas de frecuencia que no requieren licencias
 - dispositivos domésticos: hondas microondas, mandos a distancia, Bluetooth, Cordless phones, etc.

Unlicensed Bands



Sub-GHz bands

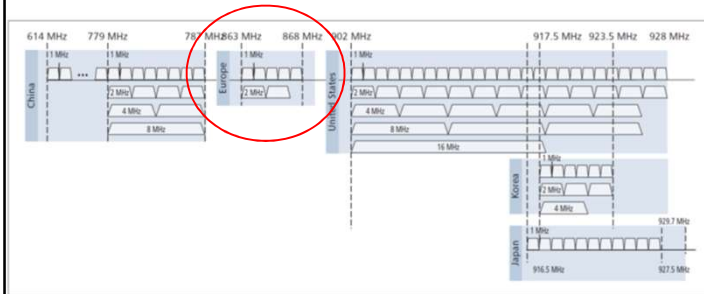


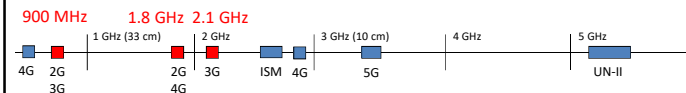
Figure 1. Frequency spectrum availabilities in sub-1-GHz and channelization in key geographies.

Licencias 3G/4G/5G en España

	Digital dividend 800 MHz 4G banda 20	Refarming 900 MHz 2G/3G banda 8		Capacity 2600 MHz 4G banda 7	
	1800 MHz 2G/4G banda 3	2100 MHz 3G banda 1	3500 MHz 5G banda 42		
Movistar	10 MHz	14,8 MHz	20 MHz	15 MHz FDD 5 MHz TDD	20 MHz 90 MHz
Vodafone	10 MHz	10 MHz	20 MHz	15 MHz FDD 5 MHz TDD	20 MHz FDD 20 MHz TDD 90 MHz
Orange	10 MHz	10 MHz	20 MHz	15 MHz FDD 5 MHz TDD	20 MHz 100 MHz
MásMóvil			14,8 MHz	15 MHz FDD 5 MHz TDD	10 MHz TDD (autonómicos) 80 MHz

Source: <https://www.adsizone.net/bandas-5G-4G-3G-2G-frecuencias-telefonía-movil>

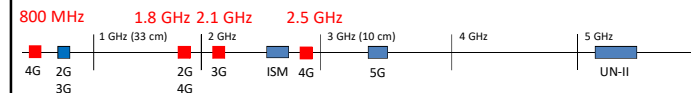
2G/3G Bands



	Año	Mecanismo asignación	Ingresos (aprox.)	Número de licencias licitadas
Alemania	2000	subasta	50000 ME	6
Austria	2000	subasta	871 ME	6
Belgica	2001	subasta	456 ME	3
Dinamarca	2001	subasta	516 ME	4
España	2000	concurso	456 ME	4
Finlandia	1999	concurso		4
Francia	2001	concurso	1200 ME	3
Grecia	2001	subasta	441 ME	3
Holanda	2000	subasta	3000 ME	5
Irlanda	2002	concurso	175 ME	3
Italia	2000	subasta	12000 ME	5
Luxemburgo	2002	concurso	9 ME	3
Noruega	2000	concurso	36 ME	4
Portugal	2000	concurso	400 ME	4
Reino Unido	2000	subasta	37000 ME	5
Suecia	2000	concurso	12000 ME	4
Suiza	2000	subasta	128 ME	4

Figura 6: Asignaciones primarias del espectro para redes 3G en Europa (Fuente: Gretel 2007³¹)

4G Bands



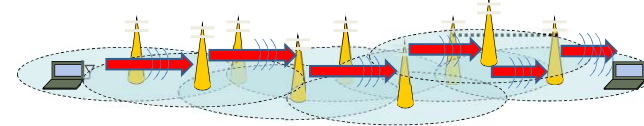
3GPP New Spectrum lining up with LTE availability			
2.5-2.6 GHz Potential for up to 20MHz per operator		2.5-2.6 GHz Early Spectrum Auction	
Capacity Band	Country	Spectrum	Date
	Austria	2500 MHz – 2600 MHz	Q2 2009
	Germany	2000 MHz – 2600 MHz	Q3 2009
	Norway	2500 MHz – 2600 MHz	Completed
Digital Dividend (800MHz) Potential for up to 10MHz per operator	Sweden	2500 MHz – 2600 MHz	Completed
	United Kingdom	2500 MHz – 2600 MHz	Q3 2009
	Hong Kong	2500 MHz – 2600 MHz	Completed
Coverage Band	Digital Dividend (800MHz) Spectrum Auction		
	Country	Spectrum	Date
	France	790 MHz – 862 MHz	2009
	Germany	790 MHz – 862 MHz	2009 – 2010
Refarming 1800MHz & 2100MHz 5-10MHz in early days & up to 15MHz eventually	Switzerland	790 MHz – 862 MHz	2013
	Sweden	790 MHz – 862 MHz	2009 – 2010
	United Kingdom	790 MHz – 862 MHz	2009 – 2012

5G in EU

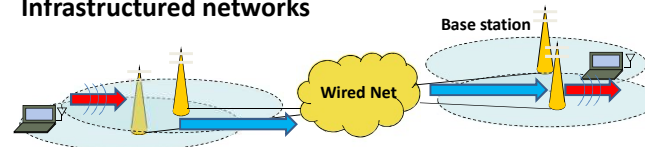
- 700 MHz, 3,5 GHz y 26 GHz
- Banda 3.5GHz en España:
 - Xfera Móviles, S.A 3420-3500 MHz;
 - Telefónica de España, S.A.U, (40 MHz) y Telefónica Móviles España, S.A.U., (60 MHz): 3500-3600 MHz
 - Orange Espagne, S.A.U., 3600-3710 MHz;
 - Vodafone España, S.A.: 3710-3800 MHz.
 - Al mismo tiempo introdujo en el mismo proceso la posibilidad de
- Posibilidad de utilizar las bandas de frecuencias de 900 y 1.800 MHz (refarming), hasta entonces de uso exclusivo por GSM.

Wireless Nets

Multihop: Small & low throughput nets

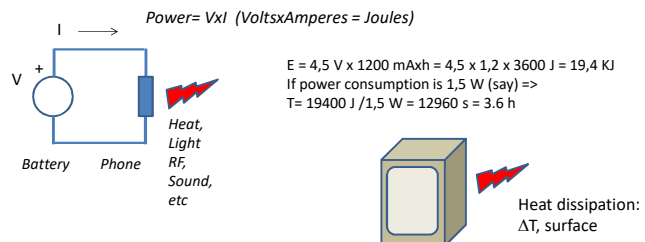


Infrastructured networks

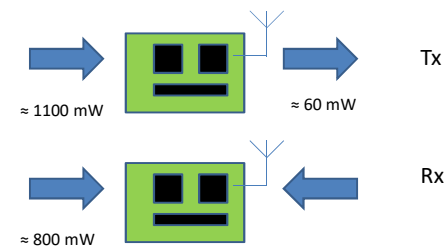


Reminder: Energy versus power

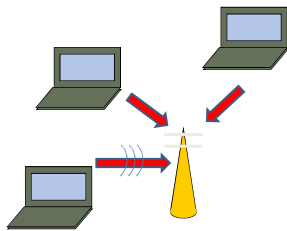
- Energy: Joules (J)
- Power: J/s = Watts (W)
- Batteries: mAxh (Note: It does not give the energy => x4.5 V)



Power consumption, e.g. WiFi

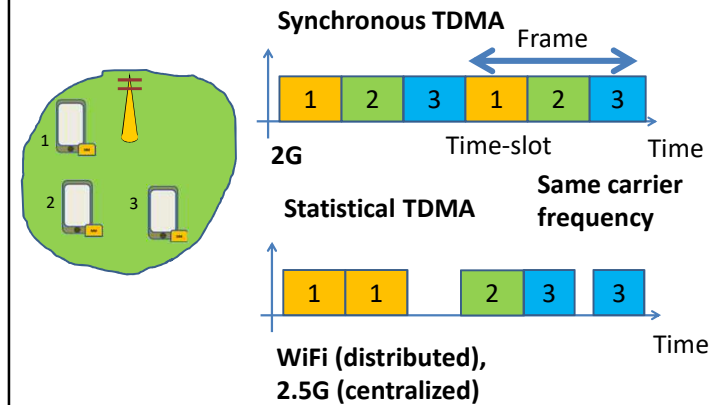


Multiple access protocols

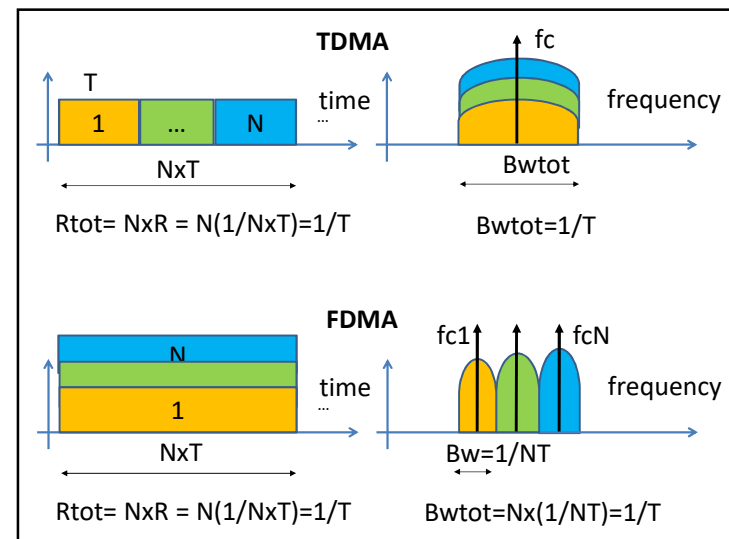
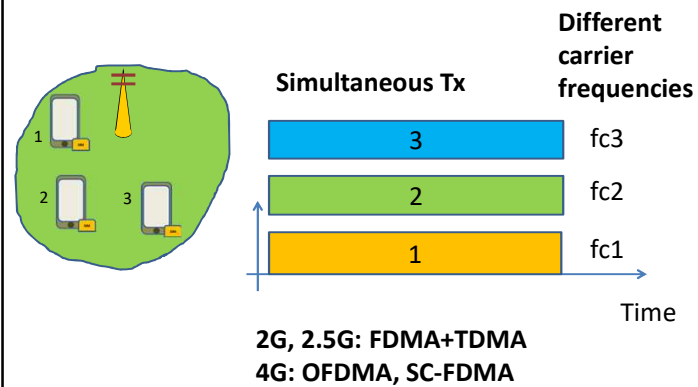


Simultaneous transmission at the same time, frequency and code, can lead to a collision

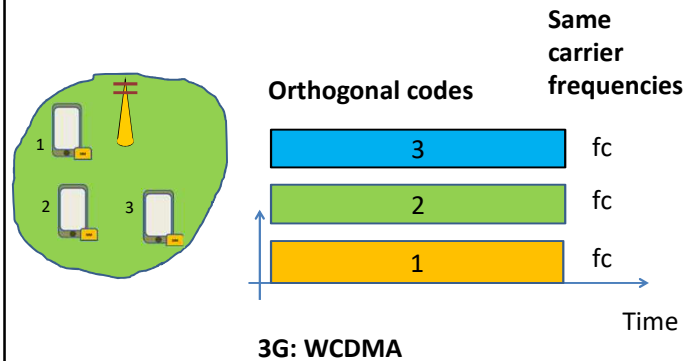
Time Division Multiple Access (TDMA)



Frequency Division Multiple Access (FDMA)

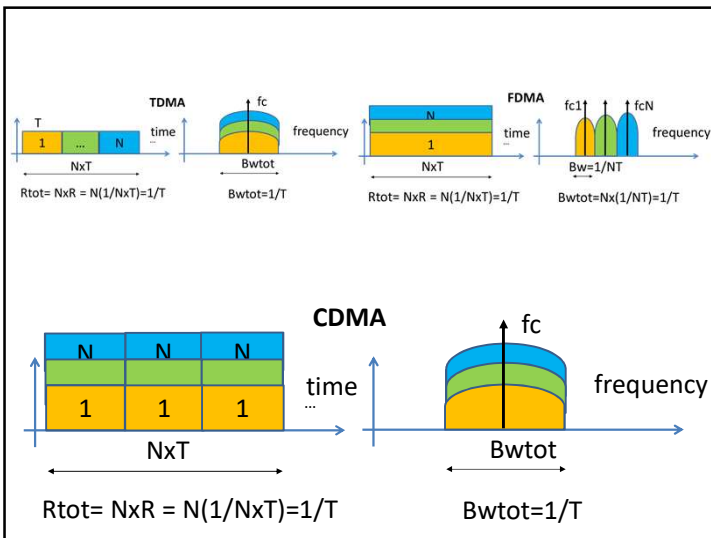


Code Division Multiple Access (CDMA)



CDMA: scalar product

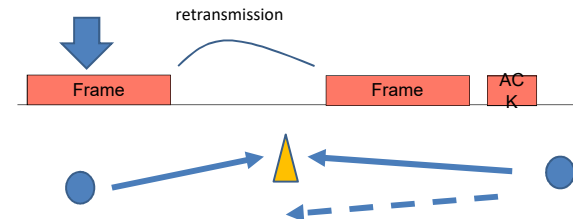
- E.g. Check that these 3 vectors are mutually orthogonal (we will call them “codes”)
 $\mathbf{c1} = \langle 1, 1, 1, 1 \rangle$, $\mathbf{c2} = \langle 1, -1, 1, -1 \rangle$, $\mathbf{c3} = \langle 1, 1, -1, -1 \rangle$
- Try to find $\mathbf{c4}$ which is orthogonal to $\mathbf{c1}$, $\mathbf{c2}$, and $\mathbf{c3}$
 (Note, as $N=4$, we cannot find more than N mutually orthogonal codes)



Access protocols Aloha (e.g. LoraWAN class A device, RFID, RACH)

The node sends a packet and wait for an ACK. IN case no ACK is received, it waits a random time and retransmits the frame

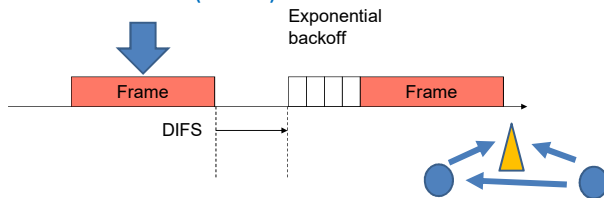
NO! Carrier sense (CSMA), simple!



Access protocols CSMA/CA (e.g. 802.11, 802.15.4)

If the medium is busy, the station must wait until it becomes idle for longer than a DIFS, and then it performs an exponential backoff...

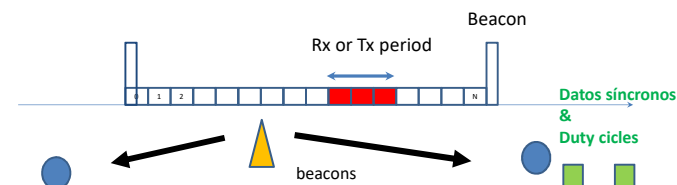
Carrier sense (CSMA)



In the first attempt, the Contention Window (CW) measures CW_{min} (e.g. 7 slots). In case of collision, in attempt n -th, the CW is set to $CW_{min} \cdot 2^{n-1} - 1$, until reaching a maximum value of CW_{max} ...

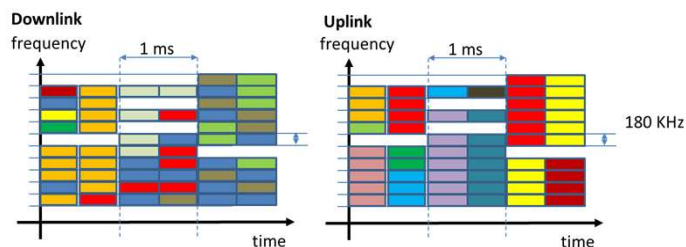
Scheduled transmission (Beacons) 2G, LoraWan Class B, 802.15.4)

- A dedicated network coordinator transmits beacons in predetermined intervals
- All nodes switch awake/sleep periods (net nodes need to receive periodic beacons)

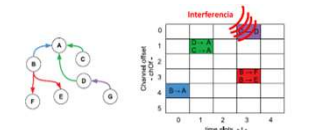


Frequency diversity frames (2G, 4G, 15.4 TSCH)

4G

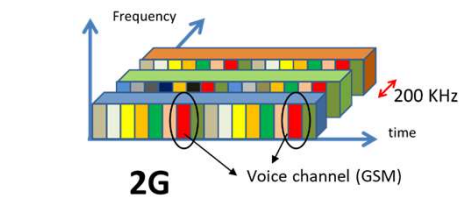


Frequency diversity frames (2G, 4G, 15.4 TSCH)

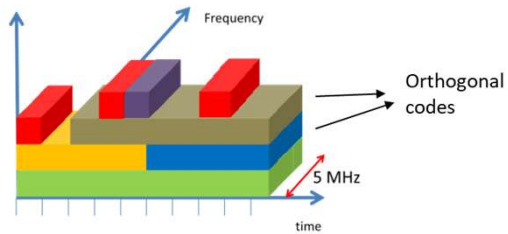


IEEE 802.15.4 TSCH

Fig. 2. Dedicated and shared links.



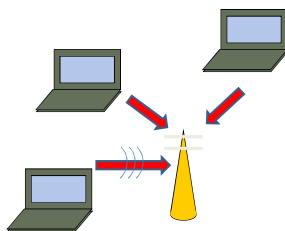
CDMA (3G)



Mobile Internet

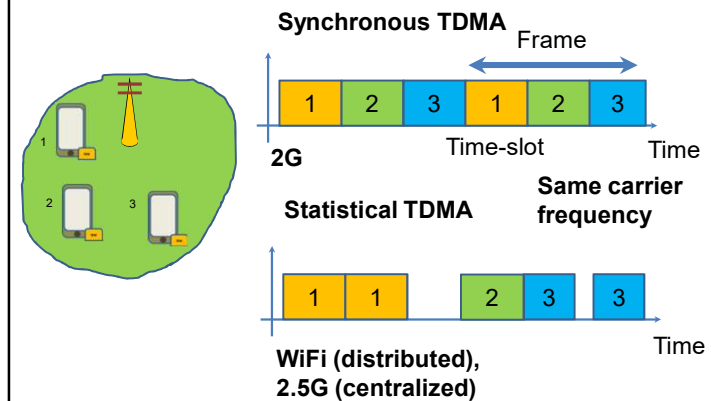
4.- Cellular Networks:
2G (GSM & GPRS), 3G(UMTS & HSPA), 4G (LTE), 5G

Multiple access protocols

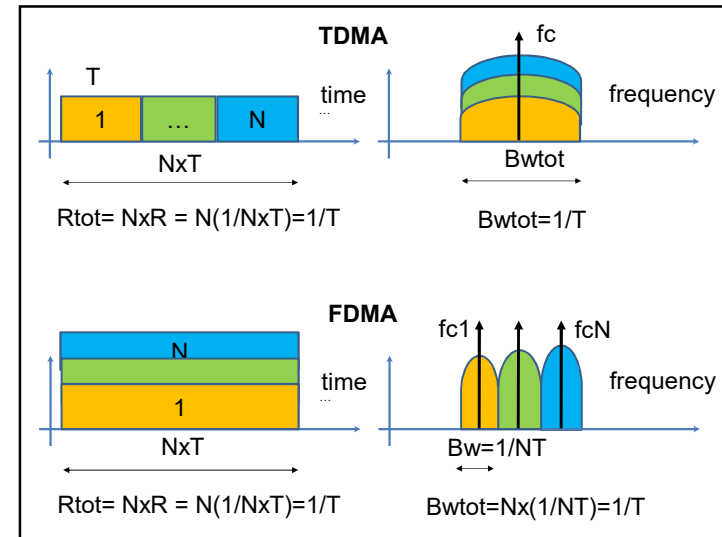
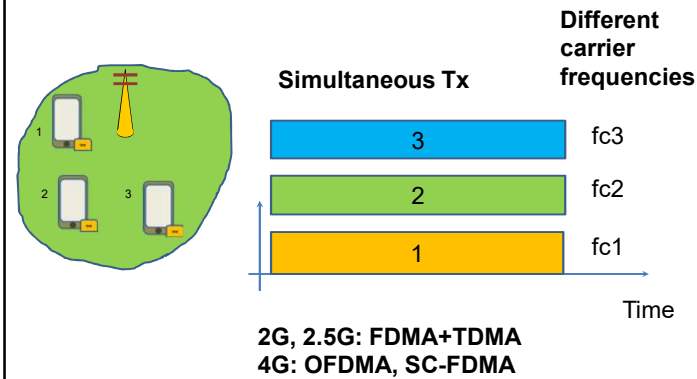


Simultaneous transmission at the same time, frequency and code, can lead to a collision

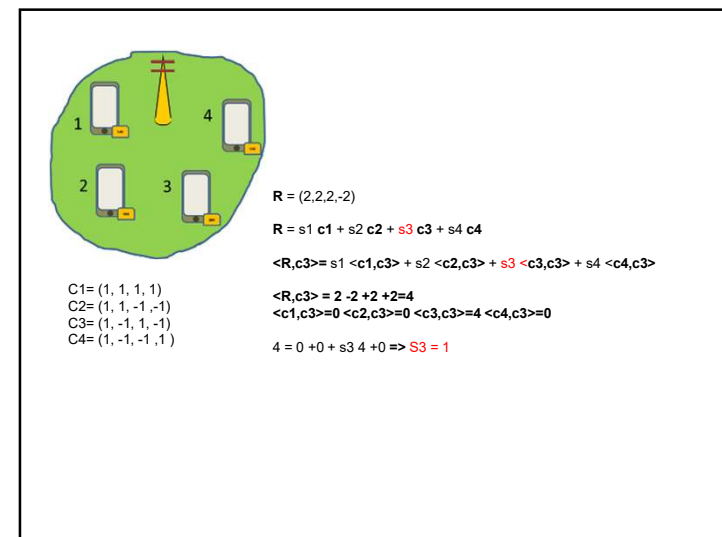
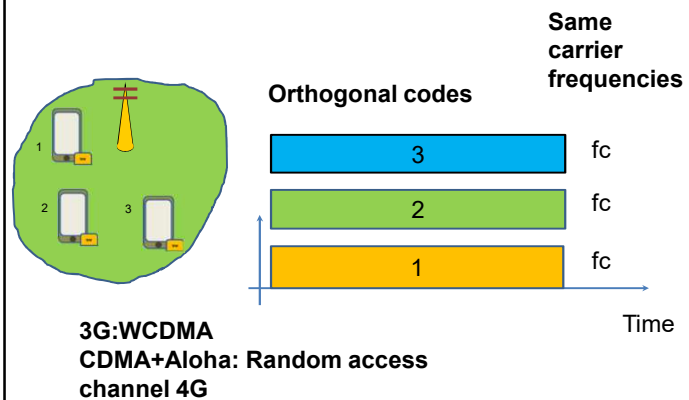
Time Division Multiple Access (TDMA)



Frequency Division Multiple Access (FDMA)



Code Division Multiple Access (CDMA)

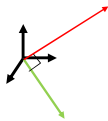


CDMA: scalar product

- Reminder

Assume $\mathbf{x}$ and $\mathbf{y}$ are two vectors of N components $\mathbf{x}=(x_1, x_2, \dots, x_N)$ and $\mathbf{y}=(y_1, y_2, \dots, y_N)$

Scalar product: $\langle \mathbf{x}, \mathbf{y} \rangle = x_1 y_1 + x_2 y_2 + \dots + x_N y_N$



Fundamental properties:

- $\langle \mathbf{x}, \mathbf{y} + \mathbf{z} \rangle = \langle \mathbf{x}, \mathbf{y} \rangle + \langle \mathbf{x}, \mathbf{z} \rangle$
- $\langle \mathbf{x}, a\mathbf{y} \rangle = a \langle \mathbf{x}, \mathbf{y} \rangle$
- If $\mathbf{x}$ and $\mathbf{y}$ are orthogonal, $\langle \mathbf{x}, \mathbf{y} \rangle = 0$

CDMA: scalar product

- E.g. Check that these 3 vectors are mutually orthogonal (we will call them “codes”)
 $\mathbf{c1} = \langle 1, 1, 1, 1 \rangle$, $\mathbf{c2} = \langle 1, -1, 1, -1 \rangle$, $\mathbf{c3} = \langle 1, 1, -1, -1 \rangle$
- Try to find $\mathbf{c4}$ which is orthogonal to $\mathbf{c1}$, $\mathbf{c2}$, and $\mathbf{c3}$
 (Note, as $N=4$, we cannot find more than N mutually orthogonal codes)

Scalar product, trick

- Check that $\mathbf{c4} = \langle 1, -1, -1, 1 \rangle$ is a solution of the question in previous slide
- Think on 4 numbers, a, b, c, d (restrict to values $+1$ or -1 , although the trick will work in general)
- Find now $\mathbf{x} = a\mathbf{c1} + b\mathbf{c2} + c\mathbf{c3} + d\mathbf{c4}$
- For instance, if we choose $a=1, b=-1, c=1, d=-1$, we have:
 $\mathbf{x} = 1\mathbf{c1} + (-1)\mathbf{c2} + 1\mathbf{c3} + (-1)\mathbf{c4} = \mathbf{c1} - \mathbf{c2} + \mathbf{c3} - \mathbf{c4} = \langle 0, 4, 0, 0 \rangle$
- Question: how to find a, b, c and d if you only know $\mathbf{x}$?
- E.g. find a, b, c , and d if $\mathbf{x} = \langle 0, 0, 0, -4 \rangle$

Answer

If you can find a, b, c , and d solving a 4×4 linear system of equation, but we can use a trick that makes things much easier:
find the scalar product between one of the codes and the vector $\mathbf{x}$:

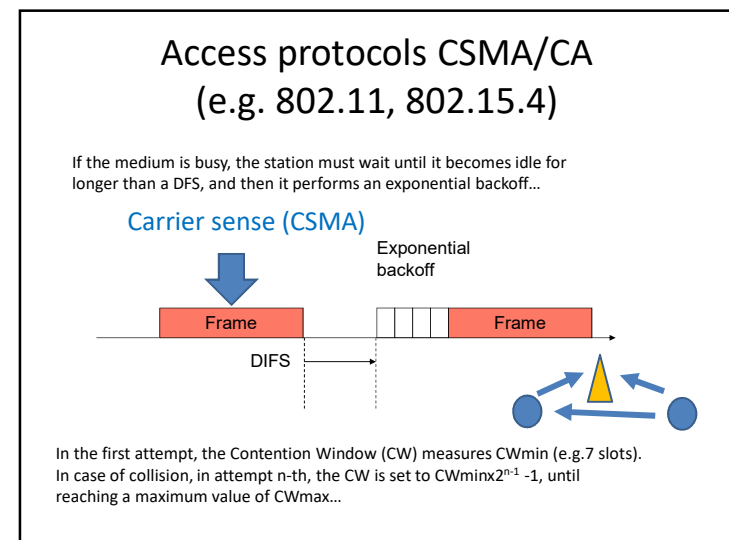
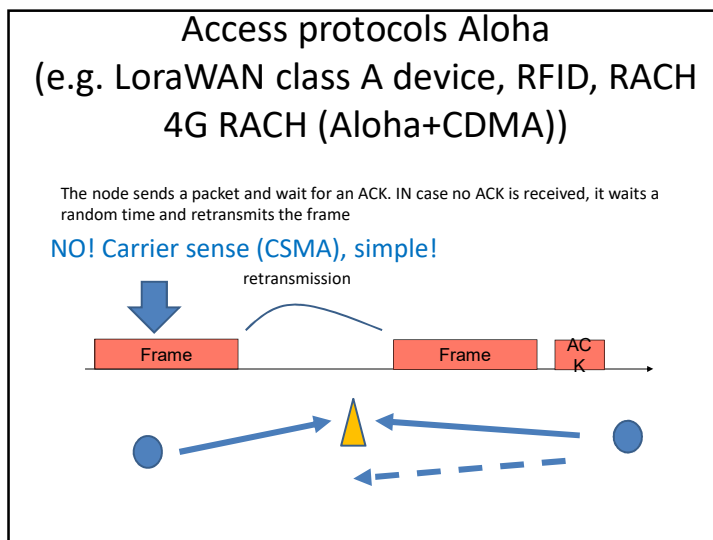
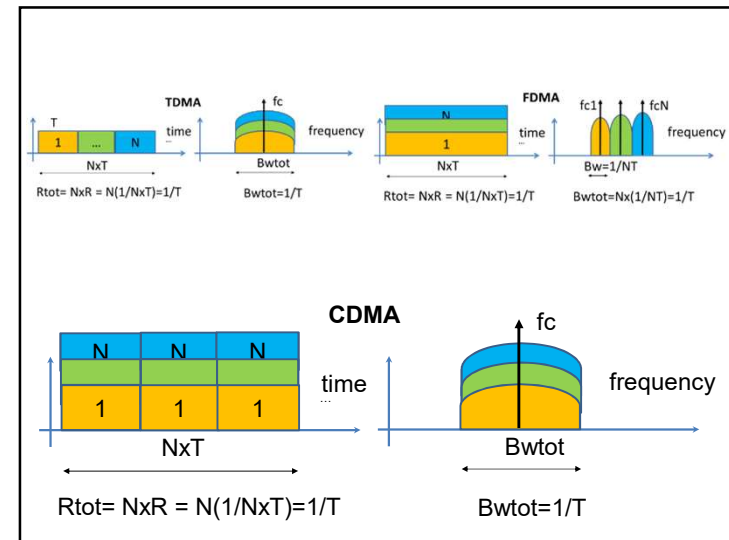
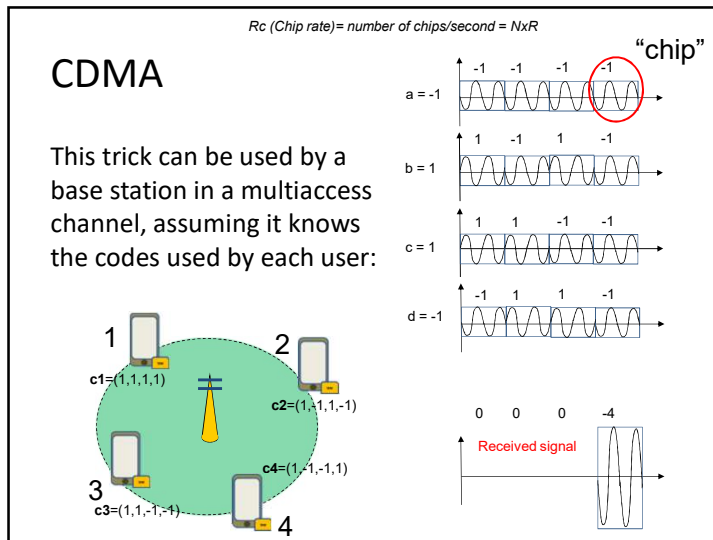
$$\mathbf{x} = a\mathbf{c1} + b\mathbf{c2} + c\mathbf{c3} + d\mathbf{c4}$$

$$\langle \mathbf{c1}, \mathbf{x} \rangle = a\langle \mathbf{c1}, \mathbf{c1} \rangle + b\langle \mathbf{c1}, \mathbf{c2} \rangle + c\langle \mathbf{c1}, \mathbf{c3} \rangle + d\langle \mathbf{c1}, \mathbf{c4} \rangle =$$

$$a\langle \mathbf{c1}, \mathbf{c1} \rangle + 0 + 0 + 0 = a\langle \mathbf{c1}, \mathbf{c1} \rangle$$

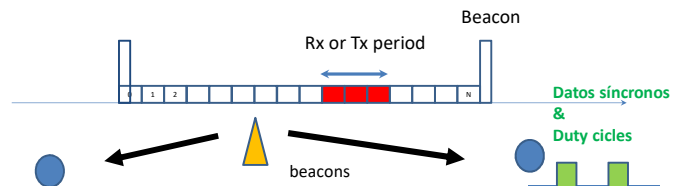
$$\text{e.g. } \mathbf{x} = \langle 0, 0, 0, -4 \rangle \Rightarrow \langle \mathbf{c1}, \mathbf{x} \rangle = -4, \langle \mathbf{c1}, \mathbf{c1} \rangle = 4 \Rightarrow a = -1$$

Correct! The same trick can be used to find b, c and d if we multiply $\mathbf{x}$ by $\mathbf{c2}$, $\mathbf{c3}$ and $\mathbf{c4}$. Check this, you should obtain $b=1, c=1, d=-1$



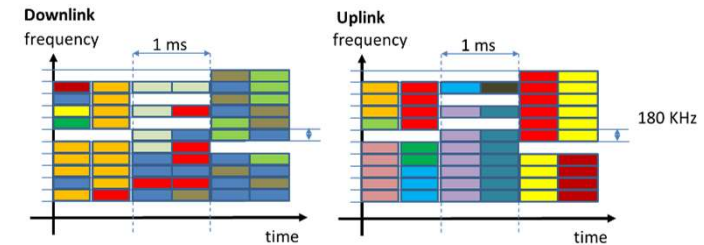
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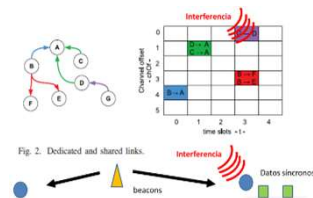


Frequency diversity frames (2G, 4G, 15.4 TSCH)

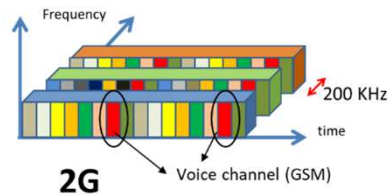
4G



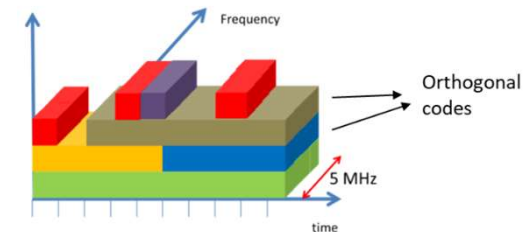
Frequency diversity frames (2G, 4G, 15.4 TSCH)



IEEE 802.15.4 TSCH

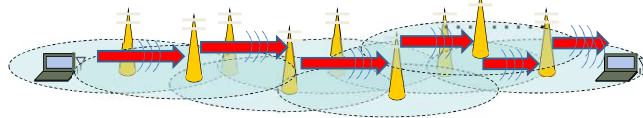


CDMA (3G)

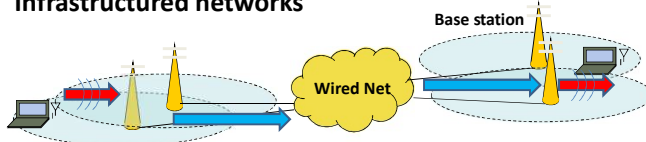


Wireless Nets

Multihop: Small & low throughput nets

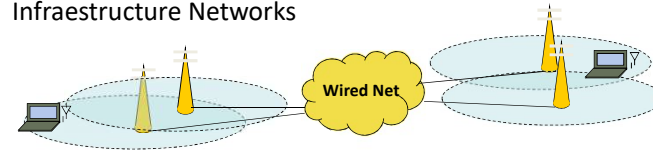


Infrastructured networks



Wireless Nets

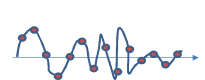
Infraestructure Networks



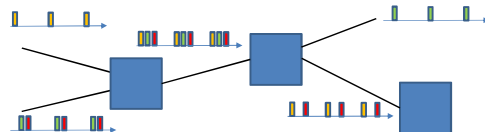
GSM (2G), GPRS (2.5G), UMTS (3G), HSPA (3.5G), LTE (4G)
LoRa/LoRaWAN
WiFi Hotspots

Circuit & packet switching

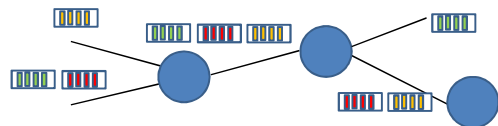
Voice,
Sampling



Quantization

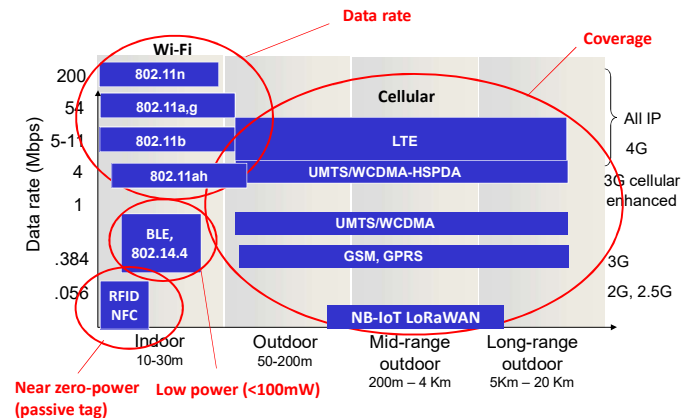


Circuit
switching

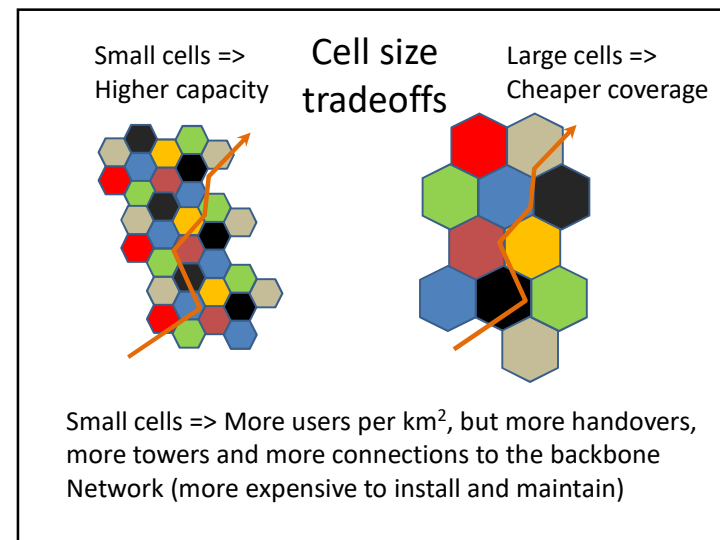
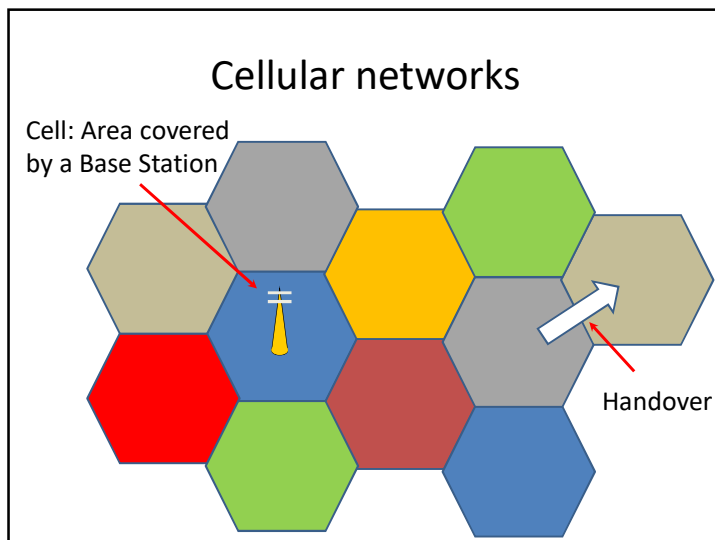
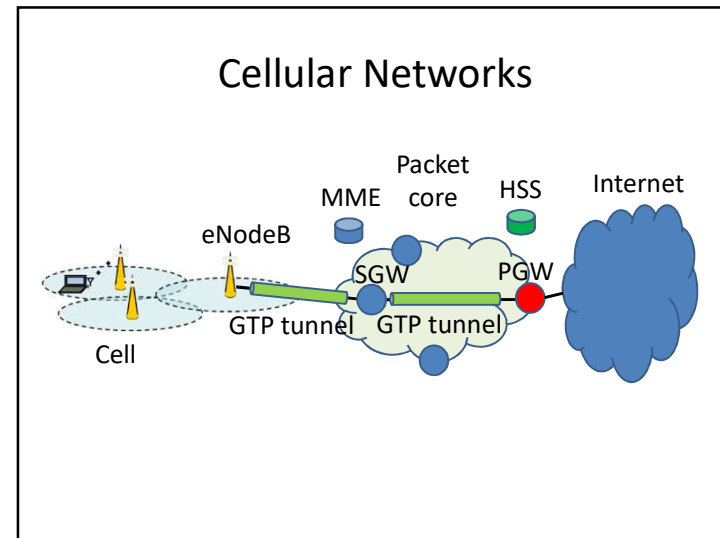


Packet
switching

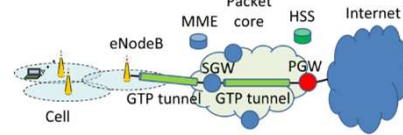
Comm: Wireless technologies



Cellular Networks

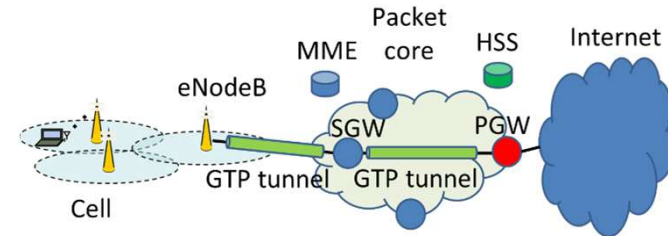


Mobile Packet Core



- **Session management**
Establishing and managing information flow of a particular service or application
- **Mobility management**
Track users as they move between cell site and route traffic accordingly
- **Security & privacy**
Authentication, encryption and user privacy.
- **Policy & charging**

LTE (Evolved Packet Core)



IP network: Data and VoLTE

(Voice over LTE)
Serving Gateway (SGW) (3g SGSN)
Packet Data Network Gateway (PGW), (3g GGSN)
Mobility Management Entity (MME) (3g VLR)
Home Subscriber Server (HSS) (3g HLR, AuC)

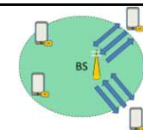
Cellular net evolution

Peak throughput · bandwidth · frequency bands (EU) · latency

GEN	STANDARD	PEAK DOWNLINK	PEAK UPLINK	CHANNEL BW	EU FREQUENCY BANDS	LATENCY
2G	GSM / GPRS / EDGE	~384 Kbps (GPRS) / 236.8 Kbps (EDGE)	~384 Kbps	200 kHz	900 MHz - 1800 MHz	300-1000 ms
3G	UMTS / HSPA / HSPA+	~42 Mbps (HSPA) / 168 Mbps (HSPA+)	~11 Mbps (HSPA)	5 MHz	900 MHz - 2100 MHz	50-200 ms
4G	LTE / LTE-A / LTE-A Pro	~1 Gbps (LTE-A Pro)	~500 Mbps (LTE-A Pro)	1.4-20 MHz (up to 100 MHz CA)	700 / 800 / 1800 / 2100 / 2600 MHz	10-50 ms (~10 ms typ.)
5G	NR (New Radio) SA/NSA	~20 Gbps (peak 4 GHz, 100-400 MHz)	~10 Gbps (peak 4 GHz, 100-400 MHz)	Up to 400 MHz (peak 4 GHz, 100-400 MHz)	700 / 1500 / 2100 / 3.5 GHz (n78) (peak 4 GHz, 100-400 MHz)	<10 ms (RRLT Target: 4 ms)
6G	IMT-2030 (proposed ~2030)	~1 Tbps (peak 100 GHz)	~1 Tbps (peak 100 GHz)	Multi-GHz (peak 100 GHz)	Sub-6 GHz + mmWave + THz (0.1 - 1 THz, not yet standardized)	<0.1 ms (Target, not confirmed)

Peak figures are theoretical maxima under ideal conditions. Real-world performance is typically 10-30× lower.

Channels

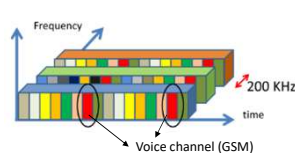


- **Traffic channel** (TCH, UD): user info
- **Random Access channel** (RACH, U), the user request use of channel, (ALOHA)
- **Access Grant channel** (AGCH, D): The network assigns channel to user
- **Paging channel** (PCH, D): The network signals the user about a new call
- **Broadcast Control Channel** (BCCH, D, broadcast): Identifies cellular network, Network, BS id, etc

Radio access

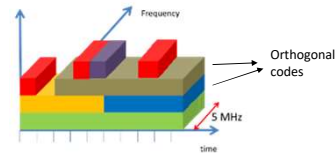
2G (GSM), 2.5 G (GPRS)

- FDMA + TDMA
- Voice: Frames of 8 time slots/200 KHz
- GSM: Voice (13 Kbps)
- GPRS: Data (< 170 Kbps)



3G (UMTS), 3.5 G (HSPA)

- WCDMA
- Bw= 5 Mhz, Rchip =3.84 Mcps
- UMTS: R < 2/0.3 Mbps,
- HSDPA/HSUPA, R< 10/2.4 Mbps



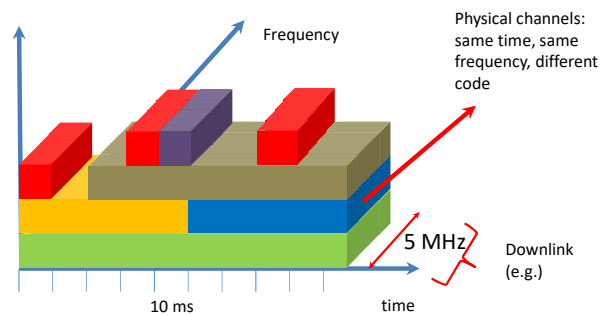
These networks (in special 3G) will probably be turned off during the next years to free spectrum for 4G and 5G

3G (UMTS)

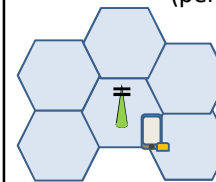
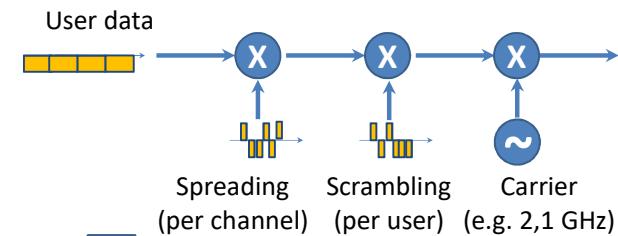
- Europe: WCDMA (UMTS FDD)
- 1900-1980, 2100-2170 and 2110-2170 MHz
- Bw= 5 Mhz
- Chip Rate 3.84 Mcps
- Data rates: <= 2/0.3 Mbps

3G (UMTS)

- Uses CDMA (same frequency & time, spreading code)
- Uplink and Downlink in different frequency bands



3G, spreading and scrambling codes

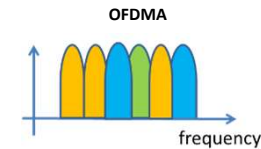


No frequency reuse!, orthogonal scrambling codes
The available number of SCs is **512**; however, they are used to identify thousands of sectors over the network. Therefore code planning is essential in the downlink direction

3.5G (HSPA)

- Physical channel for higher data rate (10/2.4 Mbps)
 - Higher order modulation (16-QAM)
 - Fast scheduling (users with more favourable channel conditions)
 - Improved ARQ (soft and hybrid ARQ)
- HSDPA: Enhanced Downlink
- HSUPA: Enhanced Uplink

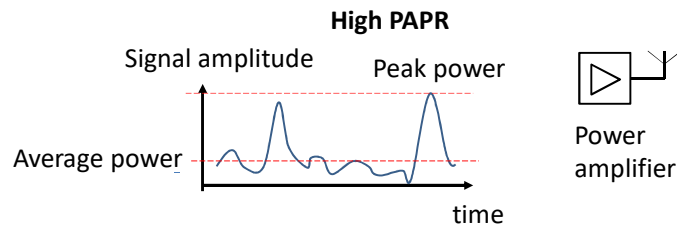
OFDMA



- Downlink channel in LTE uses OFDMA
- Long symbol times limits intersymbol interference due to multipath
- Allows flexible utilization of frequency spectrum
- Increases spectral efficiency due to orthogonality between sub-channels
- Allow optimization of data rates for users by transmitting on the sub-channels with lower fading

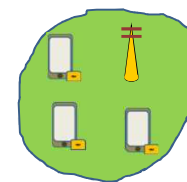
Problems with OFDMA

- The OFDMA signal has a high **Peak to Average Power Ratio (PAPR)**, meaning that the power amplifier needs a high power consumption as it must be able to amplify signals with high peaks

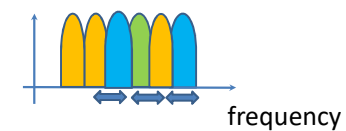


Problems with OFDMA

- In the **uplink**, the multiple channels produced by different phones are not perfectly aligned (instability in electronics, Doppler spread). Due to this offset in carriers, there is some interference among sub-channels.

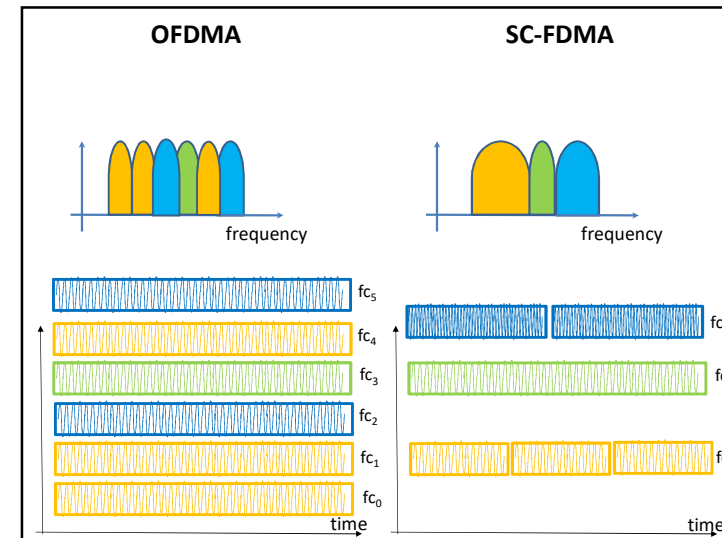
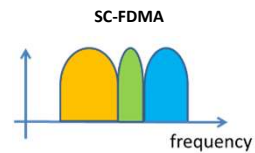


Multiple carriers

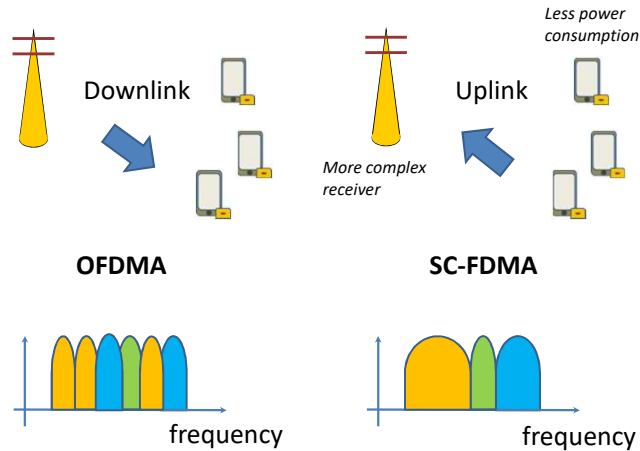


OFDMA and SC-FDMA

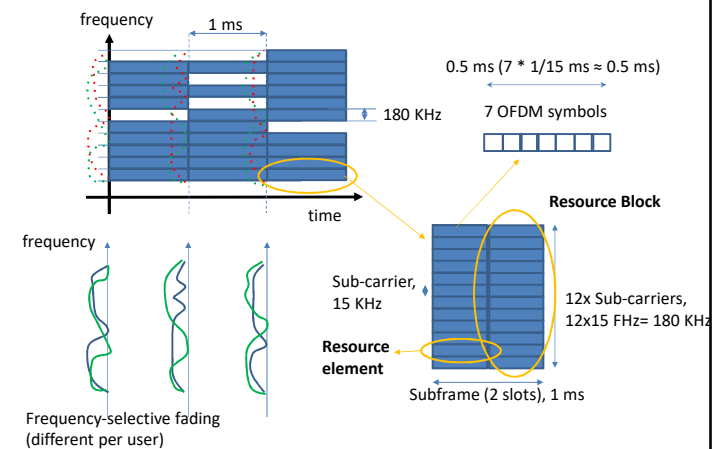
- Uplink transmission, and alternative technique called **Single Carrier- FDMA (SC-FDMA)** is used.
- A number of adjacent subchannels is allocated to each user. The **symbols are transmitted sequentially** using a single carrier per user
- The inter-symbol interference is solved by a more complex receiver at the BS.



OFDMA and SC-FDMA

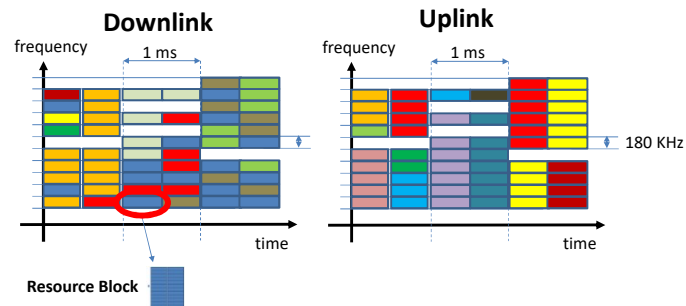


LTE PHY



RB scheduling

Scheduling of RBs is done on a subframe basis for both the uplink (consecutive RBs for same user) and uplink (non-consecutive RBs for user)



LTE channels

Downlink LTE Channels	
Name	Description
PBCH	Physical Broadcast Channel: Carries general information about the cell, like number of antennas on the base station and total bandwidth.
PDCCH	Physical Downlink Control Channel: Sends downlink control messages, e.g. for resource allocation on uplink/downlink.
PD-SCH	Physical Downlink Shared Channel: Holds downlink data meant for users in resource blocks indicated by the PDCCH.
PHICH	Physical Hybrid-ARQ Channel: Contains positive or negative acknowledgments for uplink data.
Uplink LTE Channels	
PUCCH	Physical Uplink Control Channel: Holds control information from users to base stations, e.g. ACKs, channel reports and uplink scheduling requests.
PUSCH	Physical Uplink Shared Channel: Mainly carries uplink data in resource blocks indicated by the PDCCH.

Carries scheduling decisions (which UEs tx/rx on which RBs, with what modulation)

The primary channel for delivering data to UEs. Shared among all users in the cell, the scheduler divides it into Resource Blocks allocated per subframe

The primary channel for UEs sending data to the network
Also time/frequency shared among users. When a UE has data to send, it piggybacks control info (CQI, ACK/NACK) onto PUSCH instead of using PUCCH

PRACH: Physical Random Access Channel (Uplink) Used for **initial access**: the UE's first contact with the network (random access preamble, chosen among 64 orthogonal preambles)

PCCH: Paging Control Channel (Downlink) carries paging messages over the PCH (Paging Channel) transport channel, ultimately transmitted on a Physical channel, the PD-SCH.

Access Grant: The UE sends a random access preamble on the PRACH (Physical Random Access Channel), which is equivalent to GSM's RACH. The eNodeB responds with a **Random Access Response (RAR)** carried on the PD-SCH

Control, broadcast and data channels

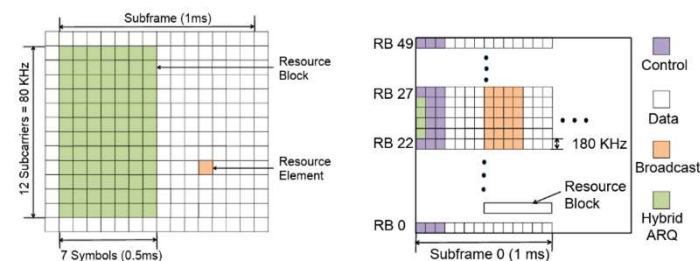
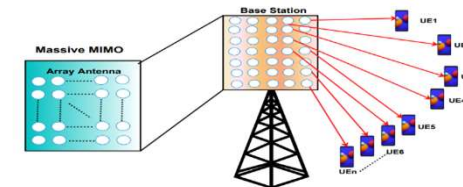


Figure 1: The Resource Grid is divided into multiple Resource Blocks.

Figure 2: Physical Channels are mapped to well defined Resource Elements in the Grid.

S. Kumar et al. "LTE Radio Analytics Made Easy and Accessible"

Massive MIMO



Non-orthogonal Multiple Access (NOMA)

(in the study/evaluation phase, strong candidate for 6G). Increases efficiency of OFDMA by letting multiple users share the same time-frequency resource simultaneously, separated instead by power levels.

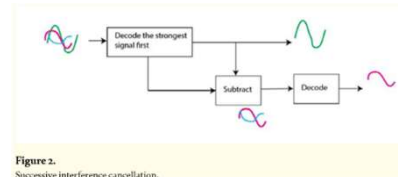
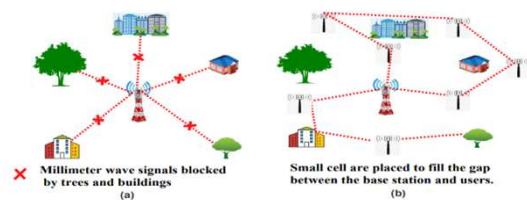
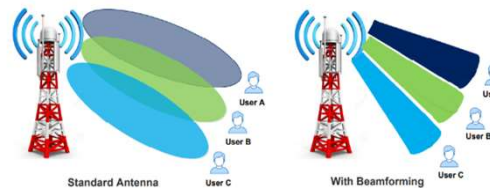


Figure 2. Successive interference cancellation.

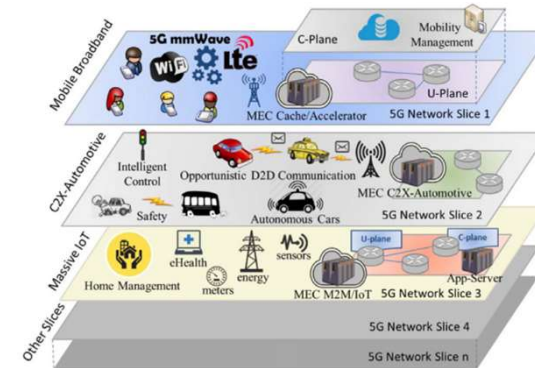
Small cells for mmWave



Beamforming

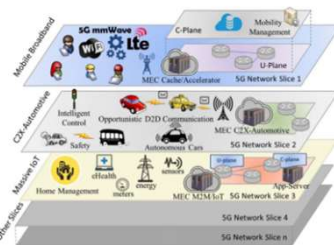


Mobile Edge Computing: reduced latency
Network slicing: create different virtual networks with specific performance



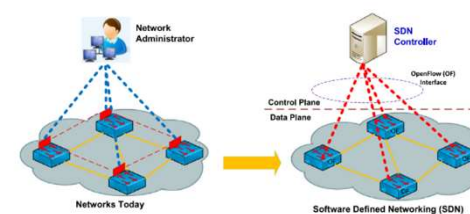
Network slicing

- The network slice is a logically separated, self-contained, independent and secured part of the network, targeting different services with different requirements on speed, latency and reliability.
- Network slice characteristics are for example low latency, high bandwidth and ultra-reliability for a critical IoT use case or higher latency and lower bandwidth for a massive IoT use case.

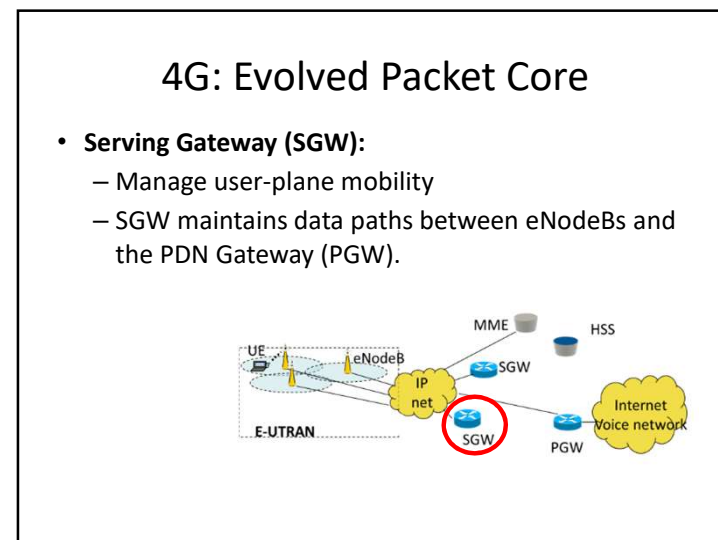
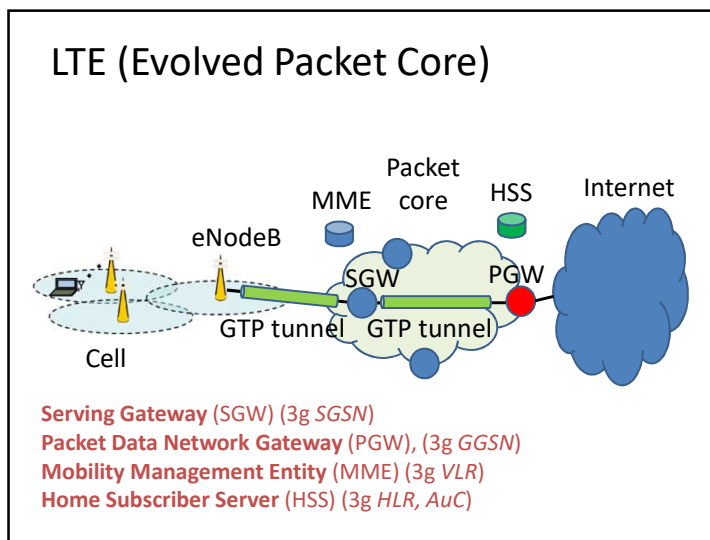
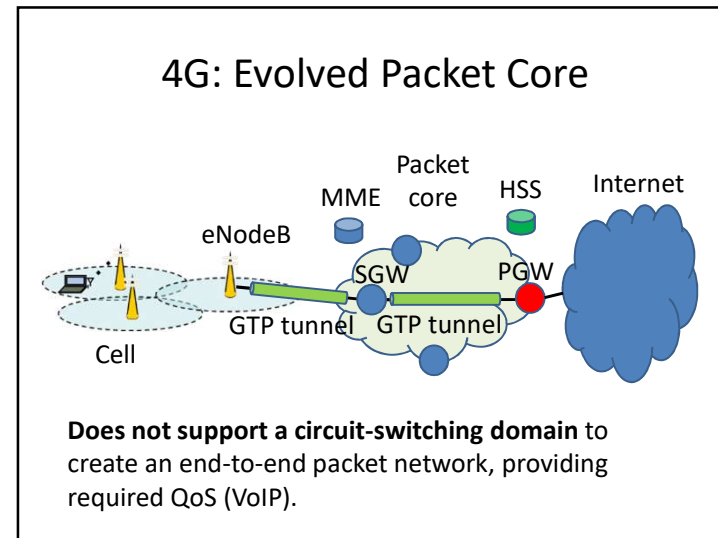
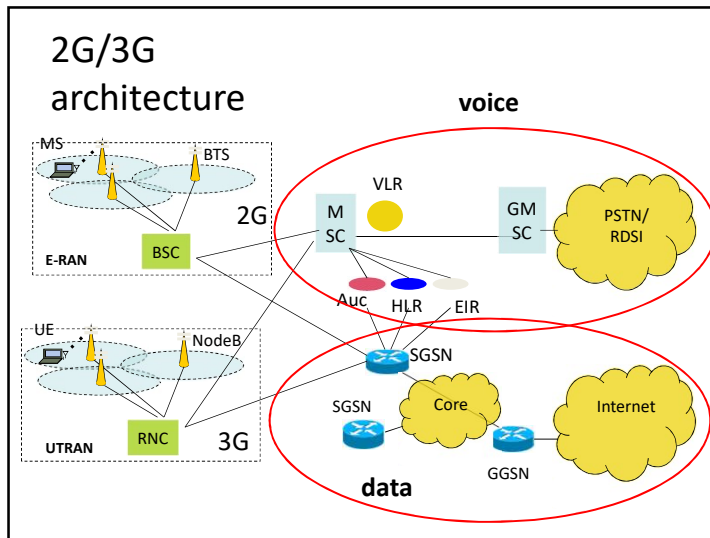


Software Defined Networks (SDN) and Network Function Virtualization (NFV)

SDN: Separate forwarding (data) and control plane

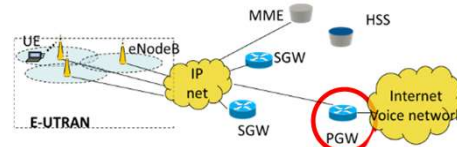


NFV: virtualize network functions (e.g., FWs, NAT, VPN, etc) on top of commodity Hw devices



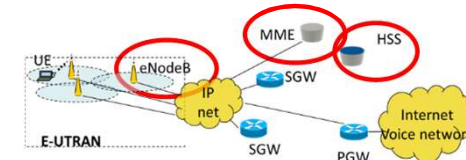
4G: Evolved Packet Core

- **Packet Data Network Gateway (PGW):**
 - Interface towards the Internet. Anchor point for sessions towards the external Internet.
 - Policy enforcement features (resource allocation and usage), Packet filtering (deep packet inspection), Charging support



4G: Evolved Packet Core

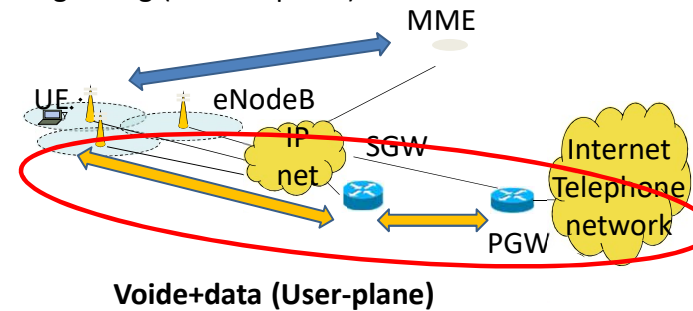
- **The Mobility Management Entity (MME):**
 - Performs the signaling and control functions to support tracking, paging, roaming and handovers.
- **The Home Subscriber Server (HSS):**
 - Performs the functions of the HLR (Home location register) and the AuC (Authentication center)

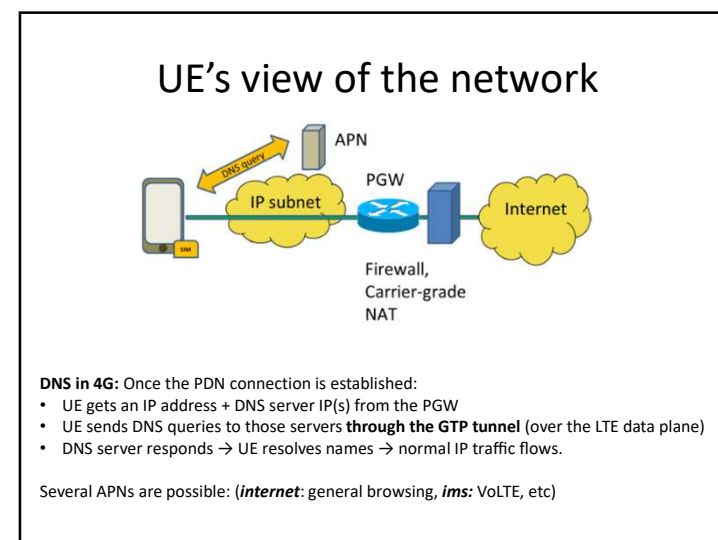
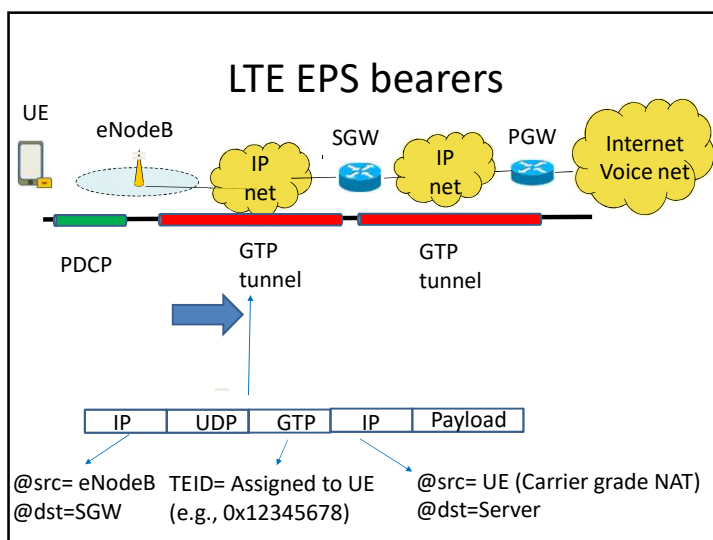
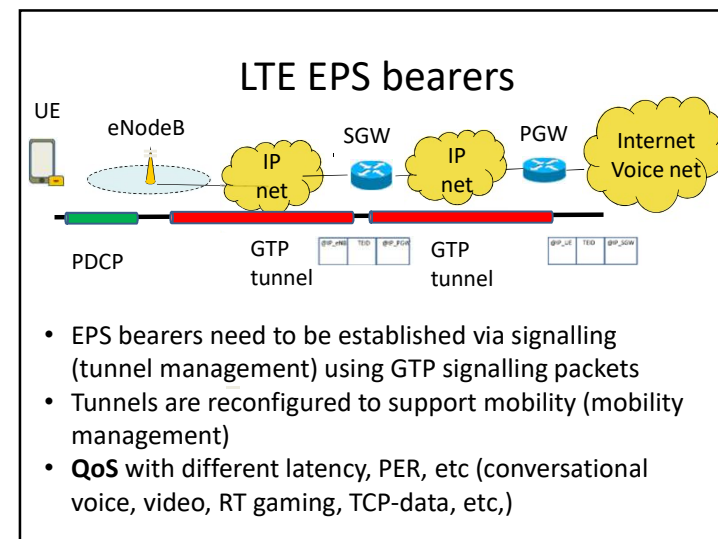
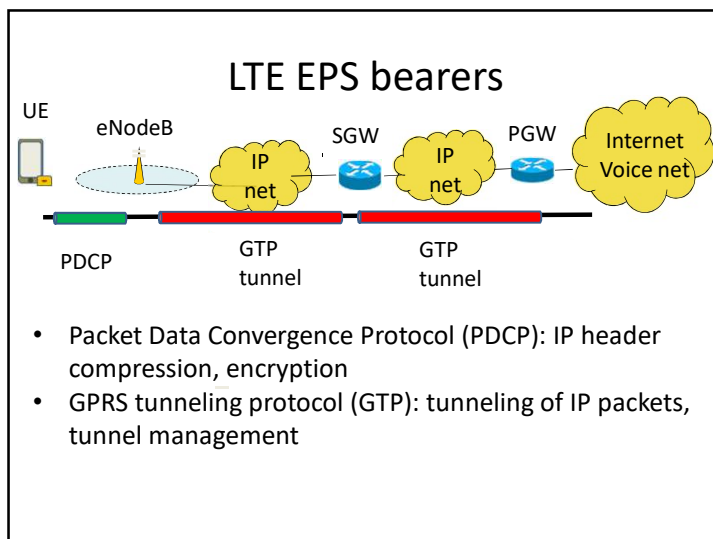


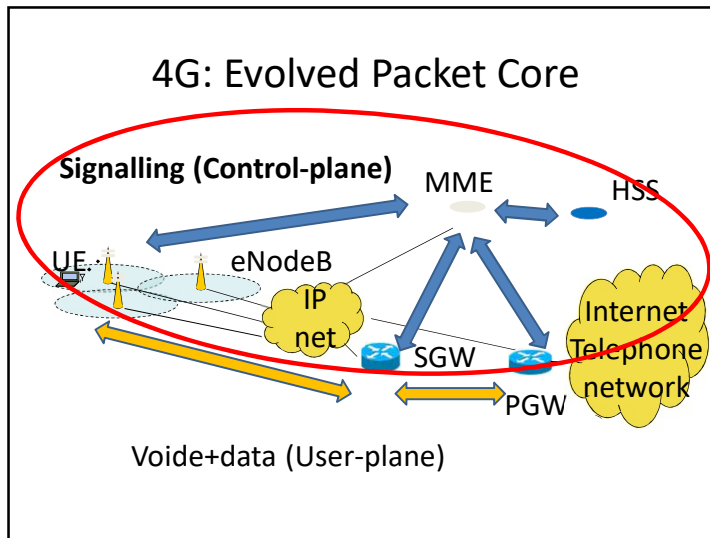
Cellular networks:
User-plane, EPS bearers

4G: Evolved Packet Core

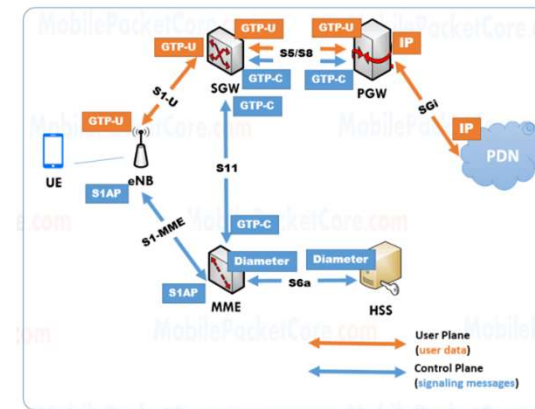
Signalling (Control-plane)







Evolved Packet Core, interfaces and protocols

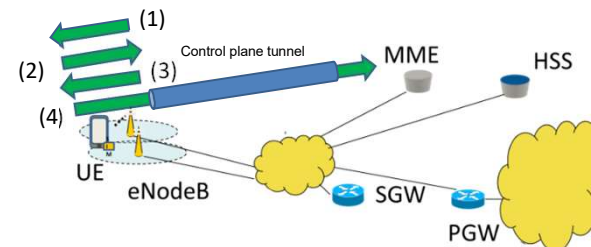


Signalling procedures

- **Radio Resources Management**
 - Establish, modify, close Radio Session
 - Handover (with/without change of SGW/MME)
- **Mobility Management**
 - Attach/Deattach to/from network
 - Tracking Area update
 - Service request/Paging
- **EPS bearer management**
 - Establish/modify/free EPS bearer

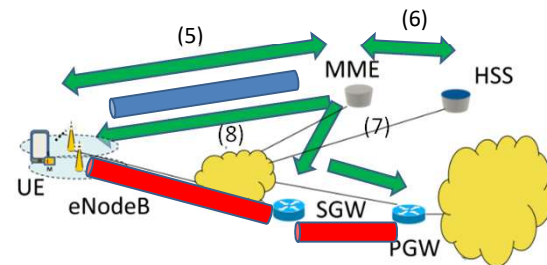
Attach (Register in network)

- (1) Listen to BCCH
- (2) Use RACH to request use of a channel
- (3) Grant use of a channel (Random Access Response (RAR) carried on the PDSCH)
- (4) Attach request (userid)



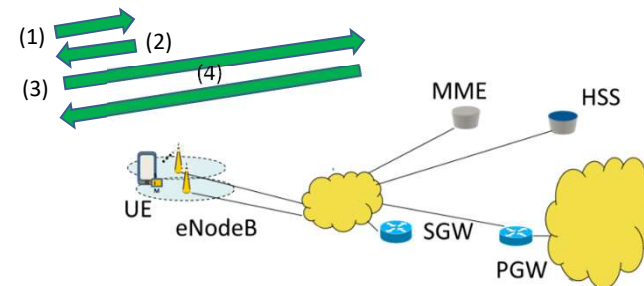
Attach (Register in network)

- (5) Authentication and key negotiation
- (6) Update IMSI location
- (7) Create default EPS bearer
- (8) Attach accept



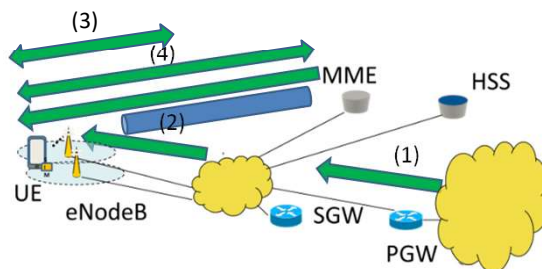
User initiated service request

- (1) Use RACH to request use of a channel
- (2) Grant use of a channel
- (3) Service request
- (4) Establishment request



Paging

- (1) Data arrives from network
- (2) Use PCH for paging user
- (3) Random Access (RACH+grant)
- (4) Establishment request + Data sent to UE



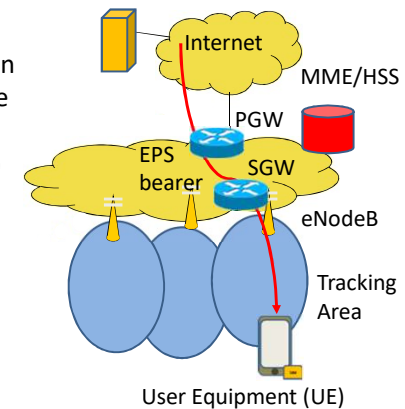
Roaming & Handovers

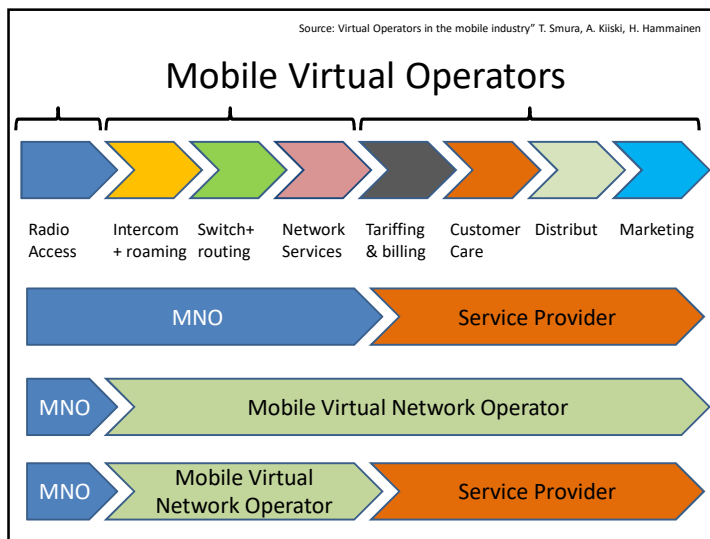
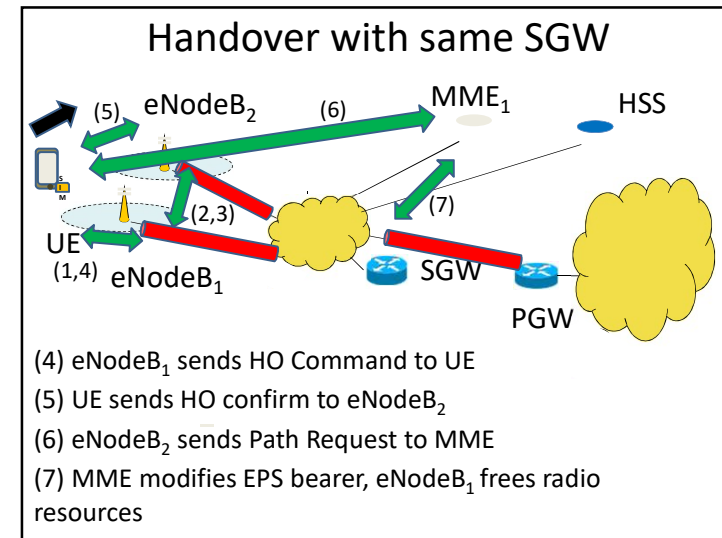
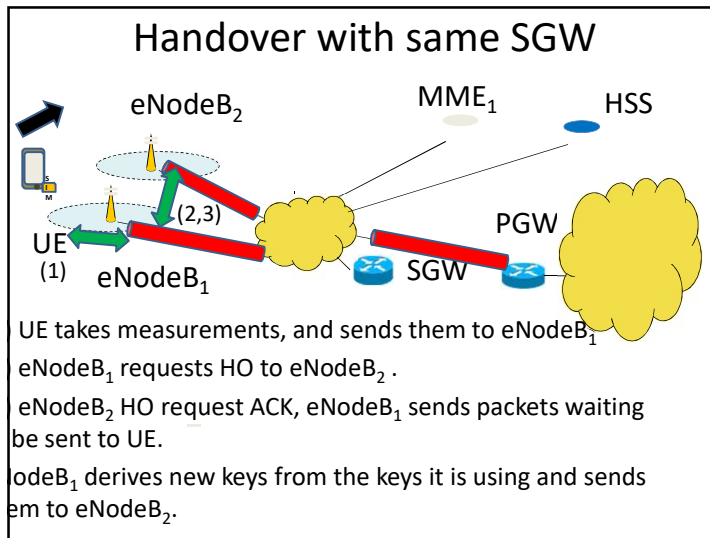
Roaming:

User changes its location
=> the system should be
informed of its current
location (**Tracking area**)

Handover:

User changes location
while engaged in a
conversation (change
eNodeB, SGW, and/or
MME)

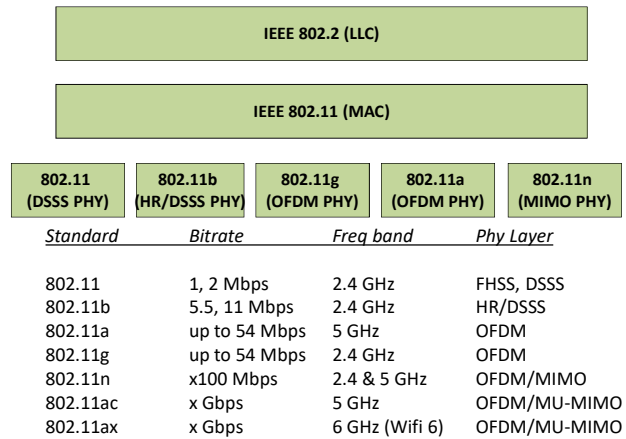




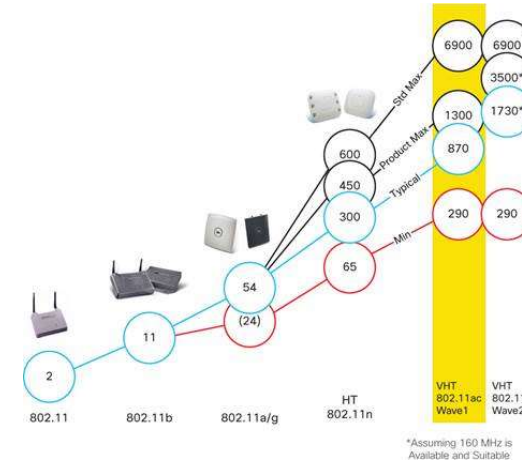
Mobile Internet

4.- WirelessLAN: WiFi

IEEE 802.11 Protocol stack



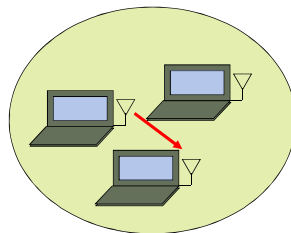
IEEE 802.11 Protocol stack



WLAN topologies

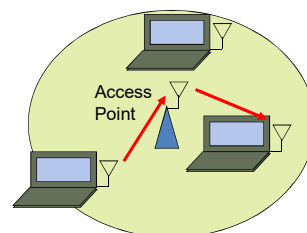
BSS: Basic Service Set

Independent BSS (IBSS/AdHoc)



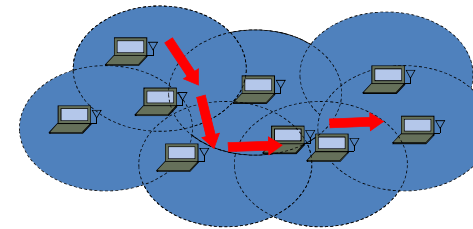
Direct communication with other stations

Infrastructure BSS (BSS)



Communication with other stations through AP

IBSS/AdHoc



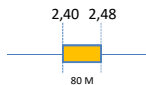
In a IBSS/AdHoc only one hop is allowed.

In a *multihop* ad-hoc network, *adhoc routing* is required, e.g.:

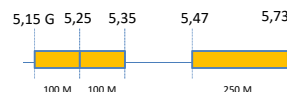
AODV (reactive, vector distance)
OLSR (proactive, link state).

Available bands

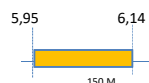
- 2.45 GHz ISM Band
 - 2,400-2,483 GHz
 - Spain: No channel 14



- U-NII 5GHz Band
 - 5.15-5.35 GHz
 - 5.47- 5.725 GHz



- U-NII 6GHz Band (Low power indoor)
 - 5.955-6.415 GHz



54 Mbps 802.11a

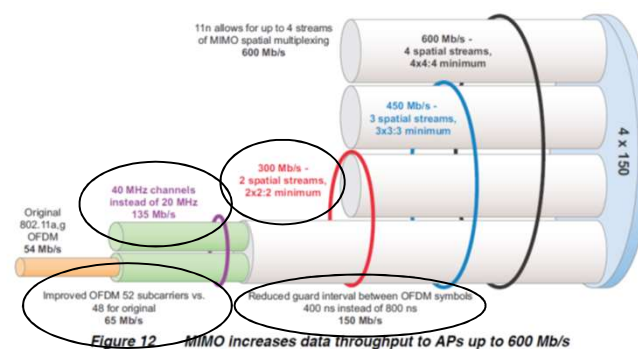
- OFDM with 64 subcarriers and 48 data subcarriers
- Coding rate: $\frac{3}{4}$ bit/codebit
- 6 coded bits per subcarrier symbol (64 QAM)
- Subcarrier symbol time: 4 μ sec

$R = 48 \text{ subcarrier} \times \frac{3}{4} \text{ bit/codebit} \times 6 \text{ codebits/subcarrier symbol} \times 1 \text{ subcarrier symbol/4 } \mu\text{secs} = 54 \text{ Mbps}$

802.11n (WiFi4)

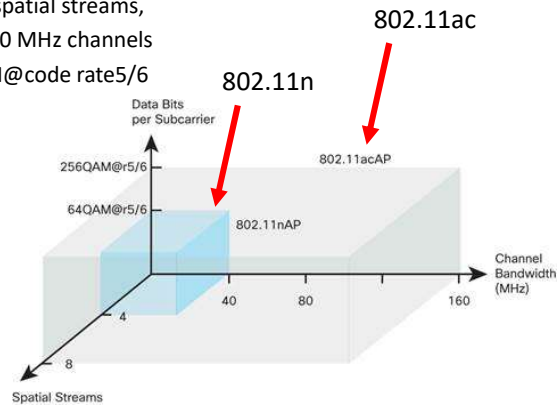
- Use MIMO to either extend range or data rate (or a combination of boths)
- Use beamforming or diversity (Mx1 or 1xM antenna cases)
- Use 2x20 MHz channels (“**channel bonding**”). UN-II band.
- **Reduce preamble times**

802.11n

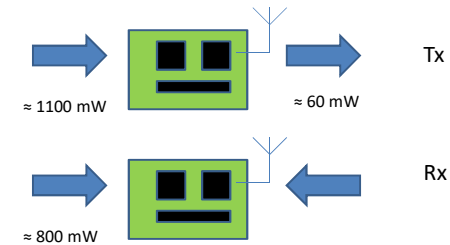


802.11ac (WiFi5)

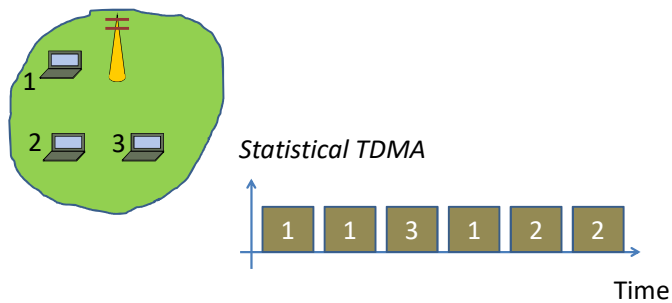
- Up to 8 spatial streams,
- Up to 160 MHz channels
- 256QAM@code rate 5/6



Power consumption, e.g. WiFi

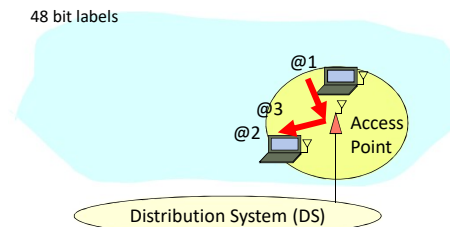
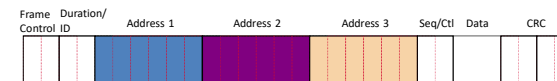


Medium Sharing: Time Division Multiple Access (TDMA)



IEEE 802.3 (Ethernet): CSMA/CD
IEEE 802.11 (EiFi): CSMA/CA

Frame format: Addresses

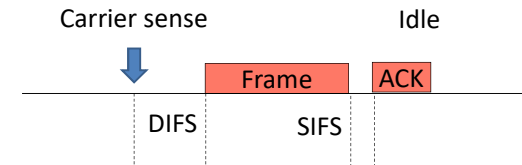


Access protocols

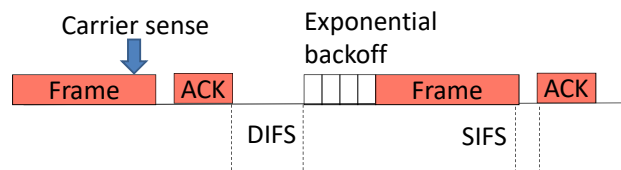
- Wireless channels: Half-duplex
- CSMA/CA is essentially the same as CSMA but ...
 - Low RF link quality => Data+ack
 - Hidden node problem => RTS/CTS
 - Difficult to listen/talk at the same time => Virtual carrier sense (NAV)

CSMA/CA: DCF

If the medium is idle, the station wait a short time (DIFS) and transmits

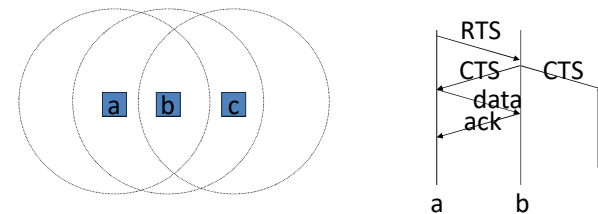


If the medium is busy, the station must wait until it becomes idle, then wait a DIFS, and then picks a random slot number in $[1, CW]$. If no transmission occurs at this slot time, it transmits the frame.



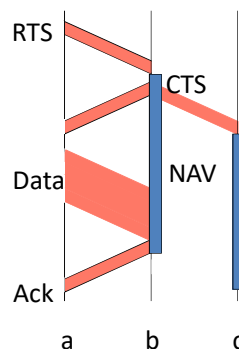
In the first attempt, the **Contention Window (CW)** has CW_{min} slots (e.g. 7). In case of collision, in attempt n -th, the CW is set to $CW_{min} \times 2^{n-1} - 1$, until reaching a maximum value of CW_{max} ...

Hidden node



- **Hidden node problem:** c cannot listen a, so it cannot detect that a is transmitting, and b can suffer a collision.
- When b receives a RTS from a, it sends a CTS so that c knows that b is busy. After this handshake, a can begin its transmission knowing that no collision will occur
- RTS threshold sets the frame length for which shorter frames do not use RTS/CTS handshaking

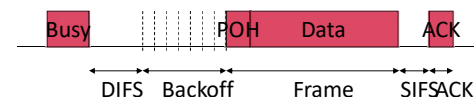
Access protocols



- For avoiding problems in carrier sense due to the high attenuation in Wireless channels, logical carrier-sense can be use instead.
- Network Allocation Vector (NAV):** it is a timer that indicates the time the network will be reserved. stations set the NAV to the time for which they expect to use the medium, including any frames needed to complete the transaction (RTS, CTS, ACKs, etc). NAV is sent in the frames header

DCF inefficiency at high data rates

- Assume that after the first backoff attempt, a frame is always transmitted without collisions...



$$\text{Throughput} = \frac{\text{Payload}}{\text{DIFS} + \text{Backoff} + \text{Frame} + \text{SIFS} + \text{ACK}}$$

Ej: in 802.11a, for Payload of **512**, if we take
 Backoff = $CW_{min}/2$ & 64 QAM 3/4... =>
 Throughput = $0.28 \times 54 \text{ Mbps} = 15.5 \text{ Mbps (!)}$

DCF inefficiency at high data rates

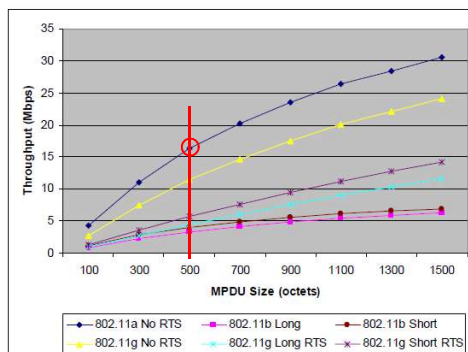


Fig. 2. Analytical maximum throughput comparison for different PHY options.

Source: Choi, DelPrado: "802.11g CP: A Solution for IEEE 802.11g and 802.11b Inter-Working"

802.11n

- Frame aggregation:** combining several frames into one, reducing the number of inter-frame gaps and acknowledgement (ACK) frames
- Aggregate frames can reach the length of 64k bytes and be composed of multiple traditional frames that can range in size from 52 bytes to 2304 bytes.
- Block ACK :** sending a single acknowledgement packet to confirm receipt of multiple frames.

Scanning

- Passive scanning:
 - The station moves from channel to the other, listen and waits for **Beacon frames**. Any received Beacon frame is stored to extract info about the BSS that sent it
 - Saves battery as it does not require transmitting
- Active scanning:
 - The station sends **Probe Request frames** to solicit responses from a network with a given name.
 - It moves from one channel to the other and waits for the ProbeDelay to expire. If any frame is detected, the channel is active and a Probe Request is sent (SSID, BSSID: broadcast)
 - **Probe Response** are generated as response to Probe Request frames.

Security

- **Authentication**: A means to decide whether a client or an AP is the system that it claims to be
- **Encryption**: Transforms the data so that it is unreadable by anyone who does not have a decryption key.
- The original 802.11 standard defined:
 - Privacy: **WEP**
 - Authentication:
 - **Open** (no authentication, relays on preconfigured WEP key to decrypt messages)
 - **Shared Key** (relays on WEP)
- **WEP is vulnerable**

Security: WEP problems

- There are implementation problems of the RC4 stream cipher. With some million frames the **keys can be obtained**
- Other attacks have been also derived that allow the **modification of frames**
- **Static WEP key management is another security problem**, as the WEP authenticates a device and not a user.

Solving 802.11 standard security problems

- Use centralized (e.g. RADIUS) **user-based auth**
 - Prevents unauthorized **users** using authorized **devs**
 - Prevents lost devices forcing re-keying the whole net
 - User-specific keys
- Enterprise and Personal mode (Pre-shared key)
- Dynamic encryption keys
 - Relieves administrator for statically manage keys
- Mutual authentication (avoid rogue APs)

WPA/WPA2/WPA3 (802.11i)

- IEEE: security amendment 802.11i
- WiFi Alliance: WPA, WPA2 (subset of 802.11i), WPA3
- WPA: Wi-Fi Protected Access
 - WPA -> 2003 (quick patch to WEP problems)
 - WPA2 -> 2004
 - WPA3, Enhanced Open, Easy Connect -> 2018
- WPA2: KRACK attack (2016)
- WPA3: some vulnerabilities found (2019)

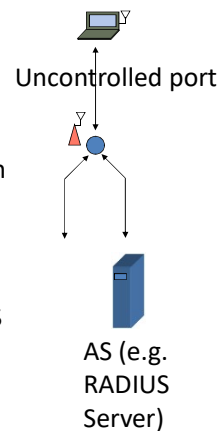
M. Vanhoef, F. Piessens "Key Reinstallation Attacks: Forcing Nonce Reuse in WPA2"

Security: WPA2, WPA3

	WPA2	WPA3
Enterprise mode	Authentication: 802.1x/EAP Encryption: AES-CCMP	Authentication: 802.1x/EAP Encryption: AES-GCMP-256, HMAC-SHA384, ECDH, and ECDSA
Personal mode	Authentication: Pre-Shared Key (PSK) Encryption: AES-CCMP	Authentication: 128-bit (SAE) Encryption: AES-GCMP-256, HMAC-SHA384, ECDH, and ECDSA It includes a WPA3-Personal transition

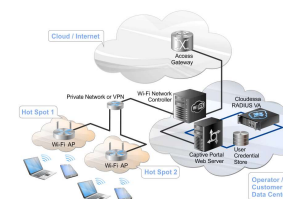
Authentication with 802.1X/EAP

- 802.1x/EAP is a framework for authentication. It does not specify any specific authentication algorithm
- Terminology:
 - Supplicant: Client
 - Authenticator: AP
 - Authenticator Server: e.g. RADIUS server

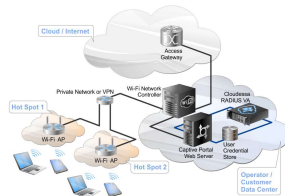


Captive portal

- A guest user connects to the guest SSID, which requires no 802.11 authentication and provides no encryption.
- When the user opens a web browser, a captive portal screen asks to enter login/passwd



Captive portal



- Authentication with the browser is protected using SSL/TLS tunnel
- After authentication, user traffic passes through the controller without 802.11 encryption.
- In many cases, authentication is based on IP and MAC addresses (easy to spoof).
- Enhanced Open OWE protect against passive eavesdropping

Mobile Internet

5.- IoT and Low Power technologies: RFID/NFC, BLE, LoRA

Constrained nodes

- **Maximum code complexity** (ROM/Flash),
- **Size of state and buffers** (RAM),
- **Amount of computation** feasible in a period of time ("processing power"),
- **Available power**, and
- **User interface and accessibility in deployment** (ability to set keys, update software, etc.).

Mobile versus IoT devices

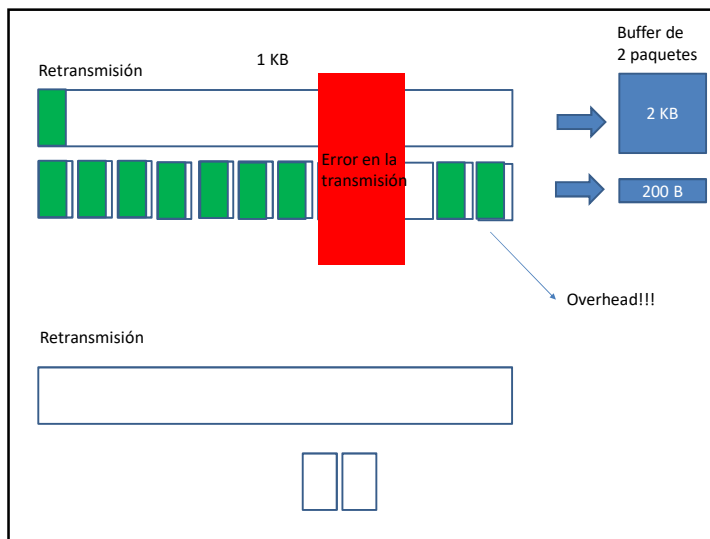
Device	CPU	RAM	Storage	Power
iPhone12	Arm Firestorm 64b, @3.1 GHz	4 GB	256 GB	Watts
Texas Instruments MCU	16b, @20 MHz	4 KB	32 KB	Active: mWatts Standby: µWatts
	x 100	x 1 M	x 8 M	x 1K

Power supply options

- **Battery power supply**, used for mobile equipment.
- **Photovoltaic power supply**, remote or outdoor applications, using photovoltaic panels to convert solar energy into electrical energy.
- **AC power supply**, used for equipment which are connected directly to the network grid.
- **DC power supply**, used when devices are connected to a battery.

Constrained networks

- **Low achievable bitrate/throughput** (including limits on duty cycle),
- **High packet loss** and **high variability of packet loss** (delivery rate),
- **Highly asymmetric link** characteristics,
- **Penalties for using larger packets** (e.g., high packet loss due to link-layer fragmentation),
- **Limits on reachability over time** (a substantial number of devices may power off at any point in time but periodically "wake up" and can communicate for brief periods of time), and
- **Lack of advanced services** such as IP multicast.



Example of a low-end IoT node

- **Micro Controller Units (MCU):** contains the **CPU**, **few kB of RAM and ROM** and **register mapped peripherals**. Typically are single core with **slow clock-rates in the order of few MHz**
- External devices such as sensors, actuators, storage, connected to the MCU via **I/O standards**, such as **UART, SPI or I2C**
- One or more network interfaces using **low-power transmission technology**

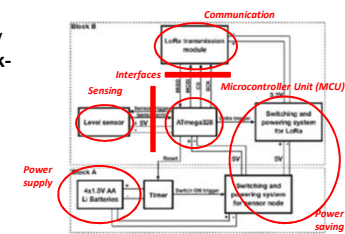


Figure 1. Block diagram of the proposed IoT node.

Classes of constrained nodes

Name	Data size (RAM)	Code size (Flash)
Class 0	<< 10 kB	<< 100 kB
Class 1	≈ 10 kB	≈ 100 kB
Class 2	≈ 50 kB	≈ 250 kB

- **Class 0**
 - Participate in Internet with larger devices acting as **proxies, gateways, or servers**.
 - **Cannot be secured or managed comprehensively**.
 - They will most likely be preconfigured with a very small data set.
- **Class 1**
 - **Cannot use a full protocol stack** such as using HTTP, TLS, and related security protocols and XML data.
 - **They are capable to use a protocol stack specifically designed for constrained nodes** (e.g. CoAP over UDP) **without the help of a gateway node**.
 - **They can provide support for the security functions required on a large network**.
- **Class 2**
 - **Are fundamentally capable of supporting most of the same protocol stacks as used on notebooks or servers**.
 - However, even these **devices can benefit from lightweight and energy-efficient protocols** and from consuming less bandwidth.

Typical code and memory requirements

Function	ROM	RAM
ND Input/Output	4800	20
ND structures	2128	238
Network interface management	1348	118
Stateless address autoconfiguration	372	16
IPv6 (headers processing, etc)	1434	44
Packet buffer (Ethernet case)	0	1296
ICMPv6	1406	16
Total	11488	1748

Table 2: Typical code and memory requirements for a complete IPv6 protocol stack.

Together with complete run-time (timers, scheduler, etc) and UDP and TCP, an OS that provides a **complete IPv6 stack can be implemented in 35 KB of ROM and 3 KB of RAM**.

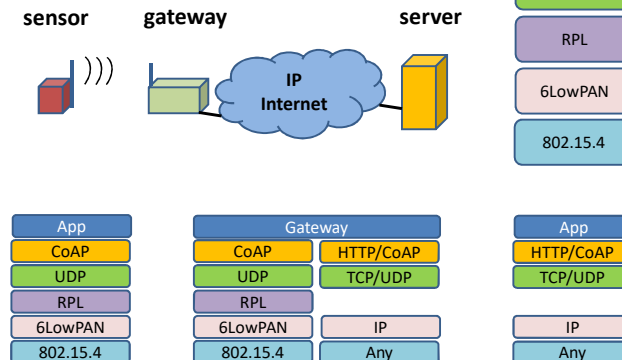
If we add the app, typical we only need 64 KB ROM and 4-8 KB RAM

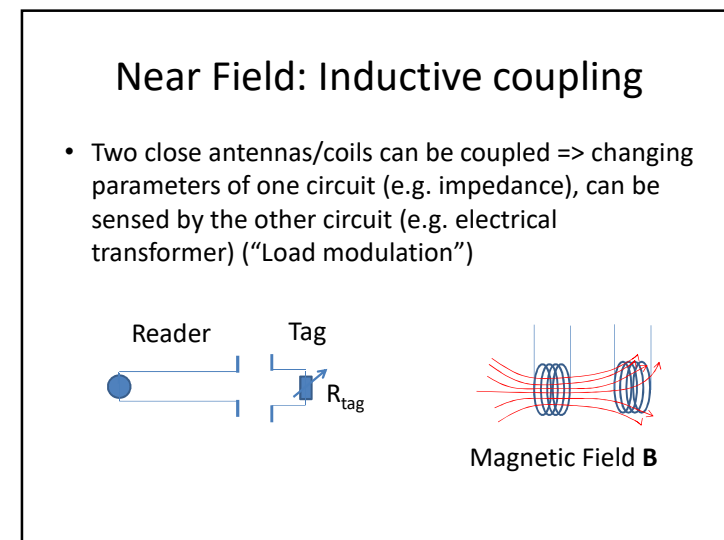
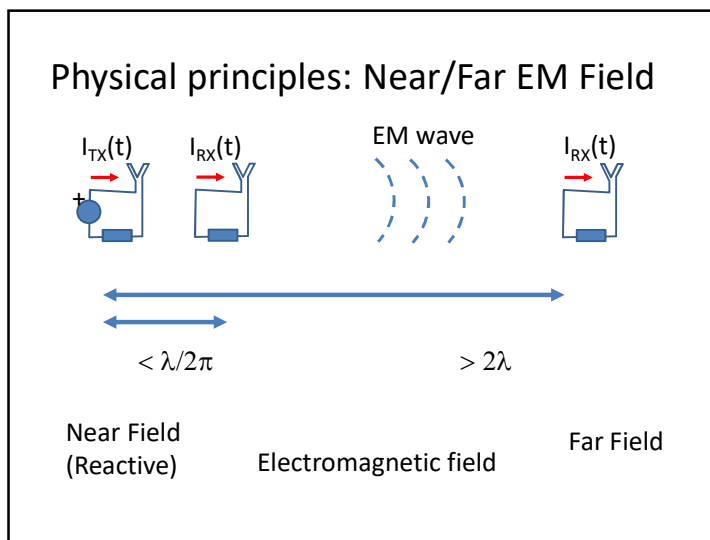
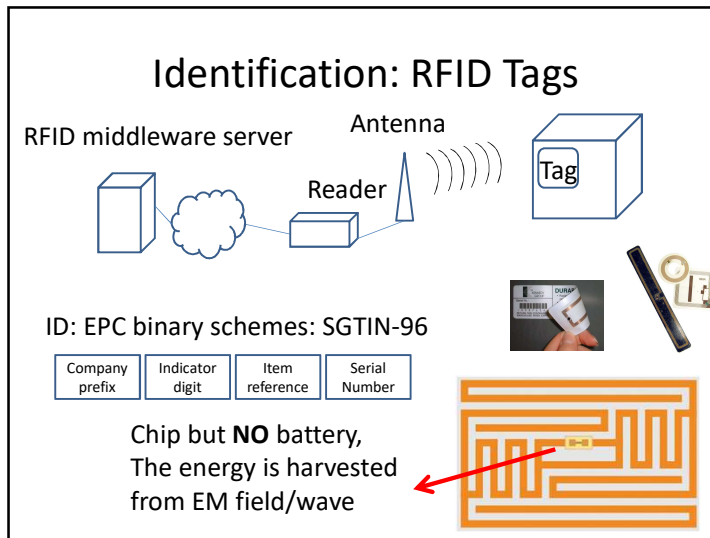
Source: "Lightweight IPv6 Stacks for Smart Objects", J Abeille, etc at IPSO

Why use IP?

- Open standard (IETF)
- Lightweight implementations (e.g. 10 to 30 KB)
- Versatile, ubiquitous
- Well supported management
- Easy interworking with the Internet.
 - Does not need application gateways.
 - It can support caches or proxies to reduce network usage
- Use IPv6 (number of addresses, address autoconfiguration)

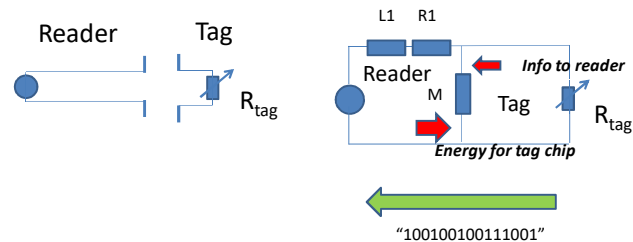
IETF RESTfull stack





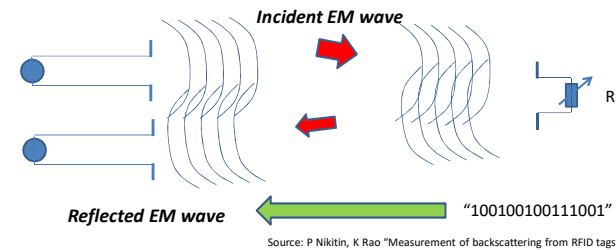
Equivalent circuit

Equivalent circuit: Changes in R_{tag} can be sensed by the reader



Far field: Backscattering

- The RFID reader sends EM waves. The RFID tag antenna changes its parameters (impedance). For each impedance value, the RFID tag presents a certain **Radar Cross Section**.



	PASSIVE	SEMIPASSIVE	ACTIVE
TRANSDUCERS	Inductive Coupling (125/134 KHz, 13,56 MHz) Backscatter (869 MHz, 2.4 GHz, 5.7 GHz) Surface Acoustic Wave (2.4 GHz)	Inductive Coupling (125/134 KHz, 13,56 MHz) Backscatter (869 MHz, 2.4 GHz, 5.7 GHz)	
TRANSCIVER			Conventional Tx/Rx (869 MHz, 2.4 GHz, 5.7 GHz)

Source: J. Griffin, "The Fundamentals of Backscatter Radio and RFID Systems"

NFC

- NFC is an evolution of Radio Frequency Identification (RFID) technology, created mainly for use by mobile devices.
- NFC communicates with a shorter range of around a maximum distance of $d = 14$ cm (**which in reality is less than 4 cm**),
- and a maximum transfer velocity of $R = 424$ Kbps of communication in the unregulated radio frequency ISM band of 13.56 MHz.

NFC

- Two modes of communication:
 - Active mode of communication:** Both devices are active and can transmit and receive data.
 - Passive communication mode:** Only one of the devices generates the radio frequency field. The second device, the passive one (named NFC Tag), acts as passive tag.

NFC



- There are three types of communication:
 - Reader/Writer (RFID compatibility)** is used to modify, store or read data from a passive element or NFC tag
 - Card emulation** is used when the NFC device works as a contactless card and can be used to manage payment systems
 - Peer to Peer** is specifically for active NFC devices, establishing a link between two devices, generally mobile phones (smartphones), which allows an exchange of information between the two.

NFC Data Exchange Format (NDEF)

- NDEF messages provide a standardized method for a reader to communicate with an NFC device.
- A NDEF message consists in a number of NDEF records.
- NDEF records contains:
 - the type (e.g. MIME type, SmartPoster or URI)
 - length of the payload,
 - optional identifier,
 - and the payload.

Example: MIFARE Ultralight Type2 NFC tag

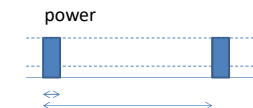
RFID reader understands this

```
[04][A3][2B][C1]: Tag serial number
[78][00][00][00]: Serial number continued
[E1 (nfc)][10 (version 1.0)][06 (mem size)][00]: Capability Container (CC)
[03 (start)][12 (length)][D1][01 (bytes)]: NDEF message start
[0E(h)][5(t)5(t)][01][65]: URI record: https://example.com
[78][61][6D][70]: "xamp"
[6C][65][2E][63]: ".le.c"
[6F][6D][FE(end)][00]: "om" + NDEF terminator
```

IEEE 802.15.4

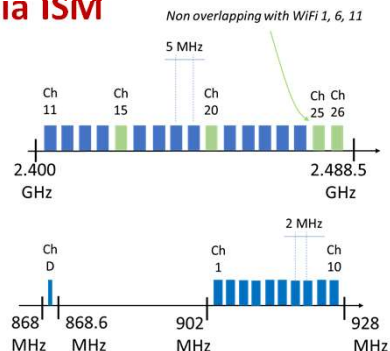
- **IEEE 802.15.4** is a technical standard that defines the operation of a **low-rate, low power, low cost Wireless Personal Area Networks (LR-WPAN)**.
- PHY and MAC layer for LR-WPANs.
- Maintained by the IEEE 802.15 working group, standard in 2003.
- **IEEE 802.15.4**: Original version (rev: **2006**; .4a 2007; .4c, .4d 2009, **2015**)
- **IEEE 802.15.4e**: (**2012**) Amendment to existing standards which adopts channel hopping against external interference and persistent multipath fading, (**TSCH**)
- Used by: **Zigbee**, **Thread** (6LowPAN), **JupiterMesh** (6lowPAN, TSCH), Z-wave

802.15.4 PHY&MAC



- 868 MHz ISM Band (1x Ch, Bw= 0.6 MHz)
(US: 10x Ch, Bw= 2 MHz, 902 MHz ISM Band)
- 2.4 GHz ISM Band (can choose 16x Ch, Bw= 2 MHz, separated 5 MHz)
Phy bit Rate: 250 Kbps (DSSS => 8 chips / bit) => 2 Mcps
- NO multirate
- Data rate up to 115 Kbps, Range: 10-20 m (typ)
- 1 mW output (or larger up to 60 mW)
- Power: 0.1 W during ON (30 mA @ 3.3 V) (*Duty cycle!*)
- CSMA/CA (non-beacon mode) and slotted CSMA/CA (beacon mode)

IEEE 802.15.4 PHY: Bandas de frecuencia ISM



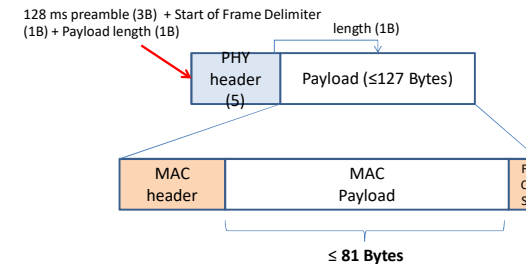
802.15.4 MACs

- 2003, 2006 { **802.15.4 non-beacon** (CSMA/CA)
802.15.4 beacon (CSMA/CA + TDMA)
- 2015 (802.15.4E) Non real time { RFID (RFID integration, Blink mode)
AMCA (Multichannel)
- 2015 (802.15.4E) real time { DSME (Deterministic latency, high reliability)
LLDN (Low latency deterministic networks)
TSCH (High reliability, time critical assurances)

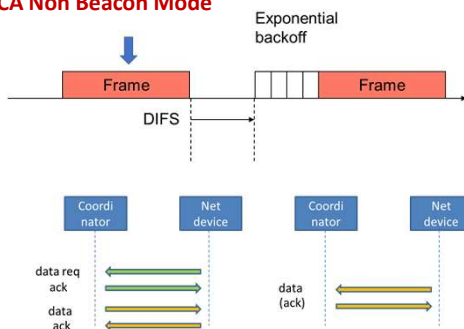
802.15.4 MAC

- The frame length is limited to **127 bytes**:
 - to ensure reasonably low packet error rates when bit-error rates are non-negligible and
 - reflects microcontrollers' limited buffering capabilities.

IEEE 802.15.4 MAC: Tamaño de trama y campo de datos



IEEE 802.15.4 MAC: CSMA/CA Non Beacon Mode



802.15.4E TSCH

- Mainly intended for the support of process automation applications with a particular focus on equipment and process monitoring (oil and gas industry, chemical & pharma products, water/waste water treatments, green energy production, climate control)
- Supports the **Frequency Hopping** mechanism, which mitigates the effects of interference and multipath fading at a considerable scale.
- Shares most of its principles with **WirelessHART**, although it presents minor differences

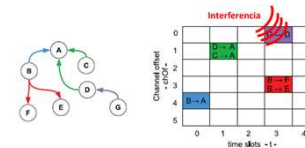
802.15.4E TSCH

- Combines time slotted access with multi-channel and channel hopping capabilities.
 - Time slotted access increases the potential throughput**, by eliminating collision among competing nodes, and provides **deterministic latency** to applications.
 - Multi-channel** allows more nodes to exchange their frames at the same time using different channel offsets. Hence, it **increases the network capacity**. In addition, channel hopping **mitigates the effects of interference and multipath fading**, thus improving the communication reliability.

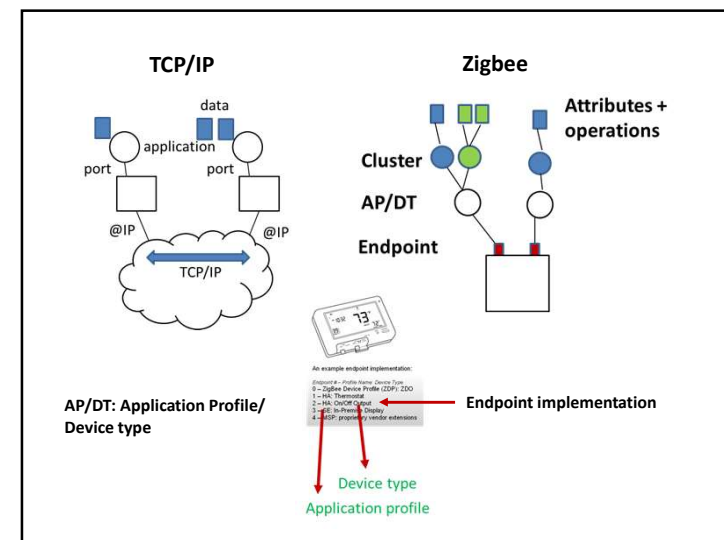
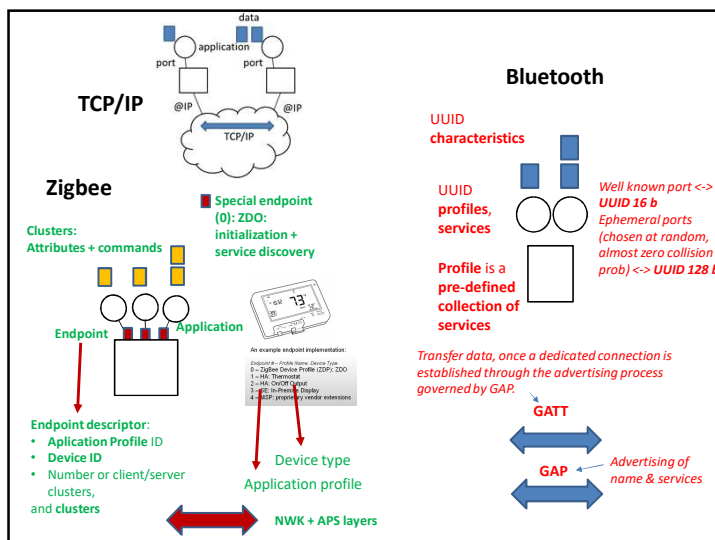
IEEE 802.15.4 MAC: Agilidad de frecuencia y TSCH

Agilidad de frecuencia:

- Cuando se configura inicialmente una red ZigBee, se elige automáticamente el canal de la banda de radio con menos ruido
- La función opcional de agilidad de frecuencia que permite trasladar toda la red a otro canal si el canal operativo cambia sus condiciones.



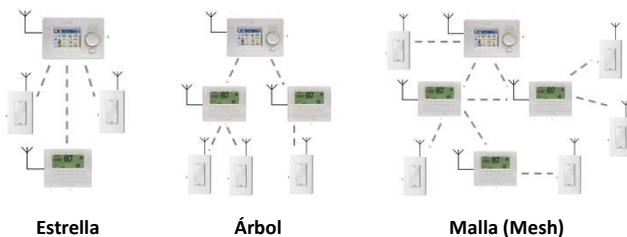
IEEE.802.15E TSCH:
Rotación de canales y scheduling para transmisión en tiempo real



Zigbee NWK layer: Topología de red

Los **nodos coordinadores** y **encaminadores** están típicamente alimentados por corriente.

Solo los **nodos finales** están alimentados por baterías



Zigbee NWK layer: Encaminamiento

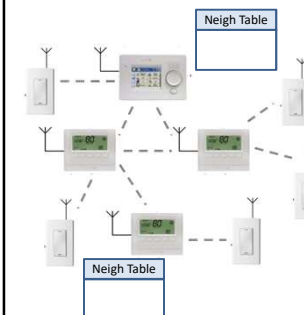


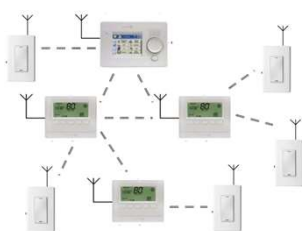
Tabla de vecinos:

Los coordinadores de red y los encaminadores mantienen una tabla con los nodos con los que hay comunicación a un salto.

Se puede fijar un número máximo de entradas en dicha tabla. Por ejemplo, si dicho número es pequeño, se fuerza a tener una red con topología estrecha y alta, "long, thin network".

Zigbee NWK layer: Direcciones e identificadores

Node ID: Net@ (16 b) + Mac@ (64 b)



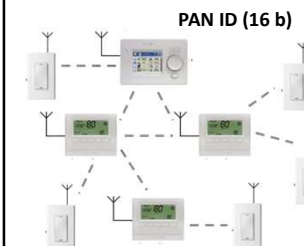
Identificador único de nodo:

Es el resultado de concatenar dos direcciones: **dirección IEEE MAC**; de **64 bits**, asignada de forma única por IEEE; y la **dirección de red**, de **16 bits** asignada como un valor aleatorio de por el padre cuando el nodo se une por primera vez a la red.

Existe un mecanismo para detectar y resolver automáticamente conflictos por dirección de red repetida.

El **nodo coordinador** siempre tiene la dirección de red **0x0000**.

Zigbee NWK layer: Direcciones e identificadores



Identificador de red (PAN ID):

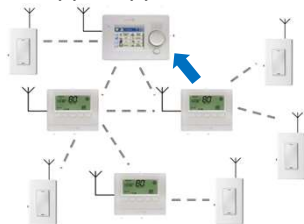
Cuando se inicia la red, el coordinador selecciona **aleatoriamente** un **identificador de red**, de **16 bits**.

Cuando otros nodos se unen a la red, lo aprenden. En caso de conflicto, se genera otro PAN ID aleatorio. Para crear o modificar la red se usa el **PAN IDs extendido, EPID**, de **64 b**, que se fija a un valor aleatorio o a la dirección MAC del coordinador de red.

Mientras que el PAN ID está pensado para ser un valor de 16 bits único para cada red, el EPID se suele utilizar como el campo SSID de las redes WIFI para darles una designación fácil de usar, que a menudo no es tan aleatoria y la establece el fabricante o el instalador.

Zigbee NWK layer: Creación de la red

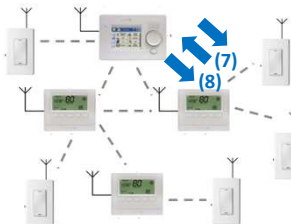
(1) EPID, (2) Net@=0x0000
(3) EDS, (4) PAN ID



1. El coordinador de red selecciona un **EPID**, y asigna su **Net@=0x0000**.
2. El coordinador selecciona un canal de radio (**Energy Detection Scan**).
3. El coordinador selecciona un **PAN ID**.
4. El coordinador puede recibir **request** de otros nodos de la red.

Zigbee NWK layer: Creación de la red

(5) scan, (6) selección, (7) Request
(8) Response, (9) Adquisición Net@

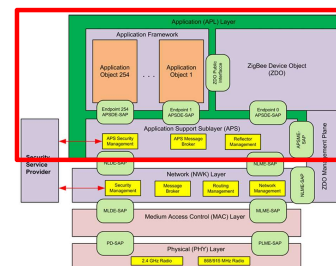


5. Encaminadores y dispositivos finales realizan un **scan**, buscando por nodos padres con algún criterio fijado por la aplicación (ej: **EPAN ID predeterminada**).
6. El nodo selecciona el nodo padre más cercano del coordinador.
7. El nodo envía un mensaje request
8. El nodo padre determina si el nodo puede ser admitido, y elige un Net@ 16b aleatorio
9. El nodo adquiere PAN ID, EPAN ID, Net@.

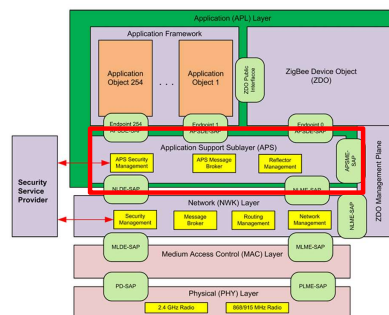
Zigbee NWK layer: Seguridad

- **Lista de acceso:** Fija qué nodos pueden unirse a la red.
- **Encriptado basado en llave:** Utiliza un sistema de encriptado AES-128 de llave simétrica basado en una llave ("network link") compartida por todos los nodos de la red (aunque se pueden usar entre dos nodos de la red, "link key").
- **Message Integrity Code (MIC):** Usando la llave simétrica se hace un hash de la cabecera y payload de la trama para generar un Código de integridad
- **Distribución de llaves:** Las llaves se pueden preconfigurar en los nodos por el fabricante, fijar durante la instalación, o usar un nodo "Trust Centre", típicamente localizado en el nodo coordinador
- **Contadores de trama:** para evitar ataques de "message reply".

Application layer



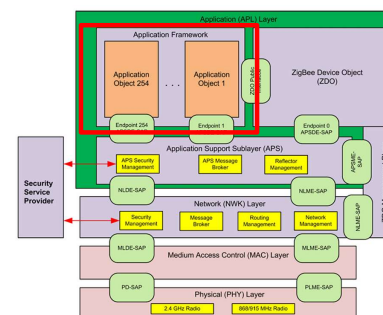
Zigbee APP layer: Application Support Sublayer



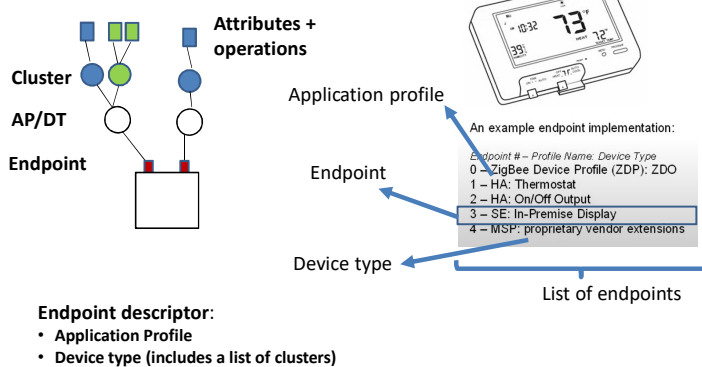
APS

- Filtrado de paquetes para endpoints no registrados, paquetes duplicados, etc
- Generación de ACKs de extremo a extremo con reintentos
- Mantenimiento de la local binding table, local groups table, local address map, etc

Zigbee APP layer: Application Framework



Zigbee APP layer: Application Framework



Zigbee APP layer: Application Framework

Application profile:

For different types of applications, define a list of **devices**, that must support a set of functionalities, defined in terms of **clusters** (mandatory or optional) selected from those defined in Zigbee Cluster Library (ZCL)

Examples of Application profiles:

Home Automation (HA), Commercial Building Automation (CBA), Smart Energy (SE), Manufacture-specific profile (MSP)

Zigbee APP layer: Application profiles

Device

6.3.4 Programmable Communicating Thermostat (PCT) Device
The PCT device shall provide the capability to control the premises heating and cooling systems.

6.3.4.1 Supported Clusters
In addition to those specified in Table 6-1, the PCT device shall support the clusters listed in Table 6-6. If a SE cluster is not listed as mandatory or optional in the following table or in the common table, then that cluster shall be prohibited on a PCT device endpoint.

ZigBee Alliance
ZIGBEE SMART ENERGY STANDARD
Application profile

Table 6-6 – Clusters Supported by the PCT

Server Side	Client Side
Mandatory	
Demand Response and Load Control	Time
Optional	
	Prepayment
	Price
	Calendar
	Metering
	Messaging
	Device Management
	MDU Pairing
Energy Management	
Alarms	
Tunneling	Tunneling

Clusters

Zigbee APP layer: Application Framework

El nodo que **almacena** los atributos es el **server**, mientras que el nodo que **manipula** estos atributos es el **client** del cluster.

- Input cluster (server cluster):** Define el cluster que se usa para **almacenar los atributos y recibir comandos** que manipulan los atributos almacenados. Por ejemplo, la aplicación puede responder a comandos que leen un cierto atributo (responder con una temperatura).
- Output cluster (client cluster):** Define el cluster que se usa para definir los atributos que manipularemos en el server y los comandos que usaremos para manipular (ej: leer una temperatura)

```

graph LR
    A[Application A  
Output Cluster (Client)] -- "Commands sent from client to server" --> B[Application B  
Input Cluster (Server)]
    B -- "Responses returned from server to client (may contain attribute values read)" --> A
    B -- "Attributes written or read, according to command" --> B
  
```

Zigbee APP layer: Application Framework

Cluster
AP/DT
Endpoint

Attributes + operations

Cluster: defines attributes + operations (Client/Server, server keeps attribute's state)

- Identified with a Cluster ID** (16b, e.g. OnOff 0x0006, Measurement 0x0402, etc)
- Defined in the Zigbee Cluster Library (ZCL)**

Zigbee APP layer: Cluster Library (ZCL) Specification

4.4.2.2.1 Temperature Measurement Information Attribute Set
The Temperature Measurement Information attribute set contains the attributes summarized in Table 4-13.

Table 4-13. Temperature Measurement Information Attribute Set

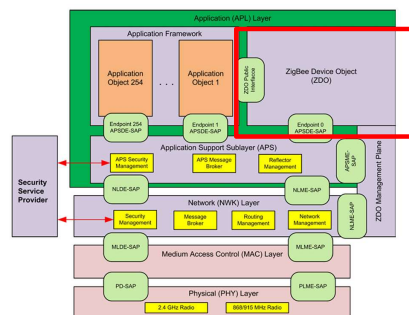
Id	Name	Type	Range	Access	Def	MO
0x0000	MeasuredValue	int16	MinMeasuredValue – MaxMeasuredValue	RP	0	M
0x0001	MinMeasuredValue	int16	0x954d – 0x7ffe	R	-	M
0x0002	MaxMeasuredValue	int16	0x954e – 0x7fff	R	-	M
0x0003	Tolerance	uint16	0x0000 – 0x0800	RP	-	O

4.4.2.2.1.1 MeasuredValue Attribute
MeasuredValue represents the temperature in degrees Celsius as follows:
MeasuredValue = 100 x temperature in degrees Celsius.
Where: -273.15°C <= temperature <= 327.67 °C, corresponding to a MeasuredValue in the range 0x954d to 0x7ffe. The maximum resolution this format allows is 0.01 °C.
A MeasuredValue of 0x8000 indicates that the temperature measurement is invalid.
MeasuredValue is updated continuously as new measurements are made.

4.4.2.2.1.2 MinMeasuredValue Attribute
The MinMeasuredValue attribute indicates the minimum value of MeasuredValue that is capable of being measured. A MinMeasuredValue of 0x8000 indicates that the minimum value is unknown.

ZigBee Alliance
Control your world
ZigBee Cluster Library Specification

Zigbee APP layer: Zigbee Device Object (ZDO)

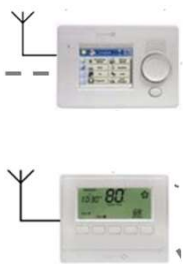


Zigbee APP layer: ZDO



- **Zigbee Device Objects, ZDO** (endpoint=0) es una aplicación especial que se utiliza para manejar los diferentes procesos que se ejecutan en una red Zigbee:
- **Define tipo de device** (coordinador, router, dispositivo final)
- **Inicializa el nodo**
- Implementa los procesos para que el coordinador inicie la red
- Da servicios de **seguridad**
- Inicia y responde las peticiones de **binding**,
- Etc...

Zigbee APP layer: ZDO Device and service discovery



- **Discovery:** Se utiliza para que un nodo descubra características de otros nodos, tales como direcciones (**Device Discovery**), tipos de aplicaciones que ejecuta, tipo de alimentación, comportamiento en periodos de sleep (**Service Discovery**), etc.
- Típicamente se ejecuta en el momento de introducir un nuevo nodo en una red.

LoRa/LoRaWAN



	Europe	North America	China	Korea	Japan	India
Frequency band	867-869MHz	902-928MHz	470-510MHz	920-925MHz	920-925MHz	865-867MHz
Channels	10	64 + 8 + 8	In definition by Technical Committee	In definition by Technical Committee	In definition by Technical Committee	In definition by Technical Committee
Channel BW Up	125/250kHz	125/500kHz				
Channel BW Dn	125kHz	500kHz				
TX Power Up	+14dBm $P_{TX} \approx 25 \text{ mW}$	+20dBm typ (+30dBm allowed)				
TX Power Dn	+14dBm	+27dBm	In definition by Technical Committee	In definition by Technical Committee	In definition by Technical Committee	In definition by Technical Committee
SF Up	7-12	7-10				
Data rate	250bps- 50kbps	980bps-21.9kbps				
Link Budget Up	155dB	154dB				
Link Budget Dn	155dB	157dB				

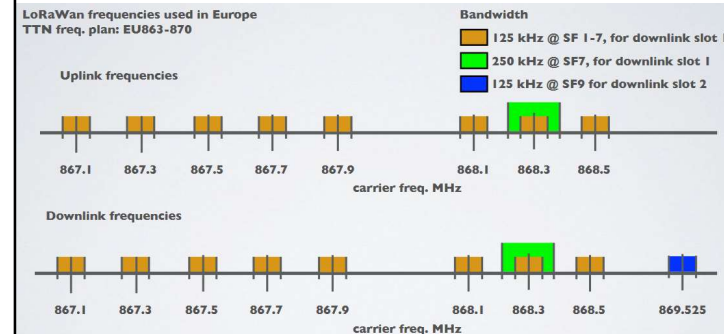
$d \approx 5 - 15 \text{ Km}$

e.g. $R= 250 \text{ bps}$, $Bw=125 \text{ KHz} \Rightarrow R/Bw=1/500$

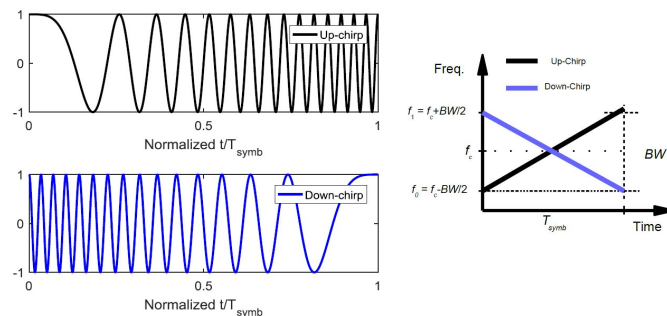
LoRa/LoRaWAN

- Good **long-range** performance but **low data rate** of the transmission modes with high **Spreading Factor (SF)**.
- Duty-cycle implies that it is **not adequate for sending one frame every few minutes**.
- Moreover it is **NOT recommended for real time streaming**.
- **Unlicensed bands**. Use a LoRaWAN operator or deploy a **private LoRaWAN network**.

LoRaWAN frequency bands

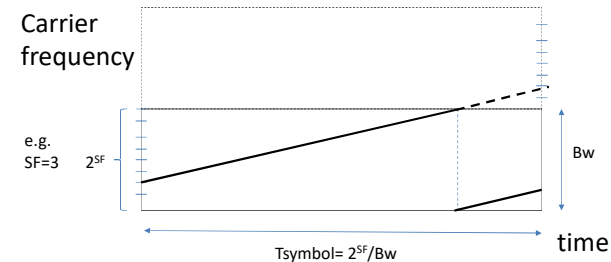


LoRa: FSK CSS (Chirp spread spectrum)

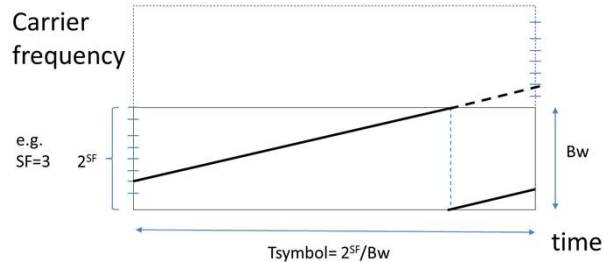


LoRa FSK CSS

The frequency ramps up from an initial value, and **increases linearly** until reaching a maximum value, in which a **frequency fold occur**



LoRa FSK CSS



- Uses a low value of spectral efficiency ($R/Bw < 1$)
- Alphabet size: 2^{SF}
- $T_{\text{symbol}} = (1/Bw) * 2^{SF}$
- $\text{CodingRate} = 4/(4+CR)$
- $R = \text{CodingRate} * Bw * SF / 2^{SF} \Rightarrow$
if SF increases by 1, R decreases by a factor of 2.

SF: tradeoff R versus d

$\text{Coding rate} = 4/(4+CR)$

TABLE I: Main transmission features for different values of the SF parameters. Packet transmission (TX) time is specified for a PHY payload of 32 bytes, explicit header mode, code rate equal to 2, and 125 kHz channel bandwidth.

SF	DR	Data rate [kbit/s]	Sensitivity [dBm]	TX time [s]
7	5	5.470	-130.0	0.740
8	4	3.125	-132.5	0.136
9	3	1.760	-135.0	0.247
10	2	0.980	-137.5	0.493
11	1	0.440	-140.0	0.888
12	0	0.250	-142.5	1.777

Tradeoff d/R

Higher values of SF $\Rightarrow$ Lower values of R $\Rightarrow$ lower values of SNR are possible

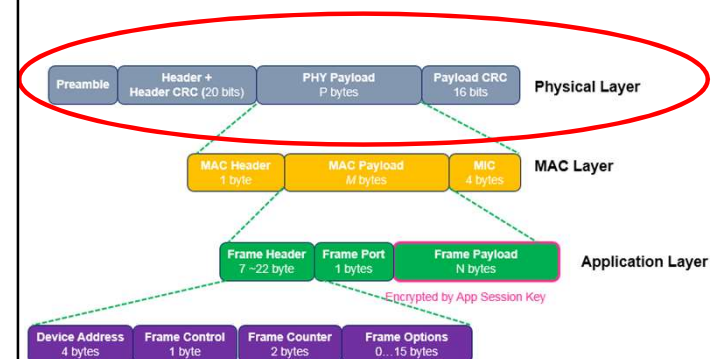
$Bw = 125\text{KHz}$, $SF = 7$, $CR = 1$

$$R = Bw * SF * \text{coding_rate} / 2^{SF} = 5470 \text{ bps}$$

ToA and Duty cycle

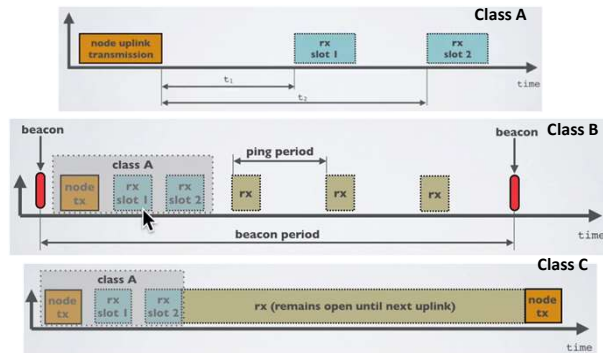
- Transmission time: **Time on Air (ToA)**.
- **Duty cycle** is the proportion of time during which a component, device, or system is operated (0.1% or 1.0%).
- E.g.: 1% duty cycle with ToA of 530ms $\Rightarrow$ after sending a message, we have to wait $99 \times 530\text{ms} = 52.47\text{s}$ before sending a new message.

LoRa Frame Format



Source: <http://www.techplayon.com/loro-long-range-network-architecture-protocol-architecture-and-frame-formats/>

LoraWAN device classes



Class A: Wakeup & send (ALOHA)
 Class B: ALOHA (Tx) + Beacon (Rx)
 Class C: ALOHA (Tx) + Rx always opened

NO association to a LoraGateway

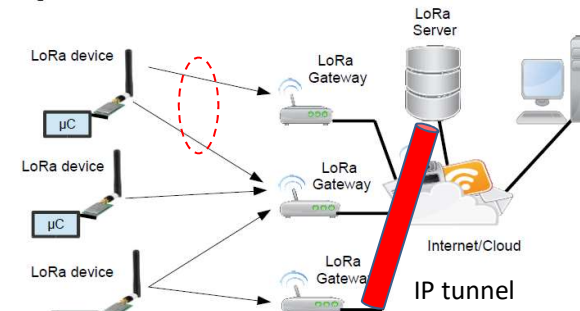
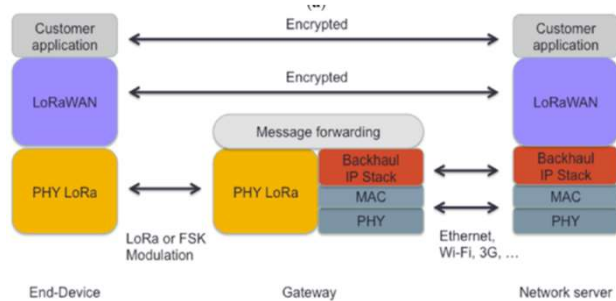


Fig. 1. LoRa simplified network concept.

Lora/LoRaWAN

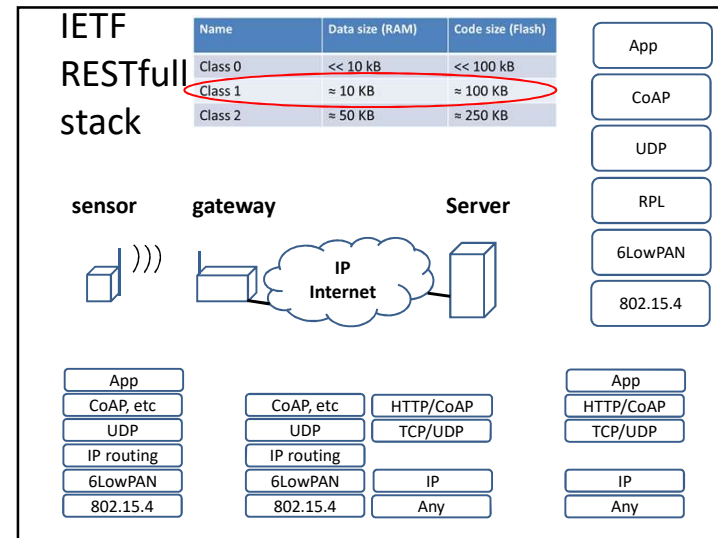


IEEE 802.15.4 PHY: Comparativa

Tech	Bw	R	R/Bw	Ptx	d
4G (2x2)	5, 20 MHz	20 Mbps	1	100 mW	1Km
802.11b	20 MHz	54 Mbps	3	60 mW	100 m
802.11n (2x2)	20 MHz	150 Mbps	7	60 mW	100 m
BLE	2 MHz	125 Kbps	0.2	10 mW	10 m
802.15.4	2 MHz	115 Kbps	0.2	1 mW	10 m
LoRa	125 KHz	50 Kbps (xDC)	0.4	25 mW DC=1%	1 Km
NB-IoT	180 KHz	60 Kbps (xDC)	0.3	100 mW DC nolimit	1 Km

Mobile Internet

5.- Protocolos para redes IoT



Thread network architecture

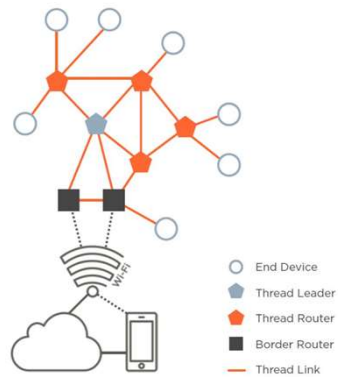
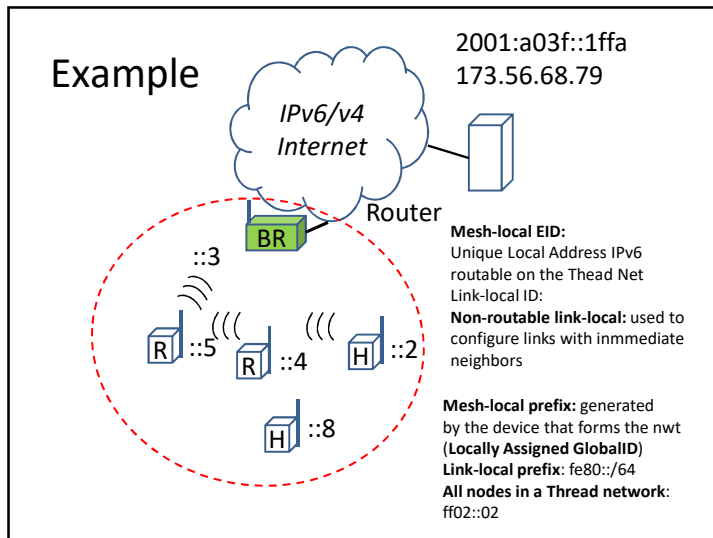


Figure 3. Basic Thread Network Topology and Devices

IPv6 header

VER	T CLASS(8)	FLOW (20)	
PAYLOAD LENGTH		N. HEADER	H. LIMIT
SOURCE ADDRESS			
DESTINATION ADDRESS			

40 bytes
(mínimo)



6Lo & SCHC

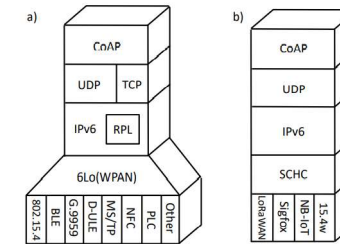


Figure 1. a) 6LoWPAN/6Lo IPv6-based protocol stack, b) LPWAN IPv6-based protocol stack

Source: "IPv6 over LPWANs: connecting Low Power Wide Area Networks to the Internet (of Things)", Carles Gomez et al.

6LoWPAN adaptation layer

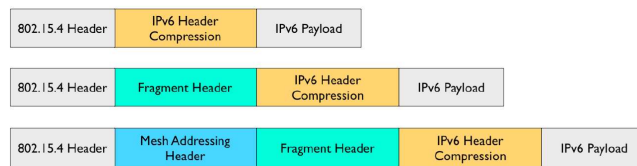


Figure 2. Typical 6LoWPAN Header Stacks.

Source: "6LoWPAN: Incorporating IEEE 802.15.4 into IP architecture", J. Hui, D. Culler, S. Chakrabarti, IPSO

6LoWPAN compression tricks

- Do not include
 - version
 - payload length
 - fe80:: network prefixes
 - Host ID if coincident with MAC UID
- Compress common values for some fields:
 - flow=0, TOS=0,
 - NextHeader: UDP, TCP, ICMP, etc

VER	T CLASS(8)	FLOW (20)	
PAYLOAD LENGTH		N. HEADER	H. LIMIT
SOURCE ADDRESS			
DESTINATION ADDRESS			

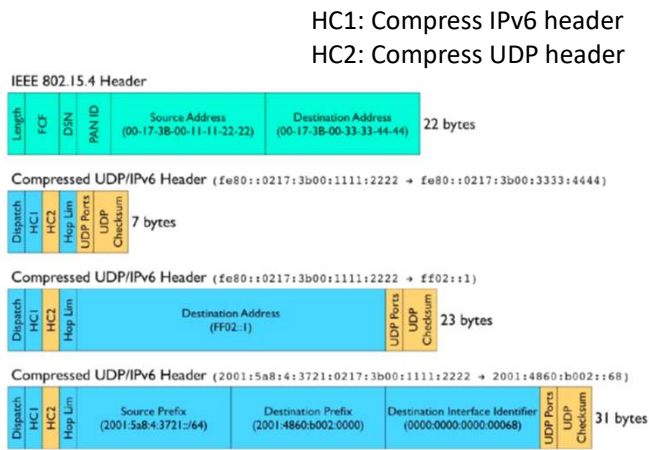
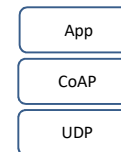


Figure 7. 6LoWPAN RFC 4944 Header Compression Examples.

Source: "6LoWPAN: Incorporating IEEE 802.15.4 into IP architecture", J. Hui, D. Culler, S. Chakrabarti, IPSO

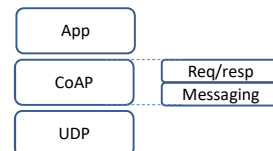
IETF CoAP draft-ietf-core-coap-13

- Similar interaction model to client/server of HTTP
- **Action** (method code) on a **resource** (URI) on a **server**.
- Server sends back a **response** (response code)
- Unlike HTTP, CoAP works on UDP



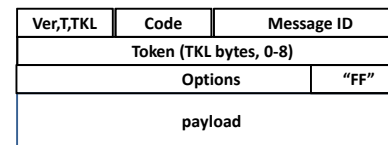
CoAP

- Messaging layer:
 - Reliability
- 4 types of messages:
 - Confirmable
 - Non-confirmable
 - Ack
 - Reset
- Request/response layer:
 - HTTP-like interactions
- Support of caching and proxies



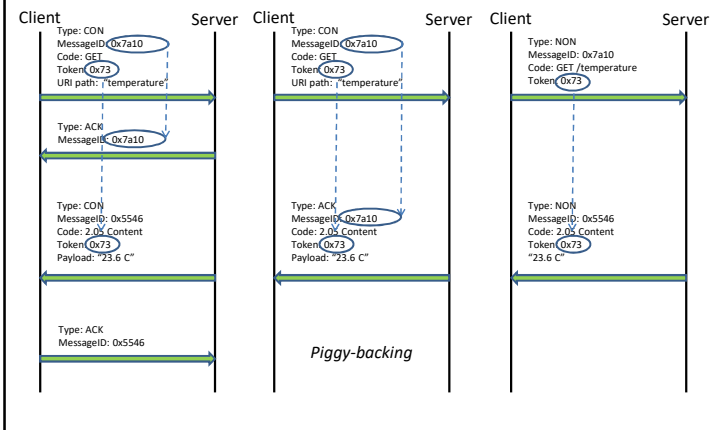
CoAP messages

- Short fixed-length binary header of 4 bytes
- Token (0-8 bytes)+options
- Payload
- Message ID for duplicates and optional reliability (match message/ACK)
- Token to match CoAP request/response



T: Conf, Non-Conf, Ack, Reset
Token Length: 0-8
"FF" (1111 1111): Only when options

Message exchange



Methods & responses

- *Methods (similar to HTTP methods):*
 - GET
 - POST
 - PUT
 - DELETE
- *Responses (similar to HTTP responses):*
 - Success: 2.XX (2.03 valid, 2.05 Content, etc)
 - Client error: 4.XX (4.03 Forbide, 4.04 Not Found, etc)
 - Server error: 5.XX (5.01 Not Implemented, etc)

CoAP versus HTTP

Experimental Evaluation of the CoAP, HTTP and SPDY Transport Services for Internet of Things

Laila Daniel¹, Markku Kojo² and Mikael Latvala²

¹ Department of Computer Science, University of Helsinki, Finland
² Mosa Consulting, Finland
 {ldaniel@cs.helsinki.fi, markku.kojo@mosa-consulting.fi, mikael.latvala@mosa-consulting.com}

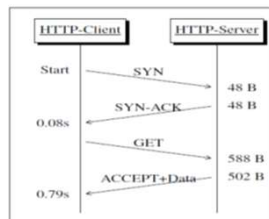


Fig. 6. Message Sequence Chart (MSC) for object download with HTTP

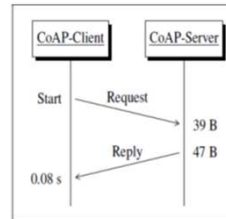
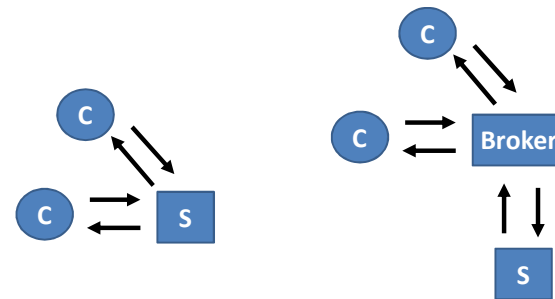


Fig. 8. MSC for object download with CoAP

Request/reply versus Pub/Subs patterns

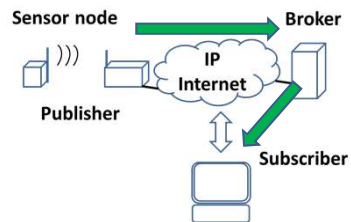


REST protocols (HTTP, CoAP)

MQTT, BLE Mesh

Pub/subs pattern

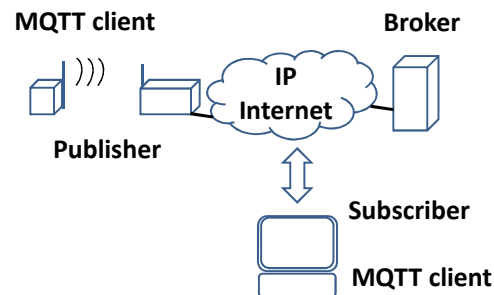
- Client nodes can be **publisher** or **subscribers**
- There is a third component, called **broker**, which is known by both the publisher and subscriber, which filters all incoming messages and distributes them.



MQTT

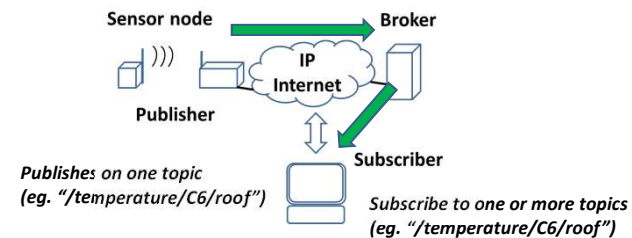
- **MQTT (Message Queueing Telemetry Transport)** uses a publish/subscribe messaging transport protocol.
- Very simple operation in the client side. It only assumes point-to-point links to a server (broker), using TCP.
- Originally developed by IBM. Now is an open ISO/OASIS standard.
- **MQTT-SN** is a modification for nodes with reduced capacities.

MQTT entities



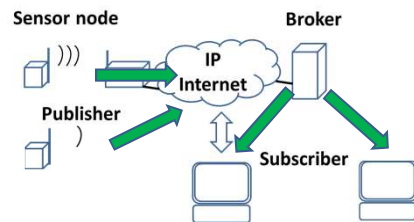
MQTT

- Information is organized into **topics**. Topics are hierarchically organized: e.g. "temperature/C6building/roof", "temperature/C6building/office212" etc

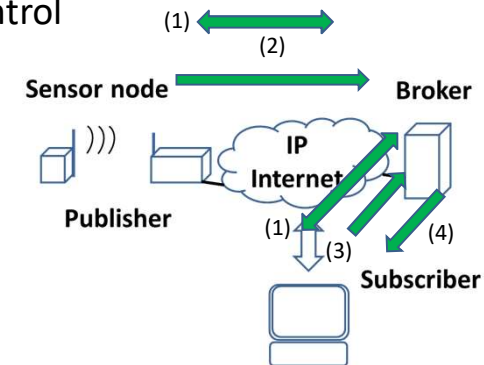


MQTT

- Publishers and subscribers don't know about the existence of one another.
- Multiple clients can receive the message from a single broker, and multiple publishers can publish topics to a single subscriber.



MQTT control messages

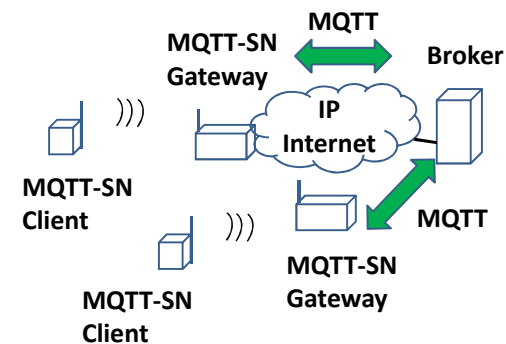


- (1) CONNECT(clientid), CONNECTACK
- (2) PUBLISH (/temperature/c6/roof 25C option:retain)
- (3) SUBSCRIBE("/temperature/c6/roof")
- (4) PUBLISH ("/temperature/c6/roof 25C")

MQTT-SN

- MQTT works on top of TCP
- MQTT topic names are often long strings which make them impractical for 802.15.4.
- The **MQTT-SN** protocol defines a **UDP** mapping of MQTT and adds broker support for **topic_id** (avoids long strings for topic names). Messages are 64 B long
- See: U. Hunkeler, H. Linh Truong, A. Stanford-Clark "MQTT-S – A Publish/Subscribe Protocol For Wireless Sensor Networks"

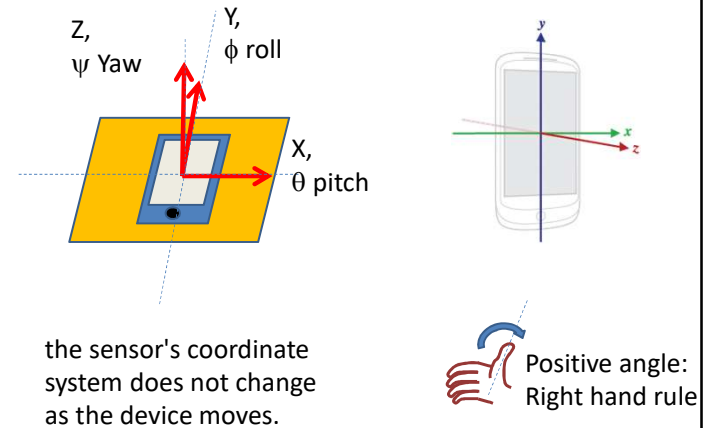
MQTT-SN



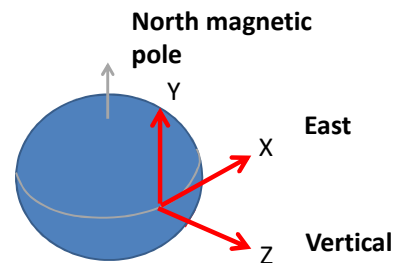
Mobile Internet

6.- Inertial Sensors

Device coordinate system

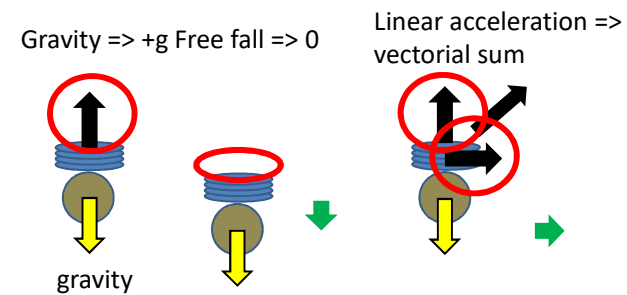


Global coordinate system

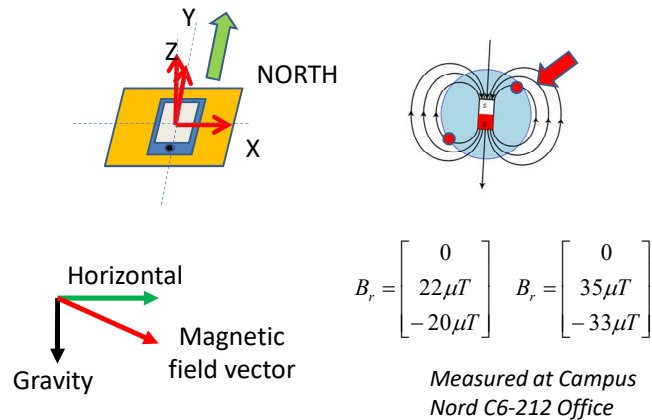


Accelerometer

Measures the **force that acts between the smartphone and the inertial mass**



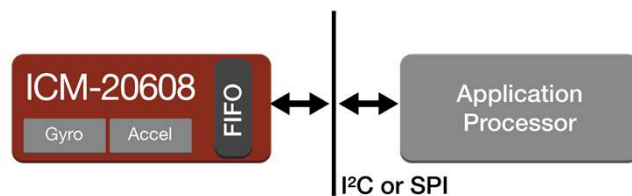
Compass



Accelerometer and gyroscope

- **Accelerometer**
 - Senses linear movement, good for tilt detection
 - Cannot differentiate between gravity and linear movement
 - Noise due to small movements (hand jitter)
- **Gyroscope**
 - Measures rotations, not linear movement
 - Does not amplify hand jitter
 - Possible constant drift
- **Acc+Gyr** = both rotation and movement tracking possible

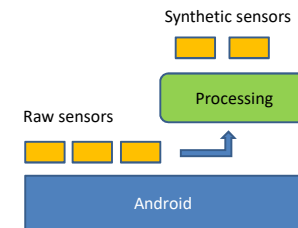
MEMS Sensing



e.g.: 6-axis accelerometer & gyroscope

Sensor types

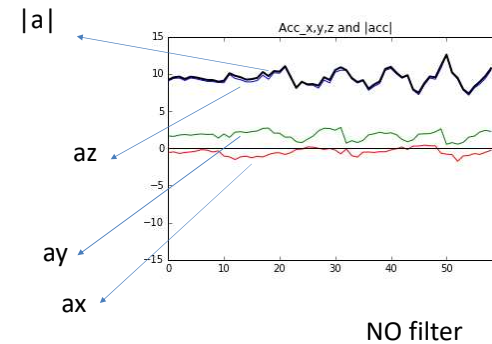
- **Raw (physical) sensors:**
 - .TYPE_LIGHT,
 - .TYPE_ACCELEROMETER,
 - .TYPE_MAGNETIC_FIELD,
 - .TYPE_GYROSCOPE
- **Synthetic sensor:**
 - .TYPE_ROTATION_VECTOR
 - .TYPE_LINEAR_ACCELERATION
 - .TYPE_GRAVITY
 - .TYPE_ORIENTATION



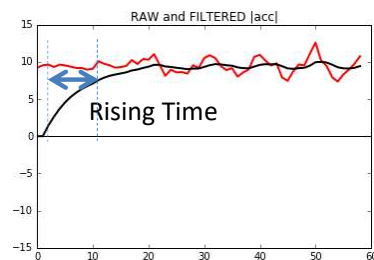
Mobile Internet

7.- Feature extraction, absolute & relative orientation

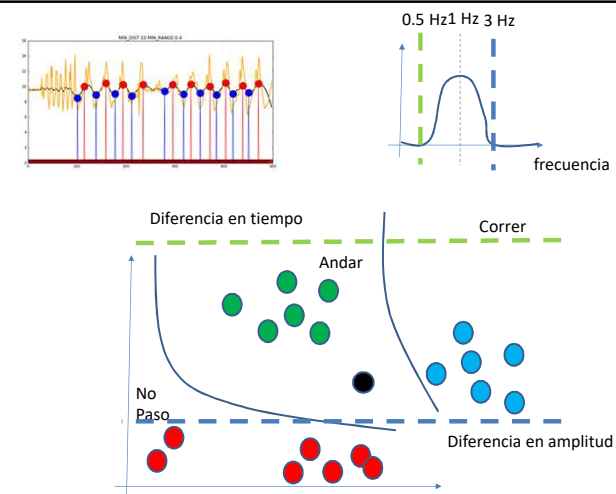
Readings from accelerometer



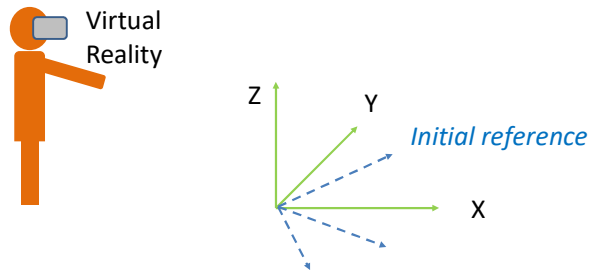
Low pass filter, alpha=0.85



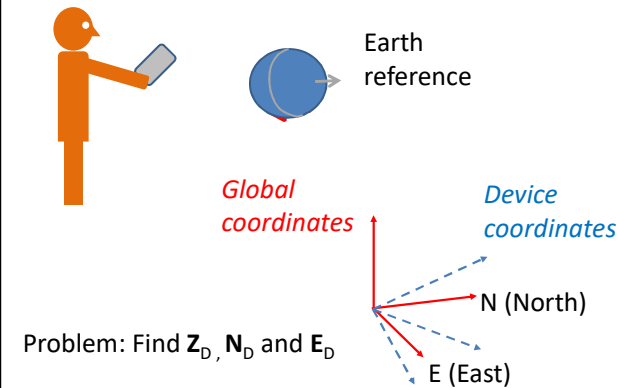
$$y[i] = \alpha * y[i-1] + (1.0 - \alpha) * x[i]$$



Relative orientation



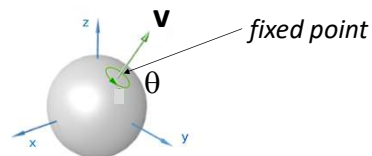
Absolute orientation



Source: https://en.wikipedia.org/wiki/Euler%27s_rotation_theorem

3D rotations: $\mathbf{v}, \theta$ (axis-angle)

Euler's rotation theorem states that, in three-dimensional space, any displacement of a rigid body such that a point on the rigid body remains fixed, is equivalent to a single rotation about some axis that runs through the fixed point.

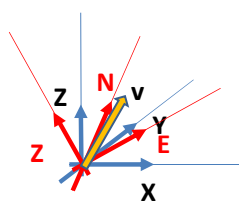


Rotation Matrix (D->E)

$$R_{E \rightarrow D} = \begin{bmatrix} E1_D & N1_D & Z1_D \\ E2_D & N2_D & Z2_D \\ E3_D & N3_D & Z3_D \end{bmatrix} \begin{bmatrix} v1_E \\ v2_E \\ v3_E \end{bmatrix}$$

East vector in Device coordinates North vector in Device coordinates Vertical vector in Device coordinates

Rotation Matrix (E->D)



$$R_{D \rightarrow E} = \begin{bmatrix} x1_E & y1_E & z1_E \\ x2_E & y2_E & z2_E \\ x3_E & y3_E & z3_E \end{bmatrix} \begin{bmatrix} v1_D \\ v2_D \\ v3_D \end{bmatrix} = \begin{bmatrix} v1_E \\ v2_E \\ v3_E \end{bmatrix}$$

x vector in Earth coordinates y vector in Earth coordinates z vector in Earth coordinates

Quaternions

- Useful way to represent 3D rotations
- Generalization of complex numbers to 4-D:

$$q = w + ix + jy + kz$$

$$i^2 = j^2 = k^2 = ijk = -1$$

$$ij = k, ji = -k, jk = i, kj = -i, ki = j, ik = -j$$

$$\text{e.g.: } (0.5 + 8j)(1 + 2k) = 0.5 + 8j + k + 16jk = 0.5 + 16i + 8j + k$$

Quaternions

- $q^* = w - ix - jy - kz$
- $|q|^2 = q q^* = x^2 + y^2 + z^2 + w^2$
- $q^{-1} = q^* / |q|^2$
- Non commutative $qp \neq pq$ in general...
- If we have two **imaginary** quaternions
 - real part of product: - dot product.
 - imaginary parts of product: cross product

Using quaternions for 3D rotations

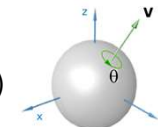
- Rotate the vector $a = (a_x, a_y, a_z)$ an angle θ around an fixed unitary vector axis $v = (v_x, v_y, v_z)$ (Euler axis):

$$a = a_x i + a_y j + a_z k$$

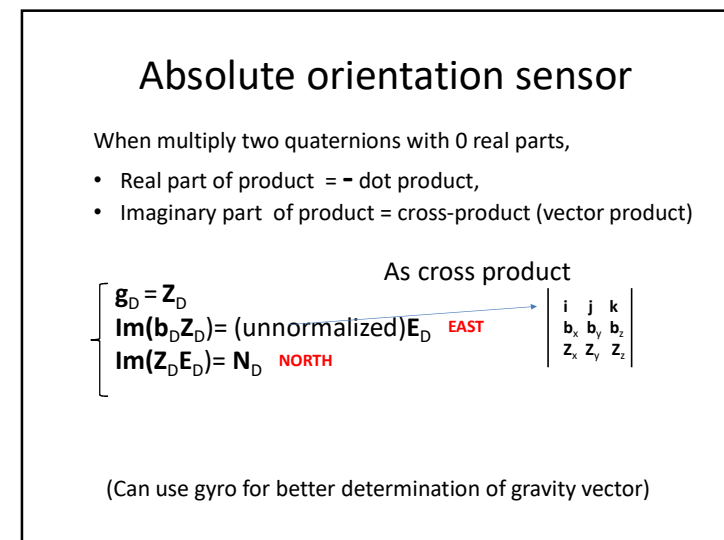
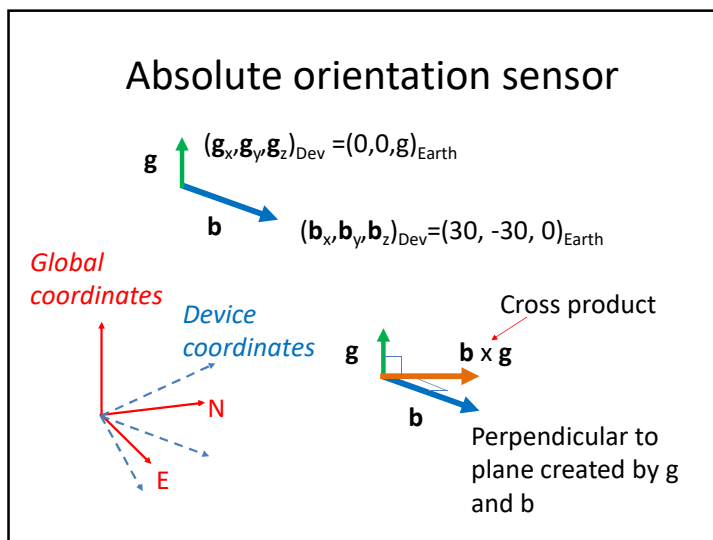
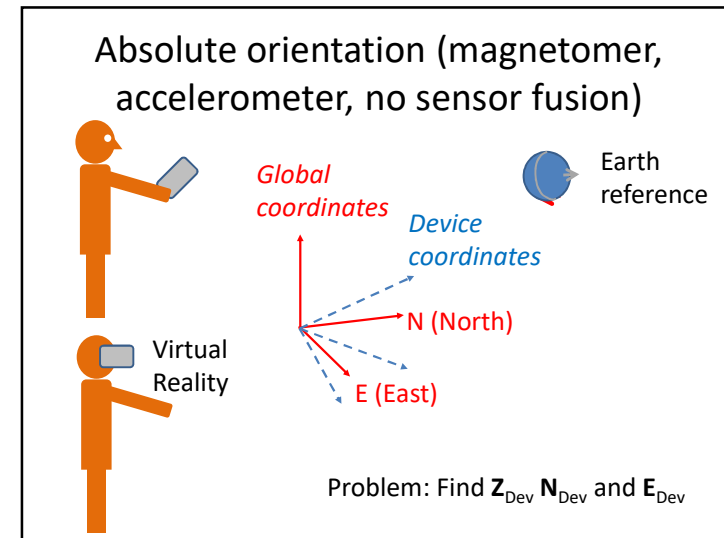
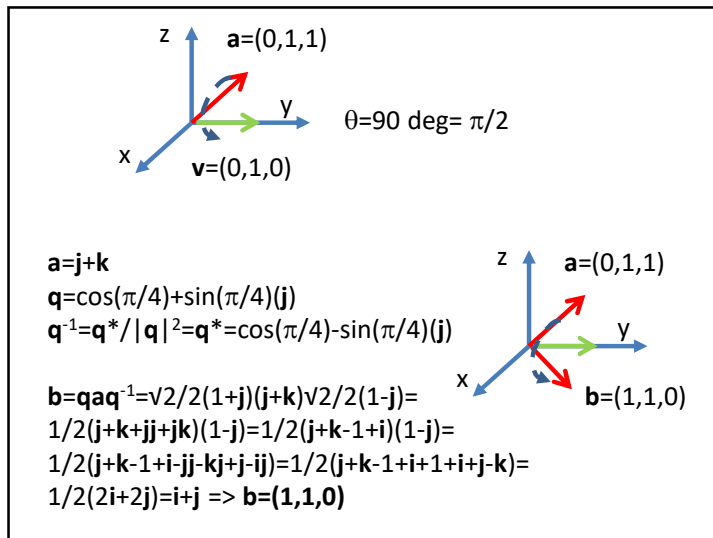
Notation: $q(\theta, v)$

$$q = \cos(\theta/2) + (v_x i + v_y j + v_z k) \sin(\theta/2)$$

$$q^{-1} = q^* = \cos(\theta/2) - (v_x i + v_y j + v_z k) \sin(\theta/2)$$



$$\text{Rotation} = q a q^*$$



Rotation matrix $R_{E \rightarrow D}$

- The rotation matrix from Earth to Devices coordinates, can be found as:

$$R_{E \rightarrow D} = [E_D \ N_D \ Z_D]$$

- Once we know this matrix, we can change from vectors in Earth coordinates to their expression in Device coordinates, e.g.

$$\begin{aligned} E_D &= [E_D \ N_D \ Z_D][1 \ 0 \ 0]^t \\ \mathbf{v}_D &= R_{E \rightarrow D} \mathbf{v}_E \\ \text{Device to Earth} \quad \mathbf{v}_E &= R_{D \rightarrow E} \mathbf{v}_D = [x_E \ y_E \ z_E] \mathbf{v}_D = R_{E \rightarrow D}^t \mathbf{v}_D \end{aligned}$$

Sensor.TYPE_GYROSCOPE

- Returns **angular velocity around the three axis**: $\omega = (\omega_x, \omega_y, \omega_z)$ in rad/sec.
- We can **convert these measurements into a instantaneous rotation quaternion** for a small time Δt as:

$$q_{\Delta t} = \cos(\theta/2) + \sin(\theta/2)(v_x i + v_y j + v_z k)$$

with: $\theta = \Delta t |\omega|$, $\mathbf{v} = \omega / |\omega|$

Relative rotation

source:

https://stanford.edu/class/ee267/notes/ee267_notes_imu.pdf

- Start with a no-rotation gyro-rotation quaternion:

$$q^{(0)} = 1 + i0 + j0 + k0$$

- Convert 3-axis gyro measurements $\omega = (\omega_x, \omega_y, \omega_z)$ to **instantaneous rotation quaternion**:

$$q^{dt} = \cos(\theta/2) + \sin(\theta/2)(v_x i + v_y j + v_z k)$$

with: $\theta = dt |\omega|$, $\mathbf{v} = \omega / |\omega|$

Relative rotation

- Integrate to find the gyro rotation quaternion:

$$q^{(t+dt)} = q^{(t)} q^{dt}$$

- The integrated quaternion $q^{(t+dt)}$ represents the **rotation from current device coordinates to initial device coordinates**:

$$\begin{aligned} a_{\text{initial_device}} &= q^{(t+dt)} a_{\text{current_device}} q^{(t+dt)*} \\ a_{\text{current_device}} &= q^{(t+dt)*} a_{\text{initial_device}} q^{(t+dt)} \end{aligned}$$

Consequences of no using sensor fusion

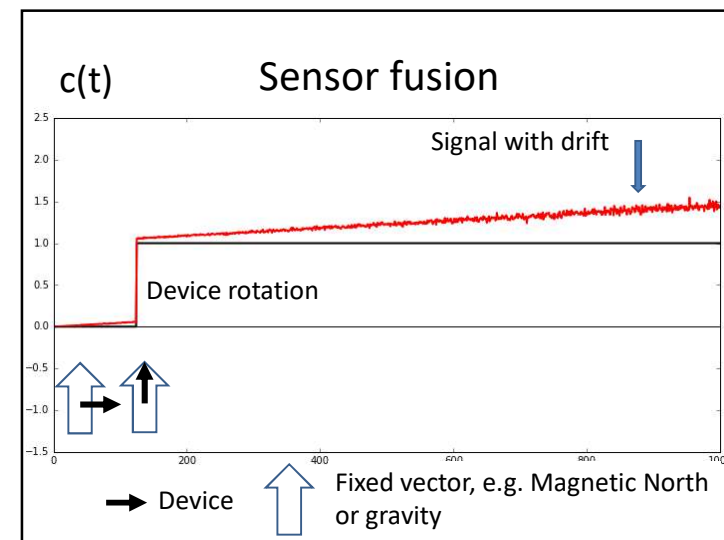
- **Absolute orientation:**
 - We have assumed that we can extract gravity
 - In situations in which the device moves this is difficult as accelerometer measures gravity + linear acceleration.
- **Relative orientation:**
 - We have assumed that the gyro works perfectly
 - Usually we can measure very well fast movements, but gyros suffer from drift.

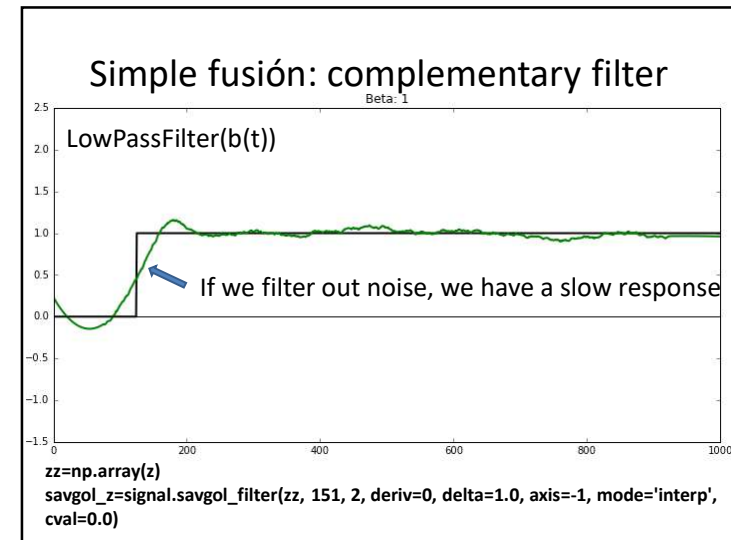
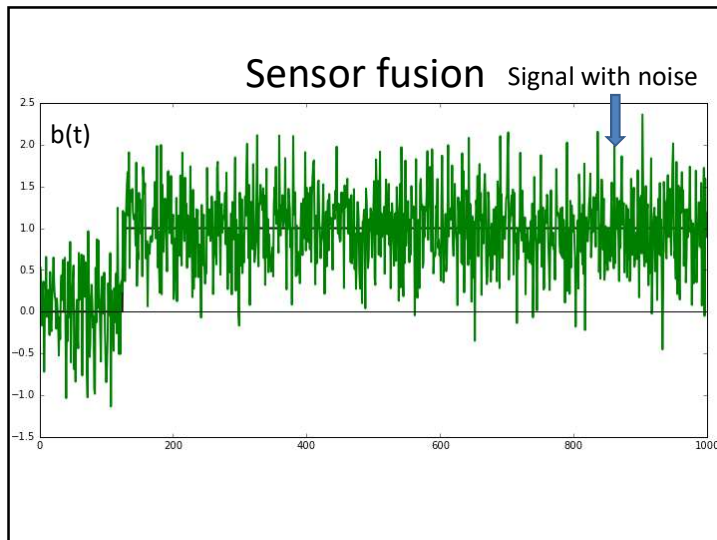
IMU sensors limitations

- **Accelerometer:**
 - We read the sum of linear acceleration and gravity. If the phone is in moving, difficult to separate these components.
 - If we low-filter the signal for extracting gravity (e.g.) we can have large delays.
 - Gravity in the earth reference system is 0 frequency signal (it is essentially constant with time), but **gravity IS NOT a low frequency signal in the device reference system** (as the device moves itself)

IMU sensors limitations

- **Magnetometer:**
 - Magnetic field fluctuates.
 - External effects (metallic objects, motors, etc)
- **Gyroscope:**
 - Reads angular velocity. To obtain angle we must integrate (integrate means low pass filtering) => follows very well fast movements, but has a drift





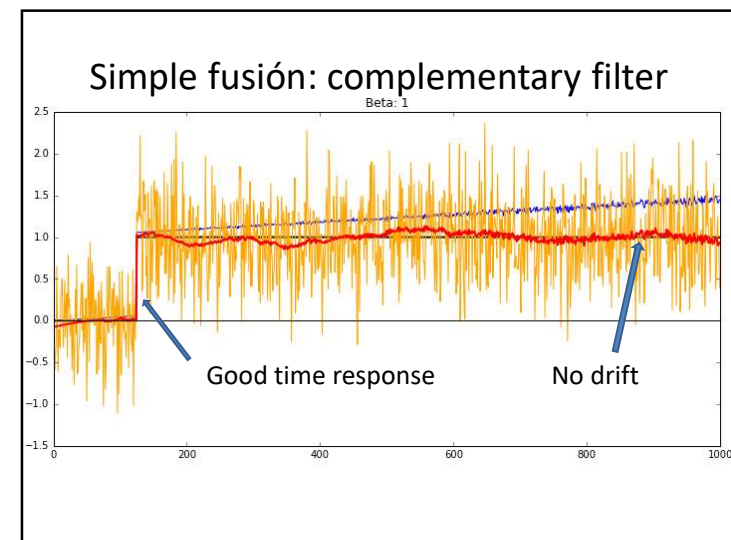
Complementary filter

- $b(t)$ = signal with noise
- $c(t)$ = signal with drift

$$s(t) = \text{LowPassFilter}[b(t)] + \overbrace{c(t) - \text{LowPassFilter}[c(t)]}^{\text{HighPassFilter}[c(t)]} =$$

$$= c(t) + \underbrace{\text{LowPassFilter}(b(t) - c(t))}_{\text{Error}(t)}$$

e.g

$$\text{LowPassFilter}[e(t)] = \alpha \text{LowPassFilter}[e(t-1)] + (1-\alpha)e(t)$$


Mobile Internet

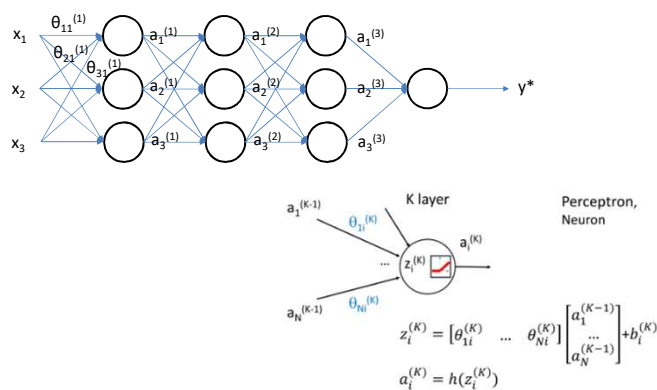
7.- Machine Learning & AI

Finding an estimate θ

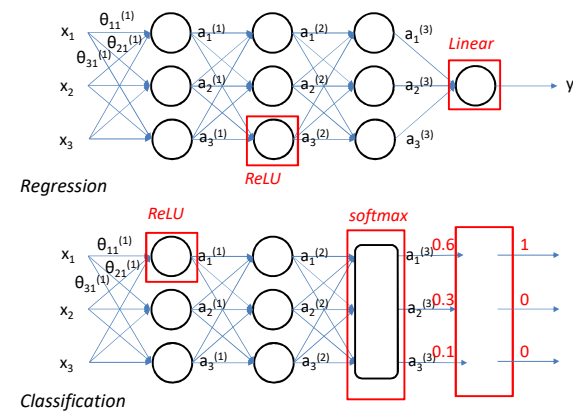
- To find the optimal value of θ , we define a **loss function** $L(g_\theta(x), y)$ and we try to find $g_\theta(x)$, that **minimizes the Risk function** $R(\theta) = E_{x,y}[L(g_\theta(x), y)]$
- When Y takes discrete values (classification problems), it is usual to use the **cross entropy loss**:

$$L(g_\theta(x), y) = -\mathbf{1}_{\{y=0\}} \log(g_\theta(x)) - \mathbf{1}_{\{y=1\}} \log(1 - g_\theta(x))$$

E.g. Neural Network



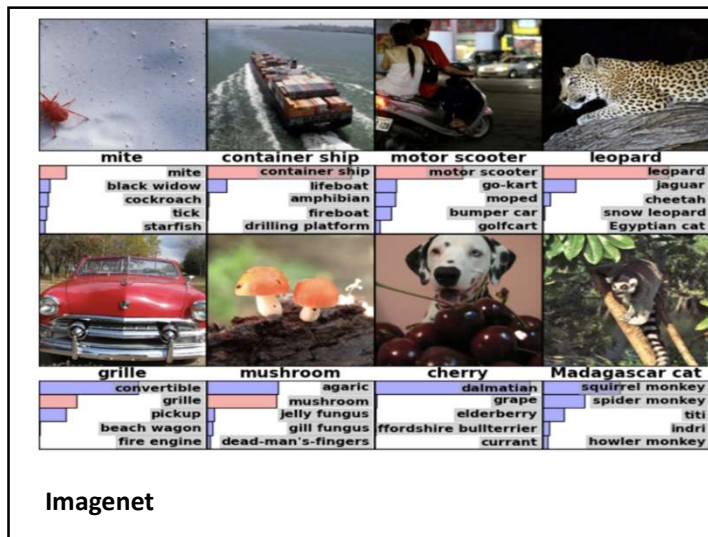
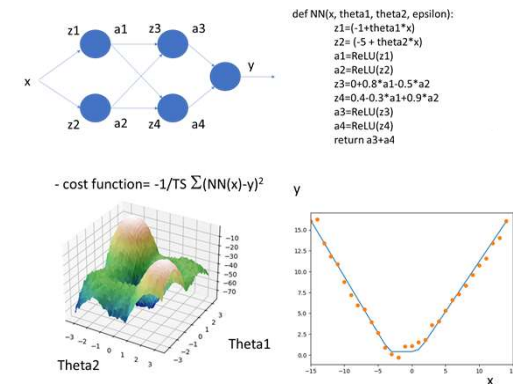
E.g. Neural Network



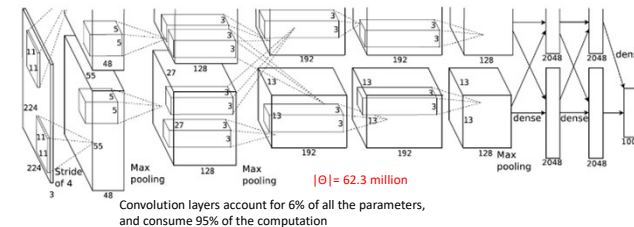
E.g. Neural network

- In this case, there is no explicit expression for the optimal value of θ . We need to use an iterative method, based on computing the gradients of the cost function $J(\theta)$ in terms of the coefficients $\theta_{ij}^{(k)}$, to find an approximate solution.
- The algorithm for finding the gradients in an efficient way is called **backpropagation**.
- Once the gradients are found, we use **Stochastic Gradient Descent** (SGD) for finding the optimal solution.

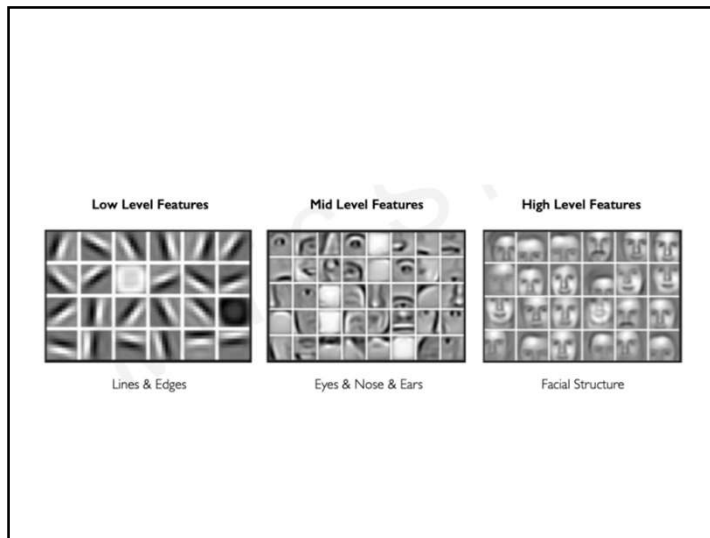
Simple Neural network for regression



AlexNet 2012 ImageNet LSVRC-2012, won the competition by a large margin
Error rates: 15.3% versus 26.2% (second place)



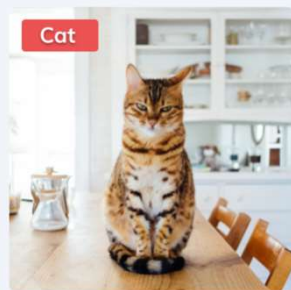
Training: 90 epochs (i.e. $5 \times |T|$)
five or six days to train on two GTX 580 GPUs.
SGD $\mu=0.01$, divided by 10 once the accuracy plateaus, momentum 0.9; weight decay 0.0005.



Computer vision

- Methods for processing and analyzing digital images for solving tasks such as:
 - **Object recognition/classification:** recognize several classes of objects,
 - **Object detection:** recognize also their 2D/3D positions in the image/scene.
 - **Identification/recognition** – recognize an individual instance of an object, for instance the specific person's face

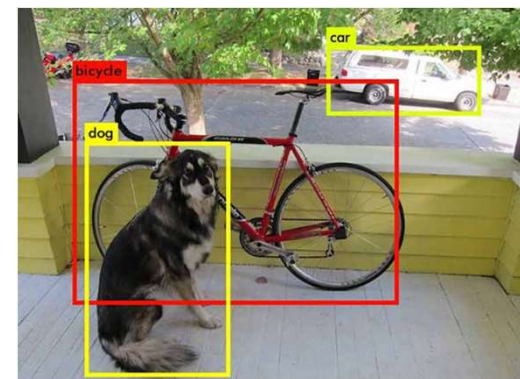
Classification

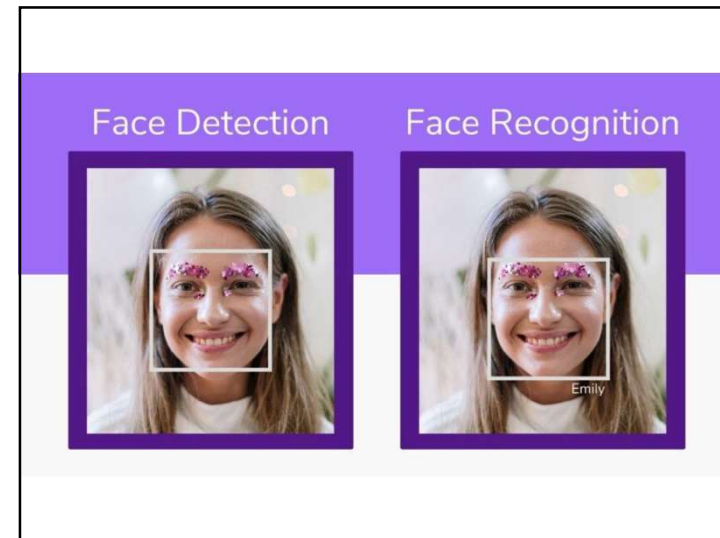
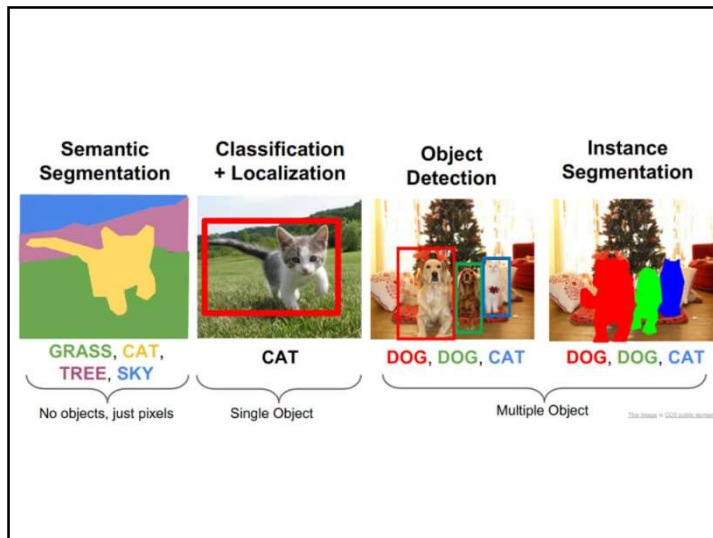


Cat

Cat

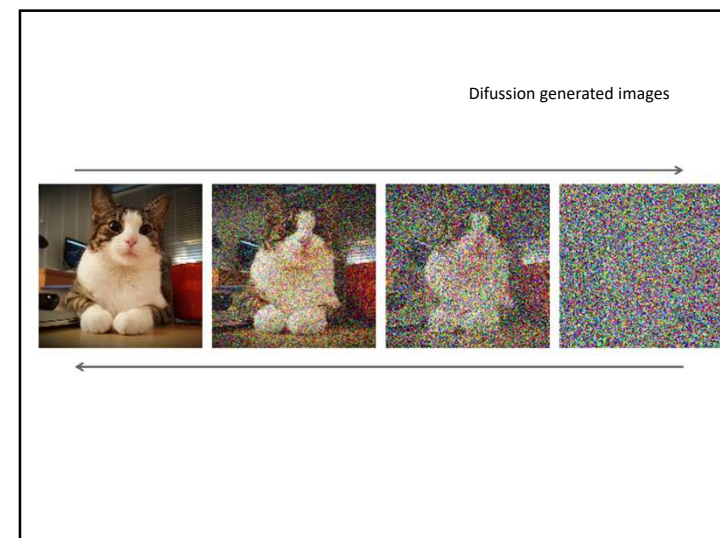
Object detection using R-CNN





CV tasks

- **Content-based image retrieval** : finding all images in a set of images which have a specific content
- **Pose estimation**: estimating the position or orientation of a specific object relative to the camera
- **Human activity recognition** : recognizing the activity from a series of video frames,



Natural Language Processing (NLP)

- Methods for processing and analyzing large amounts of natural language data for solving tasks such as:
 - **Named Entity Recognition**: Given a text, identify the proper names (such as people or places) and the type of each name.
 - **Relation extraction**: Given a text, identify relationships among NEs (e.g. A is friend of B)
 - **Coreference resolutions**: which words refer to the same object (e.g. **anaphora resolution**: matching pronouns with nouns or names, “Ella ha leído el libro que Ana le regaló”)

NLP tasks

- **Text summarization**
- **Sentiment analysis**
- **Grammatical error correction**
- **Machine translation**
- **Question/answering**
- **Text-to-image generation** (2D), to-scene (3D), to-video

Interesting reading:

“Human Language Understanding & Reasoning”, Christopher D. Manning
<https://nlp.stanford.edu/pubs/manning2022daedalus.pdf>

NLP tasks (speech)

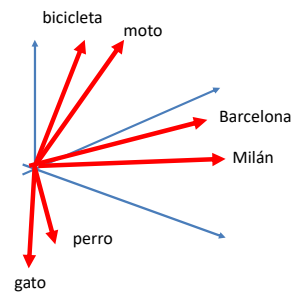
- **Speech segmentation**: given a clip of a person speaking, separate it into words
- **Speech recognition**: Given a clip of a person speaking, generate a textual representation of the speech

Distributional hypothesis in NLP

- Words that occur in similar contexts tend to have similar meanings. This link between similarity in how words are distributed and similarity in what they mean is called the **distributional hypothesis**.
- **Vector semantics** instantiates this linguistic hypothesis by learning representations of the meaning of words, called **embeddings**, directly from their distributions in texts.
- **Static embeddings** are assigned to words, while **contextualized embeddings** are assigned to words in a given context.

Word embeddings

A **word embedding** is a learned representation for text where words that have the same meaning have a similar representation



words that have similar context will have similar meanings

Similarity:

$$\langle x, y \rangle = x^T y$$

$$\langle x, y \rangle / |x| |y| = \cos \theta$$

Transformer (self attention)

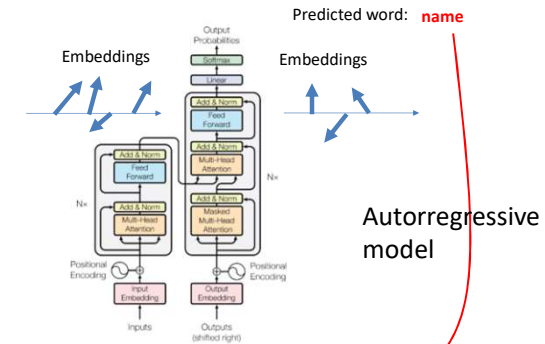


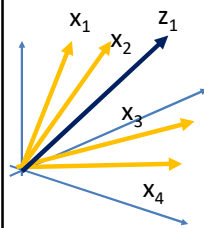
Figure 1: The Transformer - model architecture.

SOS Hola, ¿cómo te llamas? EOS *SOS Hello, what's your XXX*

Ref: A. Vaswani et al, Attention Is All You Need

Self-attention (basic form)

x_i : initial embedding



Compute scalar products of words in **same sentence** with x_1 :

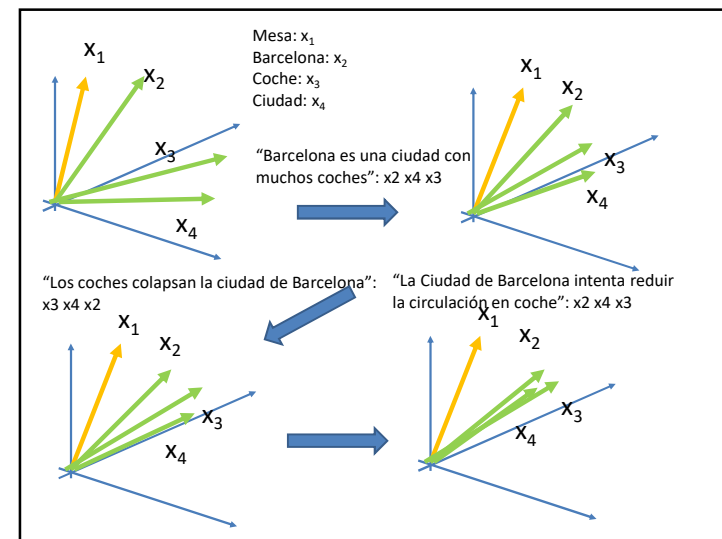
- $\langle x_1, x_1 \rangle = s_1, \langle x_1, x_2 \rangle = s_2, \dots$

Find positive coefficients that add to 1:

- $a_1 = \text{softmax}(s_1), a_2 = \text{softmax}(s_2), \dots$

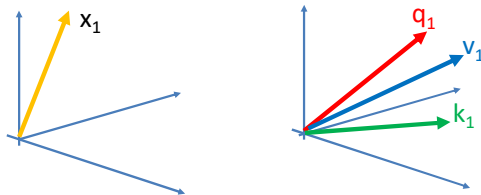
Find **new embedding z_1** by a **linear combination** of words that appear in same sentence:

- $z_1 = a_1 x_1 + a_2 x_2 + \dots$



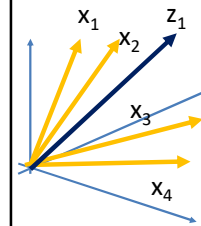
Transformers

Now map the embeddings into a different space



Self-attention (basic form)

xi: initial embedding



Compute scalar products of words in same sentence with x_1 :

- $\langle x_1, x_1 \rangle = s_1, \langle x_1, x_2 \rangle = s_2, \dots$

query

key

Find positive coefficients that add to 1:

- $a_1 = \text{softmax}(s_1), a_2 = \text{softmax}(s_2), \dots$

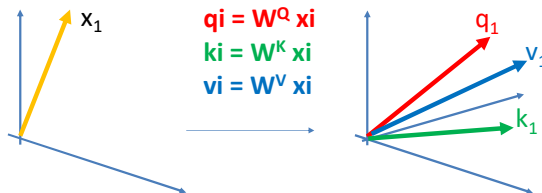
Find new embedding z_1 by a linear combination of words that appear in same sentence:

- $z_1 = a_1 x_1 + a_2 x_2 + \dots$

value

Now map the embeddings into a different space by means of matrices W

Learn these matrices from data



$$q_i = W^Q x_i$$

$$k_i = W^K x_i$$

$$v_i = W^V x_i$$

$$q_i = W^Q x_i$$

$$k_i = W^K x_i$$

$$v_i = W^V x_i$$

Compute scalar products of words with w_1 :

- $\langle q_1, k_1 \rangle = s_1, \langle q_1, k_2 \rangle = s_2, \dots$

query

key

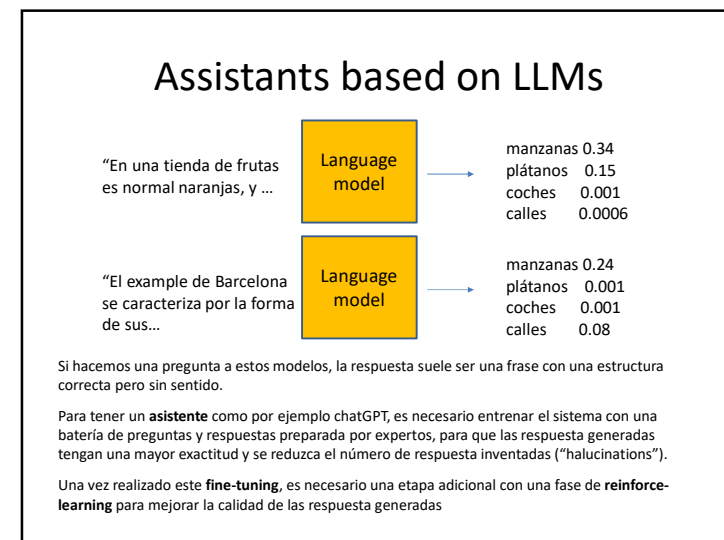
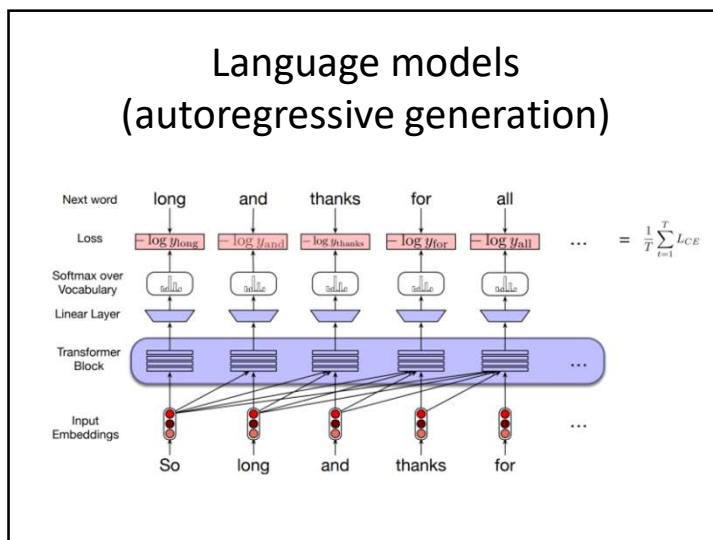
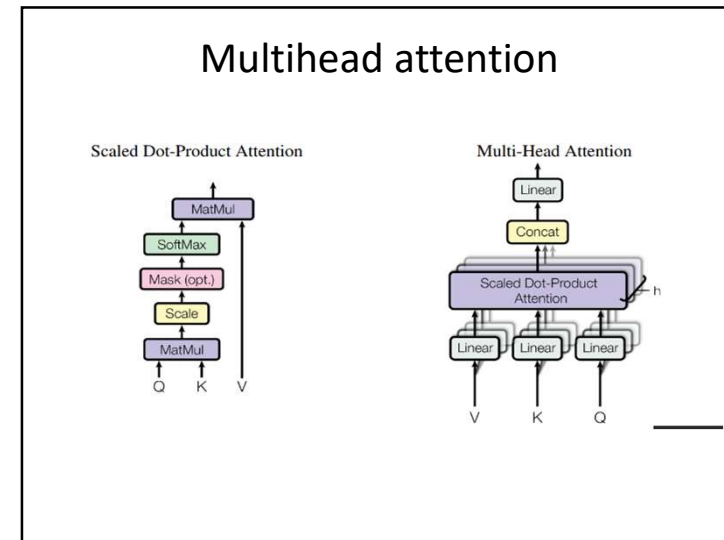
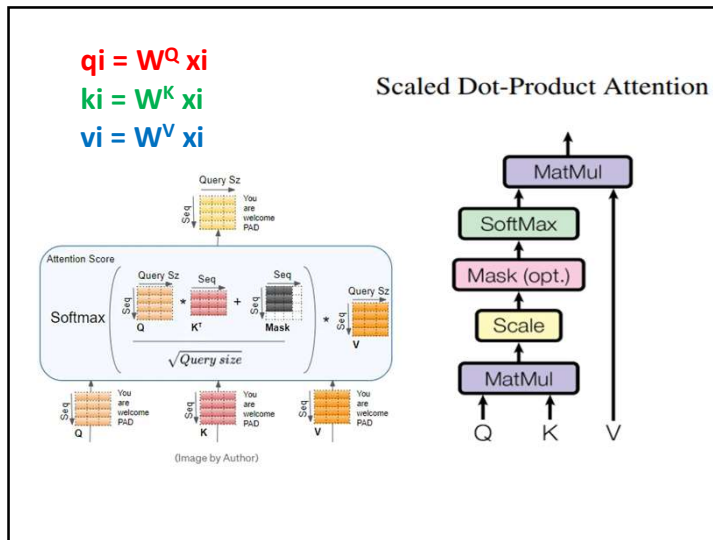
Find positive coefficients that add to 1:

- $a_1 = \text{softmax}(s_1), a_2 = \text{softmax}(s_2), \dots$

Find new embedding A_1 by a linear combination of words:

- $A_1 = a_1 v_1 + a_2 v_2 + \dots$

value



Automatic speech recognition



It's time for lunch!

ASR

- **Humans speaking to machines** (either dictating or talking to a dialogue system) **are easier to recognize than humans speaking to humans. Two humans talking to each other in conversational speech**, for example, for transcribing a business meeting, **is the hardest**.
- **Speech is easier to recognize if it's recorded in a quiet room** with head-mounted microphones than if it's recorded by a distant microphone on a noisy city street, or in a car with the window open.
- **Speech is easier to recognize if the speaker is speaking the same dialect or variety that the system was trained on.** Speech by speakers of regional or ethnic dialects, or speech by children can be quite difficult to recognize if the system is only trained on speakers of standard dialects, or only adult speakers.

English Tasks	WER%
LibriSpeech audiobooks 960hour clean	1.4
LibriSpeech audiobooks 960hour other	2.6
Switchboard telephone conversations between strangers	5.8
CALLHOME telephone conversations between family	11.0
Sociolinguistic interviews, CORAAL (AAL)	27.0
CHiMe5 dinner parties with body-worn microphones	47.9
CHiMe5 dinner parties with distant microphones	81.3
Chinese (Mandarin) Tasks	CER%
AISHELL-1 Mandarin read speech corpus	6.7
HKUST Mandarin Chinese telephone conversations	23.5

Figure 16.1 Rough Word Error Rates (WER = % of words misrecognized) reported around 2020 for ASR on various American English recognition tasks, and character error rates (CER) for two Chinese recognition tasks.

Windowing

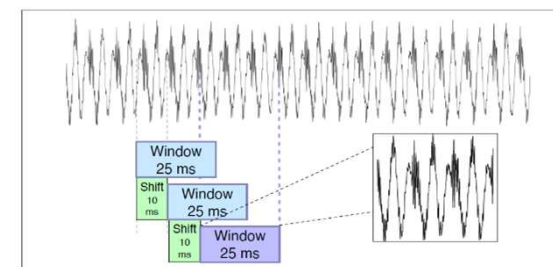


Figure 16.2 Windowing, showing a 25 ms rectangular window with a 10 ms stride.

DFT

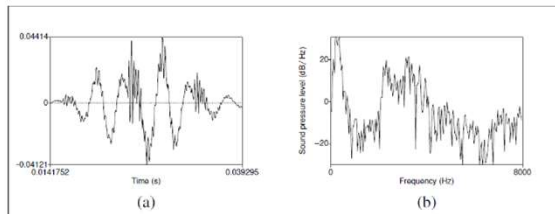


Figure 16.4 (a) A 25 ms Hamming-windowed portion of a signal from the vowel [iy] and (b) its spectrum computed by a DFT.

Mel filter

$$\text{mel}(f) = 1127 \ln\left(1 + \frac{f}{700}\right)$$

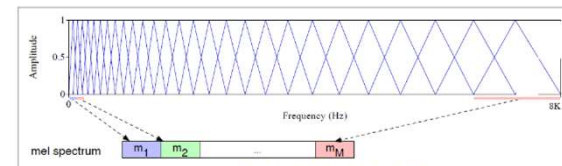


Figure 16.5 The mel filter bank (Davis and Mermelstein, 1980). Each triangular filter, spaced logarithmically along the mel scale, collects energy from a given frequency range.

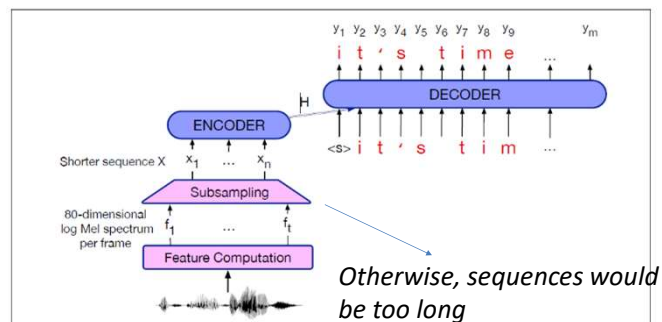
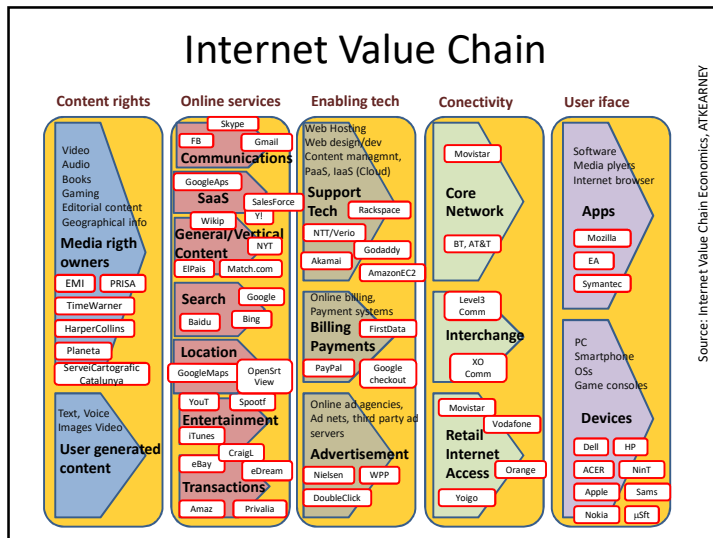


Figure 16.6 Schematic architecture for an encoder-decoder speech recognizer.

Mobile Internet

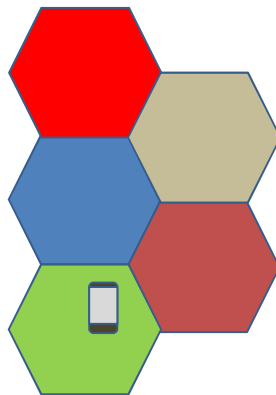
8.- Location & maps



Location systems

- Location awareness is a unique feature of mobile apps
- Location representation
 - Absolute (coordinates), relative a known reference point,
 - Symbolic (name of room, street address etc)
- **Geocoding**: provide geographical coordinates corresponding to a location.
- **Reverse geocoding**: provide a location from geographical coordinates
- Scale
 - Worldwide (works everywhere) or local.
 - **Outdoor** or **indoor**
- Active or Passive (client cooperates or not)

Cell of Origin (Cellular, WiFi, BT)



In its simplest form it only indicates cell
Easy to implement
Coarse granularity (e.g. multistory
Structure with floor-to-floor cell
overlap)

Use RSSI to better determine location

Beacons, Indoor

Identification: Beacons (Bluetooth-LP)



Launch
application

Advertisement

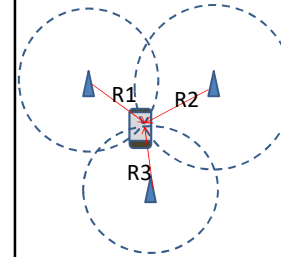
Prefix	UUID	Major	Minor
--------	------	-------	-------

Range: meters
Autonomy: months-year
Cost: 5-30 Euros

Wifi scanning



Distance-based (lateration): ToA



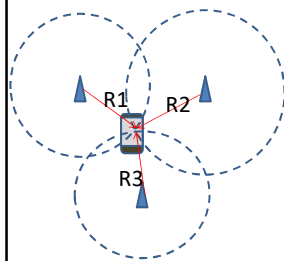
Time of Arrival (ToA) =>

$$T_{\text{received}} - T_{\text{transmitted}}$$

Requires **precise time synchronization between stations and device** (error of 100 ns => 30 m)

Problems with multipath

Distance-based (lateration): TDoA



Time Difference of Arrival

$$TDoA_{1-2} = |T_1 - T_2| = K_{12}$$

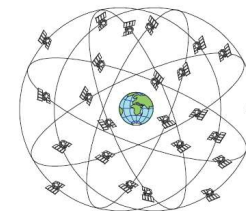
K12 is used to draw an hiperbola.
K13 is used to draw other hiperbola,
and finding the intersection

Requires precise time synchronization between stations/satellites =>
Important to control shifts in clocks

GPS

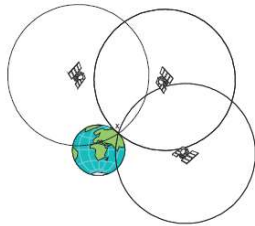
Outdoor location: GPS

- Initiated by US army in 1970 and fully operational in 1995 Trilateration with at least 4 satellites.
- Constellation of 30 satellites moving in different orbital planes



Outdoor location: GPS

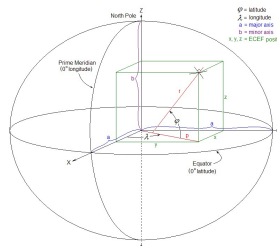
- Distance measured in terms of propagation time of a radio signal
- Extremely accurate atomic clocks in satellites



GPS

- Satellites send signals with codes identifying satellite number and satellite time
- Comparing the arrival times of signals from different satellites, and provided that the satellite orbits are known, the coordinates and height of the receiver can be computed with high accuracy

Assume ideal case



Source: <http://en.wikipedia.org/wiki/ECEF>

- Satellites follow keplerian orbits with no perturbations
- Earth perfect 3D ellipsoid
- c constant (no atmosphere)
- Accurate clock in receiver, with possible time offset dt .

Assume ideal case

$\Delta t_k = t_{r_k} - t_{t_k}$

$\sqrt{(X^k - X)^2 + (Y^k - Y)^2 + (Z^k - Z)^2} = c(\tau - \Delta t_k) = r_k$

$k = 1 \dots 4$

Rx time of signal of satellite k according to local clock
Satellite coordinates at tx time (keplerian orbits + correction sent in ephemeris)
Tx time of signal of satellite k according to satellite clock (recorded in the received signal)
Offset of local clock with satellite clock (unknown)

- Find (X,Y,Z) (coordinates) and τ (clock offset)
- If more than 4 equations are available, find least square solution.
- Using geometric model of earth, find latitude, longitude and height

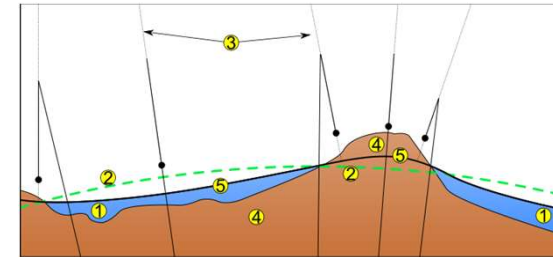
Source: https://www.courses.psu.edu/aersp/aersp055_r81/satellites/gps_details.html

Real situation

- Satellites do not follow exact keplerian orbits
 - correction: ephemeris for each satellite, sent by the satellite through a low speed channel)
- Atmosphere modifies light speed and direction of rays (ionosphere, troposphere)
- Obstacles and multipath of satellite signals
- Inaccurate clock in receiver
- Induced errors
 - Selective Availability, not in use today

Real situation

- Earth not a perfect 3D ellipsoid



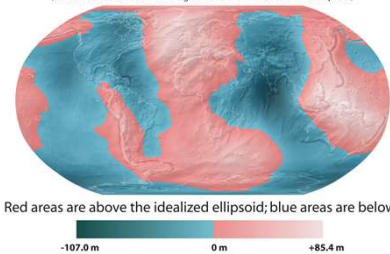
Source: <http://en.wikipedia.org/wiki/Geoid>

Geoid: theoretical sea level following plumbing line (5)

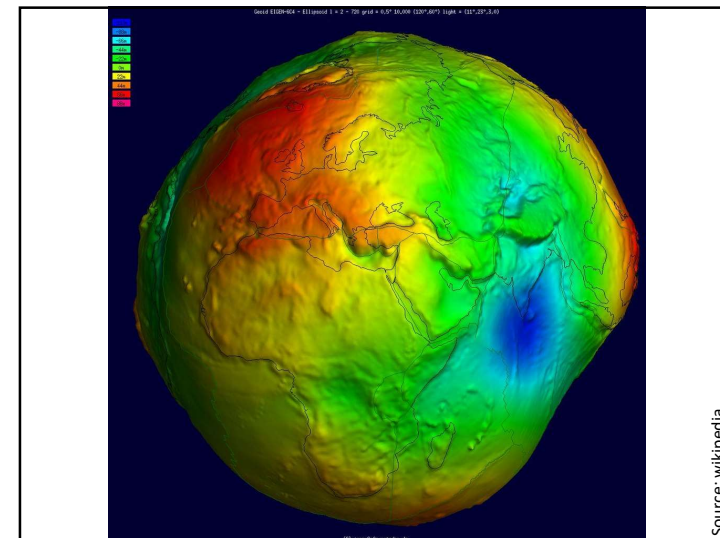
Real situation

- Correction: e.g. Datum of EGM96 geoid. Use spherical harmonics to approximate the shape of the geoid. (i.e., use coefficients of a truncated Fourier series)

Deviation of the Geoid from the idealized figure of the Earth
(difference between the EGM96 geoid and the WGS84 reference ellipsoid)



Source: <http://en.wikipedia.org/wiki/Geoid>



Source: wikipedia

Accuracy & adquisition time

- Around ± 15 m in Barcelona
- Adquisition time: around 1 minute in cold-start
 - Instead of downloading ephemeris from the satellite, (very slow channel) download them from a server (**Assisted GPS**)
- Losses very easily coverage in indoor
- Improve accuracy with **Differential GPS** (match measurements with measurements from a point with known exact positioning)